The Economics of a Trade Policy for Peace

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Abstract

Countries may use trade agreements to promote peaceful relations among allies and advance geopolitical interests. I study the U.S.-fostered QIZ agreement, which grants Egyptian firms tariff-free access to the U.S. market conditional on sourcing a minimum share of inputs from Israel. I examine the agreement's economic effects and how the design of its input requirement serves its geopolitical objectives. Using difference in differences strategies, I document three empirical findings. First, the agreement generated persistent increases in Egyptian exports to the U.S. and imports from Israel. Second, firms participating in the agreement also expanded exports to non-U.S. destinations. Third, these export spillovers are associated with increased international certification. To quantify its welfare effects and evaluate alternative designs, I develop a quantitative trade model where firms endogenously choose to utilize the QIZ agreement. The quantitative results suggest that lowering the Israeli input requirement increases Egyptian exports to the U.S and may increase imports from Israel by expanding firm participation in the agreement. Using model estimates in a bargaining exercise, I show why reducing the input requirement may be politically infeasible. The findings reveal a central trade-off: stricter sourcing requirements may reduce the very trade linkages they intend to promote.

Keywords: Trade policy, Geoeconomics, Industrial Policy, Development

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1 Introduction

The recent rise of geopolitical competition has increased interest in economic policies aimed at securing strategic advantage. Part of this effort is ensuring that relationships among allies remain stable and peaceful. Facilitating trade agreements can contribute to reducing the risk of conflict among allies and induce them to cooperate thus enhancing the geopolitical standing of third party governments. Yet the economic effects of such geopolitically motivated trade policies, as well as how their design shapes their effectiveness, remain insufficiently understood. I address this by studying the Qualifying Industrial Zones (QIZ) agreement, a trade agreement brokered by the United States to stabilize and strengthen relations among its allies in the Middle East.

Under the QIZ agreement, firms located within designated areas in Egypt and Jordan receive duty-free access to the U.S. market conditional on sourcing a minimum share of inputs from Israel. This design reflects the United State's geopolitical objectives in the Middle East which include maintaining peace between its Arab partners and Israel through deepening trade between them. However, the input requirement embedded in the QIZ agreement creates a trade-off. While it requires participating firms to purchase inputs from Israel, it also raises the cost of utilizing the agreement, potentially discouraging firms from participating and limiting the trade relationships the agreement seeks to promote. This makes the QIZ agreement a unique setting for analyzing how trade incentives and input-sourcing rules interact to advance geopolitical objectives. The central research question of this paper is therefore:

How do geopolitically motivated trade agreements affect trade and firms outcomes, and how does the design of input-sourcing requirements shape the effectiveness of these agreements?

In this paper, I evaluate the economic effects of the QIZ agreement in Egypt, focusing primarily on trade and firm outcomes. I then develop a multi-sector heterogeneous firm trade model and calibrate it to the Egyptian economy. The model is then used to quantify the welfare effects of the QIZ agreement on Egypt and investigate the role of input requirements in advancing the objectives of the agreement.

The Qualifying Industrial Zone (QIZ) program was authorized by the United States in 1996, with the first agreement between Jordan and Israel signed in 1997 and implemented in 1998. The agreement designated specific areas within Jordan as QIZs, allowing firms located in these areas, to export to the United States tariff free and quota free (Pelzman, 2011). This preferential access was conditional on the use of a specified minimum share of inputs sourced from Israel. In effect, the agreement extended the benefits of the United States–Israel Free Trade Agreement to eligible firms operating within designated zones in Jordan, provided they incorporated Israeli inputs into their production process. In 2004, Egypt entered a similar agreement with Israel and the US, and specific areas within Egypt were designated as QIZs. Following the introduction of the QIZ agreement aggregate

Egyptian exports to the United States increased simultaneously with imports from Israel. Although clear in the aggregate data, these developments alone cannot establish whether the agreement caused the observed expansion in trade or if it reflected broader trends in international trade.

Combining Egyptian product-level trade data, firm-level customs data, and firm surveys, I evaluate the economic effects of the QIZ agreement using difference-in-differences strategies. The subsequent empirical analysis yields three main findings:

First, the QIZ agreement led to an expansion of trade and economic activity among exposed products and industries. To identify these effects, I exploit variation in products' and industries' exposure to the benefits of the agreement, determined by tariffs which Egyptian exporters faced in the U.S. market prior to the introduction of the QIZ program. Products facing higher initial tariffs benefited more from the conditional tariff elimination offered by the agreement. These pre-agreement Most Favored Nation (MFN) tariffs provide a predetermined measure of the potential benefits from QIZ participation. Using a triple difference in differences strategy exploiting variation in exposure, exporting destinations, and the timing of the agreement (Ortiz-Villavicencio and Sant'Anna, 2025), I find that Egyptian exports of more exposed products to the U.S increased significantly after the agreement relative to less exposed Egyptian products and relative to the exports of other countries to the U.S. market. I also find that Egyptian imports from Israel increased substantially for products supplied to Egyptian industries that were more exposed to the agreement, consistent with the agreement's Israeli input-sourcing requirement. At the industry level, more exposed industries experienced substantial increases in output and employment relative to less exposed industries. Taken together, these findings suggest that the QIZ agreement expanded Egyptian exports to the U.S. and imports from Israel, while stimulating industrial activity and labor demand.

Second, I find that the effects of the agreement extended beyond the U.S. market, generating positive spillovers to firms' exports to other destinations. To investigate these firm level spillovers, I use Egyptian customs data from 2005 to 2016 and exploit the timing of firms' first imports from Israel as a proxy for entry into the QIZ program. Estimating event-study models, I find that firms beginning to import from Israel experience substantial and persistent increases in exports both to the United States and other destinations relative to control firms. These increases are accompanied by an expansion in the number of exported products and the number of destinations served. The results are robust to matching treated and control firms on pre-event outcomes, alternative regression specifications, and placebo exercises involving sectors that did not benefit from the agreement. The findings suggest that the benefits of preferential access to the U.S. market extended beyond the direct tariff savings provided by the agreement.

Third, I find suggestive evidence that international certification may have contributed to these export spillovers. Using the World Bank Enterprise Surveys, I compare changes in certification, foreign ownership, foreign-licensed technology, and fixed-asset investment across Egyptian exporting firms with different expo-

sures to the agreement. I find that international certification increases among exporting firms that are more exposed to the QIZ agreement. However, I find no corresponding positive effect of exposure to the agreement on exporting firms' share of foreign ownership, foreign technology adoption, or investments in fixed assets. The increase in certification is consistent with firms meeting international buyers' requirements, thus reducing their costs of serving foreign markets and expanding to multiple destinations.

Taken together, the three main findings indicate that the QIZ agreement did not only stimulate trade and industrial activity in the sectors directly benefiting from preferential access, but also generated broader spillovers through changes in firms' access to international markets. These findings raise the question of how the observed expansion in trade and firm activity translates into aggregate welfare gains. Importantly, these gains depend on which firms find it profitable to participate in the agreement. While preferential access through the QIZ creates incentives to expand exports, the Israeli input-sourcing requirement increases the cost of participation and may therefore limit the number of firms benefiting from the program. Thus, the same requirement designed to strengthen trade between Egypt and Israel may also constrain the economic benefits of the agreement. These considerations motivate the quantitative model that follows. The aim is to quantify the aggregate welfare consequences of the agreement and examine how alternative designs of the Israeli input-sourcing requirement can shape trade between Egypt and Israel and, ultimately, understand the agreement's effectiveness in advancing the United States' geopolitical objectives.

I therefore develop a multi-sector heterogeneous-firm model of the Egyptian economy which embeds the QIZ agreement. In the model, Egypt is a small open economy facing exogenous demand conditions from the United States and the rest of the world, while domestic consumers have constant elasticity of substitution preferences over product varieties. On the production side, a continuum of firms differ in productivity across 15 manufacturing sectors that face different U.S. most-favored-nation (MFN) tariffs. To enter the market firms must pay a fixed entry cost and in order to export, a destination specific fixed cost must be paid as well. Firms exporting to the United States choose between paying the applicable MFN tariff or obtaining duty-free access through the QIZ program. The latter saves the tariff but requires higher cost Israeli input sourcing and additional QIZ fixed costs. Therefore, whether a firm chooses to use the QIZ program depends on whether the tariff savings outweigh those costs. The model thus links QIZ participation to both U.S. exports and the broader economic gains documented in the data.

In the calibrated model, the agreement generates positive but modest aggregate welfare gains because export growth is concentrated among a small group of firms in the textiles and apparel sector. The export gains are therefore uneven across firms and sectors, while their effect on economy-wide welfare is positive but small. This concentration of benefits raises the question of whether a different Israeli input requirement could expand participation and increase the gains from the agreement.

The Israeli input requirement creates a policy trade-off. A higher required

input share increases sourcing for Israel per each participating firm but raises the cost of complying to the program's requirement, thereby discouraging other firms from utilizing it. Consequently, a more stringent requirement does not necessarily increase aggregate imports from Israel, as this depends both on the amount of Israeli inputs sourced by participating firms and on how many firms choose to participate. To examine this trade-off, I use the calibrated model to investigate the effects of alternative levels of the Israeli input requirement. I find that lowering the requirement from 10.5% to 9% increases Egyptian exports to the U.S. by 6.8% and increases Israeli imports by 1.1%. In this counterfactual, sourcing by the additional participants that choose to utilize the agreement after lowering the requirements offsets the lower required sourcing per firm, so Egypt's aggregate imports from Israel increase. Therefore, relaxing the requirement by 1.5 percentage points can improve Egyptian welfare while preserving the geopolitically motivated trade link the agreement seeks to promote.

Finally, I use the model's counterfactual outcomes in a bargaining exercise over the Israeli input requirement. The aim of this exercise is to investigate whether reforms to the QIZ agreement can be agreed upon by the three parties. In this exercise, I assume that Egypt values its exports, Israel values sales of its inputs to Egypt, and the United States values the increased bilateral trade between Egypt and Israel while disfavoring greater Egyptian exports to the U.S. Given each country's objectives, Egypt and Israel bargain over the input requirements subject to U.S. approval. Although relaxing the requirement may increase both Egyptian exports and Israeli input sales, it can fail to gain U.S. approval if the additional Israeli sales to Egypt are too small relative to the increase in Egyptian exports to the U.S. This highlights how reforms benefiting both Egypt and Israel may nevertheless be rejected because the additional geopolitical advantage it offers the U.S. is minimal.

Taken together, the results highlight how the design of input-sourcing requirements shapes both the economic effects of trade agreements and their effectiveness in advancing geopolitical objectives. While the requirement encourages firms to purchase inputs from Israel, its costs can limit participation and the broader welfare gains generated by the agreement. Relaxing the requirement can therefore improve economic outcomes without weakening the intended bilateral trade relationship. However, it might be infeasible to enact such a reform because of the diverging objectives of the countries participating in the agreement.

This paper contributes to four strands of literature:

First, the paper contributes to the literature on rules of origin and the design of preferential trade agreements. Earlier work links the costs of complying with origin rules to the use of trade preferences (Anson et al., 2005; Sytsma, 2021). Conconi et al. (2018) show that NAFTA rules of origin reduced intermediate-input imports from nonmember countries relative to imports from agreement partners. Related preliminary firm-level evidence finds that rules of origin under the EU–UK Trade and Cooperation Agreement distort input sourcing and raise production costs (Conconi et al., 2026). Sytsma (2022) shows that relaxing restrictive sourcing rules can substantially increase firm exports. While Head et al. (2024) show

that regional sourcing responds non-monotonically to rules of origin requirements because firms will not utilize the preference when compliance with its sourcing rules becomes too costly. I contribute by examining this trade-off in an agreement whose sourcing requirement serves an explicit geopolitical objective. My quantitative model evaluates how changes in the Israeli input requirement affect Egyptian firm utilization of the agreement, exports, and welfare, alongside imports from Israel. This allows me to assess how the requirement's stringency shapes its ability to generate the bilateral trade links it is intended to promote.

Second, the paper contributes to the literature on export spillovers across destinations (Bernard et al., 2003, 2010, 2011; Bustos, 2011; Das et al., 2007). Defever et al. (2015) show that firms' entry into an export market can facilitate subsequent entry into neighboring markets. Defever and Ornelas (2026) find that removing textile quotas by the United States and Europe increased Chinese firms' exports into other markets unaffected by this policy change. I contribute by documenting how preferential U.S. access under the QIZ agreement affected Egyptian firms' exports to other destinations. I provide survey evidence consistent with international certification as a possible channel for these wider export gains.

Third, the paper contributes to the literature on geoeconomics (Caldara and Iacoviello, 2022; Mohr and Trebesch, 2025). Clayton et al. (2026) analyze theoretically how governments use economic relationships to pursue their geopolitical objectives. Limão (2007) studies preferential access exchanged for cooperation on nontrade issues such as environmental cooperation, while Mattoo et al. (2024) examine how geopolitical considerations alter trade cooperation and bargaining outcomes. I contribute by studying a trilateral arrangement where the United States uses access to its market to encourage trade between two of its regional allies. I connect the economic effects of the sourcing requirement to a bargaining exercise in which Egypt and Israel negotiate subject to U.S. approval. The exercise examines how bargaining power, political objectives, and economic preferences shape the negotiated input requirement.

Fourth, the paper relates to the literature on industrial policies (Abagna et al., 2025; Barteska and Lee, 2024; Bianchi and Giorcelli, 2023; Kline and Moretti, 2014; Neumark and Simpson, 2015; Siegloch et al., 2025). Manelici and Pantea (2021) and Lane (2025) document how targeted interventions to specific sectors expand firms and industries. Research on special economic zones in developing countries has also examined how hese zones induce local structural change and industrial expansion, including Wang (2013) for China and Gallé et al. (2024) for India (Lu et al., 2019; Tafese et al., 2025). I contribute by examining how place-based access to a foreign market generates economic gains. Although the QIZ agreement does not target specific industries, differences between U.S. tariffs across industries and the cost of satisfying the QIZ sourcing requirements concentrate participation in textiles and apparel. Combining firm and industry evidence, I study how these incentives affected these exposed firms and industries.

The paper also builds on earlier studies and reports about the QIZ agreement in Egypt and Jordan (Yadav, 2007; Nugent and Abdel-Latif, 2010; Saif, 2006; Pelzman, 2011). I extend this work by combining Egyptian customs microdata with

variation in exposure to the agreement across industries and firms to estimate trade responses, investigate firm-level export spillovers to third market destinations, and evaluate the aggregate welfare effects of the agreement and its alternative sourcing requirements. The quantitative model and bargaining exercise connect these responses to the design and negotiation of the agreement.

The remainder of the paper is organized as follows. Section 2 provides background on the history of the QIZ agreements and presents some descriptive statistics. Section 3 describes the data. Section 4 outlines the empirical strategies. Section 5 presents the results. Section 6 looks into the mechanisms enabling export expansion to destinations beyond the U.S. Section 7 introduces a multi-sector heterogeneous-firm model embedding the QIZ agreement. Section 8 uses the model to quantify the welfare effect of the agreement, study alternative Israeli input requirements, and investigate the geopolitical value of the agreement through a bargaining exercise. Section 9 concludes.

2 Institutional Context and History

The Arab–Israeli conflict in the 20th century involved multiple wars between Arab countries and Israel. During the conflict, particularly after the 1960s, Israel was aligned with the United States of America, while several Arab republics were allies of the Soviet Union (Khalidi, 2020). In 1978, Egypt and Israel signed the Camp David Accords, which paved the way for the 1979 Egypt–Israel Peace Treaty, formally ending the state of war between them following the October War of 1973. Later, in 1994, Jordan and Israel signed the Wadi Araba Peace Treaty, following the 1993 Oslo Accords between the Palestine Liberation Organization and Israel. Both peace treaties were brokered by the United States, which had a strategic and geopolitical interest in aligning these Middle Eastern countries more closely within its sphere of influence.

The conflict inflicted significant economic and political costs on the Arab states involved. To further consolidate this regional alignment and strengthen the cold peace between Israel and its Arab neighbors, the U.S. established the legal framework for the QIZ program in 1996. Jordan's first QIZ was designated in 1998, while Egypt joined the program in 2004. The agreement states that upon an agreement between Israel and its Arab partner (Egypt or Jordan), certain regions within each Arab state would be designated as QIZs and would benefit from special trade privileges with the U.S. Specifically, firms producing within these QIZ regions would be eligible to export to the U.S. tariff-free and quota-free upon sourcing a fixed amount of inputs from the country itself and from Israel. Moreover, a substantial transformation of the product must take place within the QIZs. For both Jordan and Egypt, qualifying input and processing costs should total at least 35% of the appraised value of the product upon entry into the U.S. For Jordan, of these 35%, at least 8% should come from Israel (7% for certain high-technology products), 11.7% from Jordan itself, and the remaining 15.3% from any combination of eligible sources, including Jordanian QIZs, Israel, the U.S., and the West Bank and

Gaza Strip (Congressional Research Service, 2013). For Egypt, of these 35%, at least 10.5% should come from Israel, another 10.5% from Egypt itself, and the remaining 14% from eligible sources, including Egyptian QIZs, Israel, and the U.S. (Congressional Research Service, 2013).

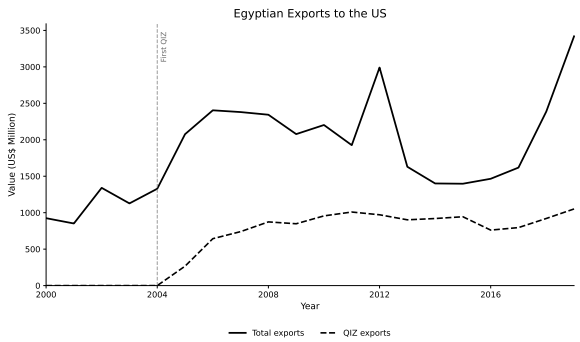
Egypt's entry into the agreement came in 2004, much later than Jordan's participation, which began with the designation of its first QIZ in 1998. Initially, Egypt was skeptical of joining the QIZ arrangement, in part because its textile and apparel exports were operating under the World Trade Organization (WTO) Agreement on Textiles and Clothing (ATC) (WTO, n.d.). However, the expiration of the ATC at the end of 2004 pushed Egypt to explore alternative mechanisms to maintain preferential market access, including the QIZ agreement (Congressional Research Service, 2013). The observed success of the QIZs in Jordan, particularly in boosting exports and creating new jobs, made the idea more appealing, provided it did not provoke public backlash. To preempt such concerns, the Egyptian government secured the support of the Islamic institution of Al-Azhar before formally entering the agreement (Raya, 2006).

QIZs were designated in certain areas of Jordan and Egypt successively in a staggered manner. A key difference between the two countries lies in the spatial design of the zones. In Jordan, QIZs were typically fenced industrial areas located in rural regions. In contrast, Egypt designated entire administrative districts as QIZs, meaning that any firm operating within those districts could potentially benefit from the agreement. Figure A1 in the appendix shows the geographic distribution of QIZ regions across Egypt.

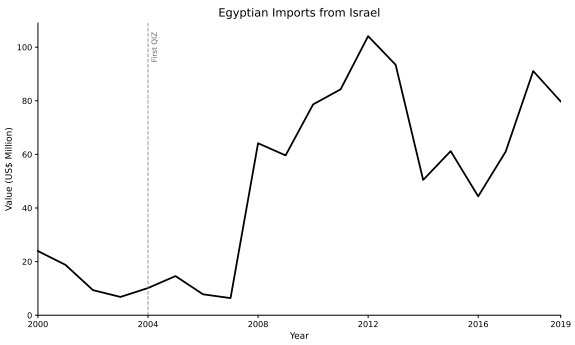
Another important aspect is that, initially, firms located within Egyptian QIZs needed individual approval from a joint QIZ committee in order to qualify for the agreement. This committee included representatives from Egypt, Israel, and the United States (QIZ Egypt, n.d.). In 2013, the United States agreed to grant QIZ status to all eligible firms and facilities located within the designated QIZ districts upon applying.

The daily administration of the program is overseen by an official unit within both Egypt and Jordan, known as the QIZ Unit. These units are responsible for registering firms, issuing and renewing certifications, and facilitating all administrative procedures related to participation in the program and ensuring that its requirements are met. In addition, according to the QIZ protocols, a joint committee composed of Israeli and Egyptian or Jordanian representatives is required to meet on a quarterly basis to coordinate and oversee the implementation of the agreement (QIZ Egypt, n.d.). However, since Jordan signed its own Free Trade Agreement with the United States in 2000—which entered into force in 2001 and was fully implemented by 2010—the QIZ agreement in Jordan became less attractive to exporters. Unlike the QIZ framework, the U.S.–Jordan FTA does not impose input sourcing requirements from Israel, making it a more flexible option for Jordanian exporters (Saif, 2006; Congressional Research Service, 2013).

Figure 1, panel (a), shows in the black line total Egyptian exports to the U.S. and in the dashed line the QIZ exports out of these total exports before and after the implementation of the QIZ agreement. We observe a substantial increase in



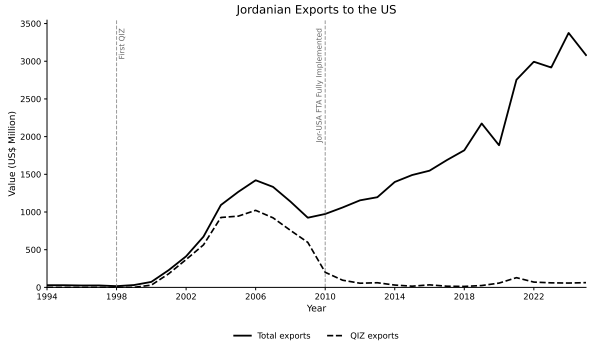
(a) Egyptian Total and QIZ Exports to U.S.



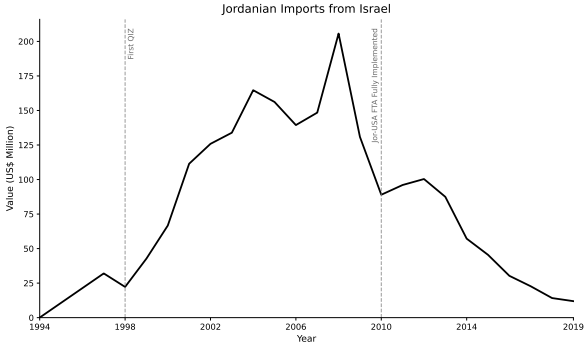
(b) Egypt's Total Imports from Israel

Figure 1

the value of QIZ exports from Egypt to the United States following the initial rollout of QIZs in December 2004–January 2005. By 2011, total QIZ exports rose to approximately $1 billion and constituted more than 50% of total Egyptian exports to the U.S. Panel (b) shows the total value of Egyptian imports from Israel over time. A sharp increase in Israeli imports after the implementation of the QIZ agreement can be observed relative to the synthetic counterpart. This rise in imports exhibits a two-year lag before accelerating, coinciding with the surge in Egyptian textile and apparel exports to the U.S. This reinforces the interpretation that the export growth was driven by the QIZ agreement. At their peak in 2012, imports from Israel reached approximately $100 million—an increase of about $85 million from 2004. Given that Egyptian firms were required to source at least 10.5% of inputs from Israel, the ratio of the increase in U.S. textile exports to the increase in imports from Israel is close to this mandated share, providing further evidence that these trade patterns were indeed a direct result of the QIZ agreement.



(a) Jordan's Total and QIZ Exports to the U.S.



(b) Jordan's Imports from Israel

Figure 2

To investigate whether the QIZ input requirement imposed a binding constraint on firms within Egypt and Jordan, taking a look at the Jordanian case is indicative. In contrast to Egypt, Jordan signed a direct Free Trade Agreement with the U.S. in 2002, which was fully implemented in 2010. This enabled Jordanian firms that wanted to export to the U.S. to choose between exporting under the QIZ

program by satisfying the Israeli input requirement or exporting under the FTA with no Israeli input requirement. Figure 2, panels (a) and (b), presents the same export and import dynamics as Figure 1, but for Jordan. The pattern is similar. QIZ exports to the U.S. increased by around $1 billion by 2004, six years after the first QIZ was designated in 1998, representing more than 95% of total exports to the U.S. Total exports to the U.S. were negligible before the QIZ agreement and rose substantially afterward. Although QIZ exports started declining after 2006 as the rollout of the FTA started affecting QIZ industries, total exports to the U.S. continued increasing well after 2010, reaching more than $3 billion. Figure 2(b) shows a corresponding rise in imports from Israel after 1998, peaking at approximately $200 million. However, these imports declined sharply after 2006 as the U.S.–Jordan Free Trade Agreement's staged tariff reductions increasingly offered exporters an alternative route to preferential market access, culminating in the agreement's full implementation in 2010. Once fully implemented in 2010, firms shifted to the more profitable FTA regime, suggesting that QIZ-based regional integration was a second-best arrangement for complying firms where trade integration was only tied to market access.

The industries benefiting the most from the QIZ agreement in Egypt are textiles and apparel. As can be seen in Figure 3, textile and apparel products constituted more than 90% of QIZ exports to the U.S throughout time.

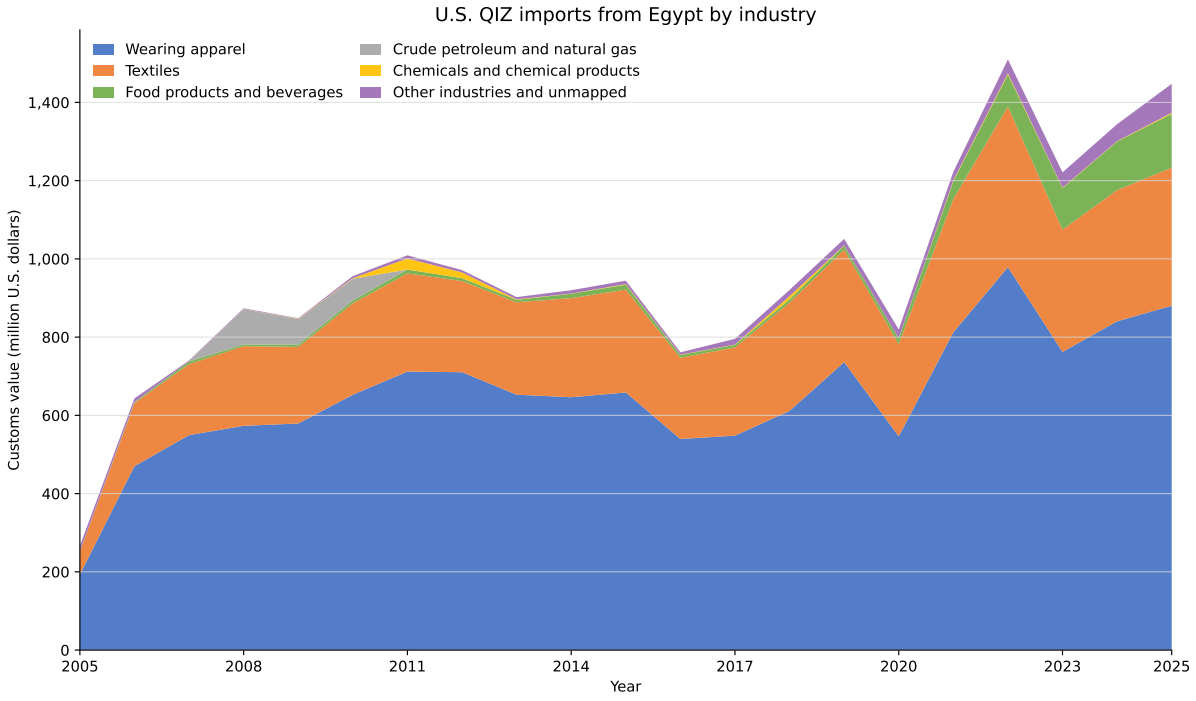


Figure 3: Egypt's QIZ Exports by Industry

3 Data

I draw on several complementary data sources to examine the effects of the QIZ agreement on Egyptian exports, manufacturing activity, and firm-level adjustment. The main analysis combines product-level trade data, industry-level manufacturing statistics, administrative customs records, and firm-level survey data. These sources allow me to examine changes in trade and manufacturing outcomes and investigate potential mechanisms of firm adjustment, including certification, input sourcing, and foreign investment linkages.

Product- and Industry-Level Data

To examine the effects of the QIZ agreement on trade, I use detailed trade data from UN Comtrade covering 1996–2020 and U.S. customs data recording imports from Egypt and the World at the six-digit (HS6) product level for 1997–2020. I complement these records with actual 2003 and 2004 applied U.S. MFN tariffs from the WTO Integrated Database. Together, these data allow me to examine changes in Egyptian export performance across products while accounting for differences in U.S. tariff protection.

For Egypt's imports from Israel, I use Israel-reported exports to Egypt and the World from UN Comtrade for 1996–2020. I combine these flows with Egypt's 2000/01 input-output table to measure how input-supplying industries are linked to downstream industries benefiting from U.S. tariff preferences. The baseline uses the pre-agreement 2000/01 production linkages; the Appendix reports results using the later 2008/09 table. The import sample retains all mapped HS6 merchandise, including nonmanufacturing goods.

To investigate whether changes in export activity are accompanied by broader adjustments in domestic manufacturing, I use industry-level data from the United Nations Industrial Development Organization (UNIDO). The industry event studies use the expanded 1998–2021 series, combining observed data with extrapolated values for missing detailed industry observations. This provides me with annual information on Egyptian manufacturing industries at the three-digit industry level. The main outcomes include gross output, value added, gross fixed capital formation, employment, and total wages and salaries. This allows me to examine changes in manufacturing production, investment, and labor-market outcomes at the industry level.

Firm-Level Data

Two firm-level datasets are used to examine firm responses to the QIZ agreement: administrative customs records from Egypt and the World Bank Enterprise Surveys. The customs records allow me to track changes in firms' trade activities, while Enterprise Surveys provide information on some firm characteristics and practices that can shed light on potential mechanisms driving the trade responses to the QIZ agreement

Administrative customs records. I use firm-level import and export data for 2005–2016 compiled by the General Organization for Export and Import Control (GOEIC) and accessed through the Economic Research Forum (ERF). These administrative records cover the universe of Egyptian firms engaged in international trade and report import and export values by firm, trading partner, HS6 product, and port used.

Using unique firm identifiers, I link import and export records and construct a firm-level panel of trading activity. This allows me to examine changes in firms' export behavior following their first observed import from Israel, a particularly relevant observation given that the QIZ agreement mandates an Israeli-input requirements. I also distinguish between incumbent exporters and new entrants.

World Bank Enterprise Surveys. Mechanisms through which the QIZ agreement affects firm behavior can be investigated using firm-level data from the World Bank Enterprise Surveys for Egypt. I exploit multiple survey rounds (2004, 2007, 2008, 2013, 2016, 2020) and their panel component. The surveys provide detailed information on firm characteristics that are not captured in the administrative customs records. I focus particularly on certification, fixed asset investments, the adoption of foreign technology, and foreign ownership of the firm as potential mechanisms facilitating trade adjustment.

4 Empirical Strategies

To identify the economic impact of QIZs, I use two different empirical strategies.

4.1 Triple-Difference Analysis of Trade Flows

I exploit variation in products' and industries' exposure to the QIZ agreement to estimate its effects on Egyptian trade and manufacturing activity. The main source of variation is the U.S. most-favored-nation (MFN) tariff that Egyptian exporters faced before the agreement. Since the QIZ program eliminated these tariffs for eligible exports, products facing higher initial tariffs stood to benefit more from the agreement. I use this variation to examine two trade responses: the increase in Egyptian exports to the United States and the increase in imports from Israel induced by the agreement's input-sourcing requirement.

Measuring exposure to the agreement. For exports to the United States, I measure each product's exposure using its pre-agreement MFN tariff:

$$\text{Exposure}_p^{\text{exp}} = \text{MFN}_{p,2004}, \tag{1}$$

where p denotes an HS6 product and tariffs are measured in percentage points. The intuition is straightforward: a higher initial tariff implies a larger potential benefit from obtaining duty-free access through the QIZ program, and therefore a stronger incentive to satisfy its sourcing requirements. I use the 2004 applied

MFN tariffs in the baseline specification and 2003 tariffs as a robustness check. I harmonize tariff and trade classifications using product concordances and take the arithmetic mean when multiple tariff lines map to the same HS6 product. The analysis covers manufacturing products in ISIC Revision 3 divisions 15–37.

For imports from Israel, the relevant exposure depends on how strongly an imported product is linked to Egyptian industries benefiting from preferential access to the U.S. market. Under the QIZ agreement, firms seeking duty-free access must source part of their inputs from Israel. I therefore expect Israeli imports to increase more for products supplied to Egyptian industries facing larger initial U.S. tariffs.

To capture these linkages, I use Egypt's 2000/01 input-output table. Let $Z_{ij,2000/01}$ denote intermediate sales from supplying industry i to purchasing industry j. I construct import exposure for each HS6 product h as

$$\text{Exposure}_h^{\text{imp}} = \sum_{q \in \mathcal{P}} \frac{s_{i(h),j(q),2000/01}}{N_{j(q)}} \text{MFN}_{q,2004}, \qquad s_{ij,2000/01} = \frac{Z_{ij,2000/01}}{\sum_\ell Z_{i\ell,2000/01}}, \quad (2)$$

where $i(h)$ is the industry supplying imported product h, $j(q)$ is the industry producing downstream product q, and N_j is the number of products with available tariffs in industry j. The weight $s_{ij,2000/01}$ measures the share of industry i's intermediate sales going to industry j, which is distributed equally across the latter's products.

This measure assigns greater exposure to Israeli products supplied by industries whose Egyptian customers benefit more from the QIZ tariff preferences. For example, an input-supplying industry selling a large share of its output to apparel producers would receive greater exposure if apparel products faced high initial U.S. tariffs. Each imported HS6 product inherits the exposure of its supplying industry, while trade flows remain measured at the HS6 level. One unit of exposure corresponds to one percentage point of the input-output-weighted tariff index.

Both measures use information determined before the agreement. Export exposure relies on pre-agreement U.S. tariffs, while import exposure combines these tariffs with input-output relationships from 2000/01. This avoids constructing exposure using trade flows or sourcing patterns that may have responded to the agreement. Moreover, U.S. MFN tariffs applied broadly across trading partners and were not set specifically in response to Egypt's participation in the QIZ program. As a robustness check, I also construct import exposure using the 2008/09 input-output table. Since these later relationships may themselves reflect adjustment to the agreement, I report these results in the Appendix.

Empirical specification. To estimate the effects of the agreement on trade, I use a triple-difference strategy that compares products with different levels of exposure, before and after the agreement, across Egypt and other trading partners. The third difference is important because changes in trade following the agreement may reflect broader product-specific shocks rather than the QIZ program itself. By

comparing Egypt with other trading partners, I can control for shocks that affect trade in the same products more generally.

I estimate the following event-study specification separately for exports and imports using Poisson pseudo-maximum likelihood (PPML):

$$\mathbb{E}[Y_{pct} \mid \mathbf{X}_{pct}] = \exp\Big[\sum_{\substack{k \in \mathcal{K} \\ k \neq -1}} \beta_k \left(\text{Exposure}_p^{\text{exp/imp}} \times D_c \times \mathbf{1}\{t - 2005 = k\} \right) + \alpha_{pt} + \gamma_{ct} + \delta_{pc} \Big]. \tag{3}$$

Here, Y_{pct} denotes trade in product p, trading-partner category c, and year t. The indicator D_c equals one for Egypt and zero for the rest of the world. For the export analysis, I use U.S.-reported imports from Egypt and compare them with U.S. imports from all other countries. For the import analysis, I use Israel-reported exports to Egypt and compare them with Israeli exports to all other destinations. The sample covers years 1997–2020. I balance the panels across products, trading partners, and years, retaining zero trade flows, which are accommodated by the PPML estimator.

The agreement entered into force in Egypt in 2005, and I use 2004 as the omitted reference year. Product-by-year fixed effects, α_{pt}, absorb shocks affecting trade in a given product that are common across trading partners. In the export analysis, these include changes in U.S. demand for particular products, while in the import analysis they capture changes in Israeli product supply. Trading-partner-by-year fixed effects, γ_{ct}, control for annual shocks common across products for a given partner, and product-by-partner fixed effects, δ_{pc}, absorb time-invariant differences in trade relationships. The coefficients β_k therefore measure how trade with Egypt changes across products with different exposure levels, relative to other trading partners and to the pre-agreement year.

In my preferred export specification, I further control for trading-partner-by-HS2-by-year fixed effects, $\eta_{c,s(p),t}$, replacing γ_{ct}. This allows Egypt and other suppliers to experience different shocks across broad product categories, so identification comes from tariff variation across HS6 products within the same HS2 chapter. For Israeli imports, I report both the baseline specification and one including trading-partner-by-broad-sector-by-year fixed effects. The broad sectors are agriculture, extraction, textiles and apparel, other manufacturing, and utilities and services. I do that, Since import exposure is common to products within the same input-output industry, and thus including partner-by-input-output-industry-by-year fixed effects would absorb the variation used to identify the effect.

I also estimate the average post-agreement effect by replacing the annual event-time interactions with a single interaction between exposure, Egypt, and a post-2004 indicator.

For the export analysis, I cluster standard errors at the HS6 product level. For imports, I cluster at the input-output industry level, where exposure varies, allowing for correlation across products within the same industry over time. The

2000/01 input-output classification contains 16 traded industries. Tables report clustered standard errors, while event-study figures show pointwise 95% confidence intervals using a $t(G-1)$ reference distribution, where G denotes the number of clusters.

To assess the validity of the identifying assumption, I examine whether more exposed products experienced differential trade changes before the agreement. I jointly test whether the pre-agreement event-study coefficients are equal to zero. Given the limited number of clusters in the import analysis, I complement conventional Wald tests with null-imposed wild cluster score-bootstrap tests following Kline and Santos (2012) and Roodman et al. (2019). The bootstrap re-estimates the fixed effects and post-treatment coefficients under the joint null, using 99,999 Webb-weight draws at the relevant clustering level. I also report results using Rademacher weights.

The identifying assumption is that, absent the QIZ agreement, changes in Egypt's trade relative to other trading partners would not have been systematically related to the pre-agreement exposure measures, conditional on the fixed effects. Although this assumption cannot be tested directly, the event studies allow me to examine whether differential trends were already present before the agreement.

Manufacturing activity. Finally, I examine whether the trade expansion was accompanied by changes in domestic manufacturing activity. Using UNIDO industrial statistics, I compare outcomes across three-digit Egyptian manufacturing industries with different levels of exposure to the agreement. I construct industry-level exposure by mapping product tariffs to industries using United Nations Statistics Division concordances and combining 2004 U.S. MFN tariffs with Egyptian exports to the United States and the world over 2000–2004..

I estimate the following event-study specification by ordinary least squares:

$$\ln Y_{it} = \sum_{\substack{k \in \mathcal{K} \\ k \neq -1}} \beta_k \left(\text{Exposure}_{i,\text{pre}} \times \mathbb{1}\{t - 2005 = k\} \right) + \theta_i + \omega_t + \varepsilon_{it}, \tag{4}$$

where Y_{it} denotes the manufacturing outcome for industry i in year t, while θ_i and ω_t are industry and year fixed effects. The agreement entered into force in 2005, with 2004 as the reference year. The coefficients β_k measure whether industries with greater pre-agreement exposure experienced larger changes in manufacturing activity following the agreement relative to less exposed industries. Unlike the trade regressions, which compare Egypt with other trading partners, this exercise exploits variation in exposure across industries within Egypt.

4.2 Firm-Level Event Study

To examine how QIZ participation affects firms' exporting behavior, I use administrative customs data covering the universe of Egyptian exporters and importers between 2005 and 2016, disaggregated by firm, HS6 product, and partner country. Since the customs records do not directly identify which firms utilize the

QIZ program, I use the timing of firms' first imports from Israel as a proxy for participation. This is motivated by the fact that Israeli sourcing is a requirement for obtaining preferential access to the U.S. market under the agreement. Moreover, since Israeli inputs are more costly than those from alternative suppliers as shown by customs record, firms have an incentive to begin sourcing from Israel only because of the QIZ program. Nevertheless, some firms importing from Israel may not actually participate in the program. To the extent that these firms do not benefit from preferential access, their inclusion in the treated group would tend to attenuate the estimated effects, making the observed export responses a potentially conservative measure of the gains among actual QIZ participants.

I define a firm as treated in year t if it begins importing from Israel in that year, with no recorded imports from Israel in earlier years. Firms that never import from Israel during the sample period constitute the comparison group. Firms already importing from Israel in 2005 are excluded from the treated group, as the beginning of their Israeli sourcing cannot be observed. I construct an annual panel of firms that exported at least once during the sample period, assigning zero exports in years without recorded export activity.

The main analysis focuses on textile and apparel firms, which account for the vast majority of QIZ exports to the United States. I identify these firms by their dominant export product, restricting the sample to products within chapters 50–63. The baseline analysis includes 110 firms whose first Israeli import occurred between 2008 and 2012 and a pool of 1,630 potential comparison firms that never imported from Israel. I also extend the analysis to all eligible firms importing from Israel between 2006 and 2015 as a robustness exercise.

To estimate the dynamic export responses, I compare firms beginning to import from Israel with firms that never do so, separately for each first-import cohort g as suggested by Sun and Abraham (2021). I estimate the following event-study specification:

$$Y_{igt} = \alpha_{ig} + \lambda_{g,s(i),t} + \sum_{\substack{k \in \mathcal{K}_g \\ k \neq -1}} \theta_{gk} D^k_{igt} + \varepsilon_{igt}, \tag{5}$$

where Y_{igt} denotes the outcome of firm i in year t in the comparison associated with cohort g. The indicator D^k_{igt} equals one when a treated firm is k years from its first recorded Israeli import and zero otherwise. The firm-by-cohort fixed effects, α_{ig}, absorb time-invariant differences across firms within each cohort comparison, while the cohort-by-HS2-by-year fixed effects, $\lambda_{g,s(i),t}$, control for shocks common to firms in the same export chapter and calendar year within each cohort comparison. The year immediately preceding the first Israeli import, $k = -1$, is omitted and serves as the reference period.

The coefficients θ_{gk} trace the evolution of firm outcomes before and after the beginning of Israeli sourcing. To account for potential heterogeneity in treatment effects across cohorts and avoid problematic comparisons between firms treated at different times, I allow the coefficients to vary by cohort and aggregate them using the interaction-weighted estimator of Sun and Abraham (2021). The aggregation

weights reflect the number of treated firms contributing to each event time. I also report conventional two way fixed effects event-study estimates that impose common coefficients across cohorts. Both specifications include all observed event-time indicators, while the figures display event years -3 through $+4$. Standard errors are clustered at the firm level which is the treatment level.

For outcome variables, I examine total exports, exports to the United States and other destinations, the number of exported HS6 products, and the number of destination markets served. For export values, I use a thresholded log transformation, $\log(\max\{x, 10{,}000\})$, assigning a common lower bound to zero and small export flows as suggested by Chen and Roth (2024) and implemented recently by Barteska et al. (2026). This threshold lies just below the weighted 10th percentile of positive pre-event U.S. exports in the matched sample. The resulting estimates capture both the entry of firms into export markets and changes in sales among existing exporters, rather than changes along the intensive margin alone. I also report results using export levels and alternative outcome specifications in the Appendix.

Coarsened Exact Matching

A potential concern is that firms beginning to import from Israel may differ systematically from firms that never do so. For example, treated firms may be larger, more internationally integrated, or already expanding their exports before entering the QIZ program. Comparing these firms directly could therefore confound the export response associated with Israeli sourcing with pre-existing differences in firm characteristics.

To address this concern, I construct a matched comparison group using Coarsened Exact Matching (CEM). Within each treatment cohort, I match firms operating in the same dominant export HS2 chapter and with similar total exports in the year preceding the first Israeli import. Export values are divided into bins using thresholds of \$10,000, \$100,000, and \$1 million. I additionally match on earlier export growth, measured between years $g-3$ and $g-2$. Treated firms receive unit weights, while comparison firms are reweighted within each matched cell to reproduce the number of treated firms. The procedure retains all 110 treated firms and 1,630 distinct comparison firms in the baseline matched sample.

Matching improves the comparability of treated and control firms based on observed pre-event characteristics, while the event-study coefficients allow me to examine whether their export trajectories differed before the first recorded Israeli import. However, the timing of Israeli sourcing is an endogenous firm decision and may not coincide exactly with the beginning of QIZ participation. I therefore interpret the estimates as changes in exporting behavior associated with the onset of Israeli sourcing, taking into account the observed pre-event dynamics.

5 Results

In this section, I investigate the economic effects of the QIZ agreement by presenting the results of the empirical analysis described in the previous section. I begin by examining how the QIZ agreement affected product-level trade between Egypt and the United States, as well as between Egypt and Israel. I then investigate whether these effects on trade are also reflected at the industry level. Finally, I examine the impact of the QIZ agreement on firms.

5.1 QIZ and U.S. Exports

Table 1 reports the pooled export estimates from equation (2). In the preferred specification in column (1), which measures exposure using 2004 MFN tariffs, the DDD coefficient is 0.0789 (standard error 0.0329; $p = 0.017$). This implies that a one-percentage-point higher baseline tariff is associated with an 8.2% larger post-agreement increase in Egyptian exports to the U.S. relative to non-Egyptian suppliers. A ten-percentage-point difference in baseline tariffs corresponds to approximately 120% greater relative export growth. These results indicate that the agreement generated larger export gains for products facing higher initial U.S. tariffs. Columns (2)–(4) show that the findings are robust to measuring exposure using 2003 MFN tariffs and clustering standard errors at the two-digit product level (HS2) rather than the six-digit level (HS6).

Table 1: Manufacturing Exports to the United States: Pooled PPML DDD Estimates

	HS6 clustering		HS2 clustering	
	2004 MFN	2003 MFN	2003 MFN	2004 MFN
	(1)	(2)	(3)	(4)
DDD coefficient	**0.0789****	**0.0776****	**0.0776****	**0.0789****
	(0.0329)	(0.0327)	(0.0304)	(0.0307)
Response to 1 pp MFN	8.2%	8.1%	8.1%	8.2%
Fixed effects				
Trade partner × HS6	Yes	Yes	Yes	Yes
HS6 × year	Yes	Yes	Yes	Yes
Trade partner × HS2 × year	Yes	Yes	Yes	Yes
Observations	148,420	148,420	148,420	148,420
Years	1997–2020	1997–2020	1997–2020	1997–2020
Clusters	4,390	4,390	89	89
Bootstrap pretrend p-value	0.640	0.609	0.966	0.976

Notes: Standard errors clustered at the level indicated in the column headings are in parentheses; $^{**}p < 0.05$. Bootstrap p-values test joint pretrends (Webb weights).

Figure 4 presents the corresponding event-study estimates. The results show

a persistent increase in relative Egyptian exports among products facing higher initial MFN tariffs. The joint test of pre-trends does not reject the null of zero pre-treatment effects (Webb score-bootstrap p-value of 0.640 and conventional Wald p-value of 0.666). By 2020, the estimated relative export response reaches 14.2% per tariff percentage point, with a 95% confidence interval of 7.8–20.9%. These results are consistent with larger gains from preferential market access among products facing higher initial tariffs. Importantly, the inclusion of exports to the U.S. from other countries allows me to control for product-specific U.S. demand shocks through product-by-year fixed effects. These absorb changes in U.S. demand for a given product that are common across exporting countries, helping isolate the Egypt-specific response to the agreement. The observed export expansion, however, may reflect both growth in existing trade flows (intensive margin) and the emergence of new trade flows (extensive margin).

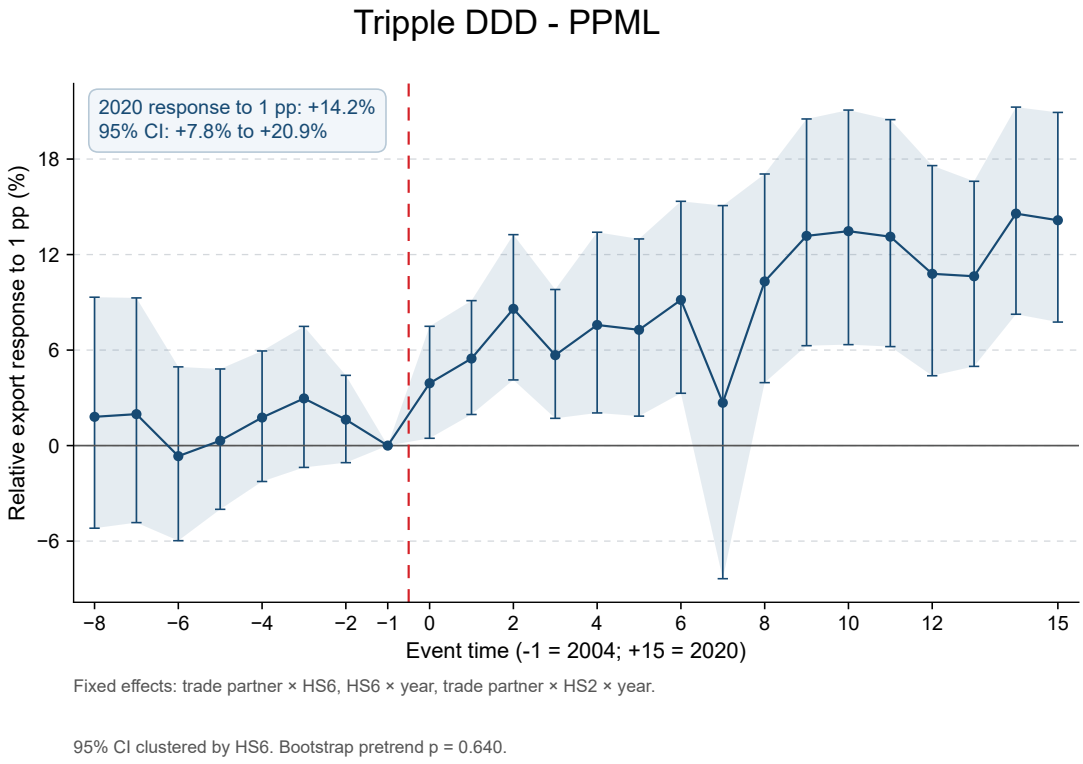


Figure 4: Triple Difference Effects on Exports to the United States

5.2 QIZ and Imports from Israel

Table 2 reports the estimates from equation (3), examining the effects of the QIZ agreement on imports from Israel using the import exposure measure defined in the previous section. The results show that imports from Israel increased more for products supplied by Israeli industries whose downstream Egyptian customers benefited from larger U.S. tariff preferences. In column (1), the pooled DDD coefficient is 0.7386 (standard error 0.0854), corresponding to a 109.3% larger

relative post-agreement increase in imports per percentage point of input-weighted MFN exposure. Including trade-partner-by-broad-sector-by-year fixed effects in column (2) yields a coefficient of 1.0125 (standard error 0.1433), corresponding to a 175.2% relative increase. These findings suggest that the QIZ agreement increased Egyptian demand for Israeli inputs, particularly those used more intensively by industries benefiting from preferential access to the U.S. market.

Table 2: Imports from Israel: Pooled PPML DDD Estimates, 2000/01 IO Exposure

	Baseline	Sector FE	Sector FE + trends
	(1)	(2)	(3)
DDD coefficient	**0.7386*****	**1.0125*****	**0.3611*****
	(0.0854)	(0.1433)	(0.0829)
Response to 1 pp exposure	+109.3%	+175.2%	+43.5%
Fixed effects and trends			
Trade partner × HS6	Yes	Yes	Yes
HS6 × year	Yes	Yes	Yes
Trade partner time effects	Partner×Year	Partner× Sector × year	Partner× Sector × year
Egypt × industry linear trends	No	No	Yes
Observations	103,005	102,729	102,729
Years	1996–2020	1996–2020	1996–2020
IO-industry clusters	16	16	16
Bootstrap pretrend p-value	0.545	0.139	–

Notes: Sector FE denotes trade partner × broad sector × year effects; column (3) adds Egypt × industry linear trends. Standard errors clustered by IO industry in parentheses; ***$p < 0.01$. Bootstrap p-values test joint pretrends (Webb weights); – denotes not applicable.

Figure 5 presents the corresponding event-study estimates with trade-partner-by-sector-by-year fixed effects. The results show a substantial increase in relative imports from Israel, reaching 457.6% per exposure percentage point by 2020. The joint pre-trend test yields mixed evidence. While the Webb score-bootstrap does not reject the null of zero pre-treatment effects ($p = 0.139$), the conventional Wald test rejects it ($p < 0.001$). This discrepancy highlights the sensitivity of pre-trend inference to the inference procedure, particularly given that the identifying variation comes from only 10 industries in the Egyptian input-output tables. The dynamic estimates should therefore be interpreted with this limitation in mind. Appendix A.2 presents the corresponding event study with trade-partner-by-year fixed effects.

These findings are consistent with the mechanism underlying the QIZ agreement: larger tariff preferences for downstream producers increase their incentives to source inputs from Israel. The estimates capture changes in imports associated with exposure to the agreement rather not direct measures of compliance by firms with the Israeli input requirement. Appendix A.3 reports robustness results using the 2008/09 input-output tables, with pooled estimates remaining positive under

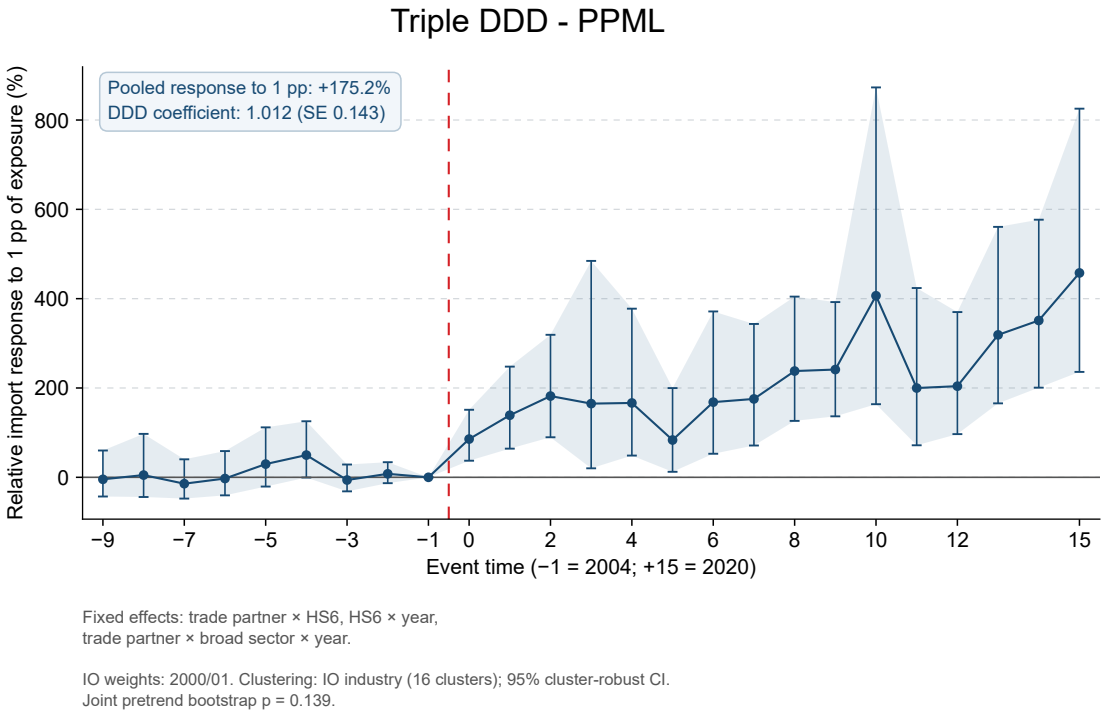


Figure 5: Triple Difference Effects on Imports from Israel

the alternative IO weights.

5.3 QIZ and Industry

I next examine whether the expansion in trade was accompanied by an increase in domestic manufacturing activity. Using the expanded UNIDO series for 1998–2021, I compare changes in output, employment, and total wages across three-digit Egyptian manufacturing industries with different levels of exposure to the agreement. Exposure is measured using pre-agreement U.S. tariffs aggregated to the industry level, with higher tariffs indicating greater potential benefits from preferential market access. This strategy exploits variation in exposure across industries within Egypt to examine whether more exposed industries experienced larger increases in economic activity following the agreement. Figure 6 presents the corresponding event-study estimates.

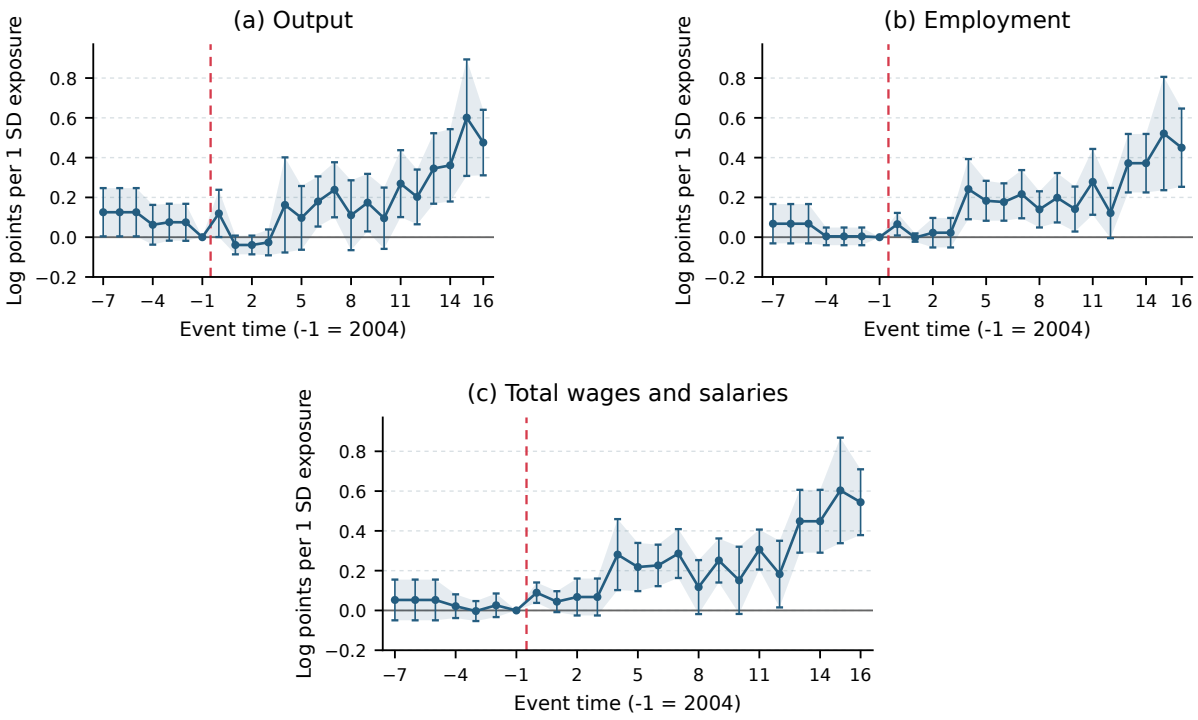


Figure 6: Industry-Level Responses of Output, Employment, and Total Wages

Notes: Fixed effects: industry and year. Pointwise 95% confidence intervals clustered by three-digit industry. Exposure is measured in standard deviations; 2004 is the reference year.

Figure 6 shows that more exposed industries experienced larger increases in output, employment, and total wages, with the differences becoming more pronounced in later years. By 2020, a one-standard-deviation increase in exposure is associated with increases of 0.60 log points in output, 0.52 log points in employment, and 0.60 log points in total wages and salaries relative to 2004. These correspond to approximately 82%, 68%, and 82%, respectively, and are statistically significant at the 1% level using industry-clustered standard errors. Since the wage measure captures the total wage bill, its increase may reflect both higher employment and higher wages per worker.

Taken together, these findings suggest that the trade expansion induced by the QIZ agreement was accompanied by broader increases in manufacturing activity and labor demand, particularly among industries benefiting more from preferential access to the U.S. market.

Additional industry-level results on labor productivity are reported in Appendix C (Figure C1).

5.4 QIZ and Firms

The preceding analysis shows that exposure to the agreement was associated with changes in U.S. exports, Israeli imports, and manufacturing activity. I now examine how individual firms' exports evolved around their first recorded import from Israel. Israeli sourcing is a useful indicator of potential QIZ utilization, although the customs records do not identify firms' QIZ registrations or preference claims.

I study textile and apparel firms whose first recorded Israeli import occurred between 2008 and 2012. The matched sample contains 110 such firms and 1,630 distinct comparison firms that never imported from Israel during the customs panel. I use coarsened exact matching within a firm's dominant export HS2 chapter, based on total exports in the year before the event and, where supported, earlier total-export growth. The event studies include firm-by-cohort and cohort-by-HS2-by-year fixed effects, with standard errors clustered by firm. Event year zero is the first recorded Israeli import, and year -1 is the omitted reference period. To address concerns about potentially misleading comparisons across treatment cohorts in staggered difference-in-differences designs, I report cohort-weighted Sun–Abraham estimates alongside conventional two-way fixed-effects (TWFE) estimates.

Total Exports

Figure 7 shows a substantial increase in total exports around the first Israeli import. In the Sun–Abraham specification, thresholded log total exports rise by 1.33 log points in event year zero and by 1.82 log points in the following year. In levels, the corresponding estimates are approximately \$1.49 million and \$3.09 million per firm. The two displayed pre-event log coefficients are jointly indistinguishable from zero ($p = 0.364$). The TWFE and Sun–Abraham post-event estimates are nearly identical in this sample, indicating that the post-event pattern is insensitive to the choice between these two estimators.

The customs measure may include firms that purchased Israeli inputs without claiming QIZ preferences. Such firms would dilute the estimated response among actual QIZ users if their exports otherwise followed those of the matched controls. Comparison firms have no recorded direct Israeli sourcing. This makes them a relevant benchmark, although purchases through intermediaries remain possible.

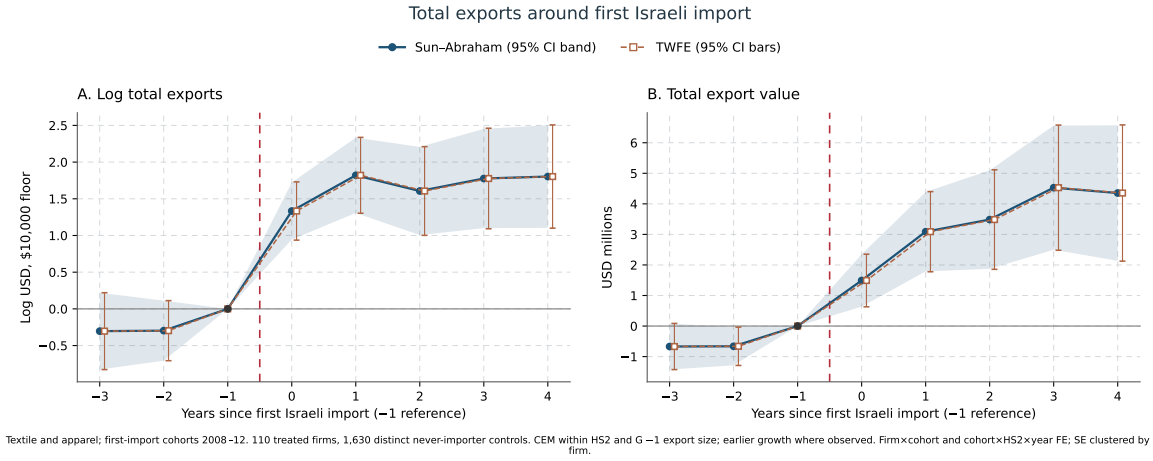
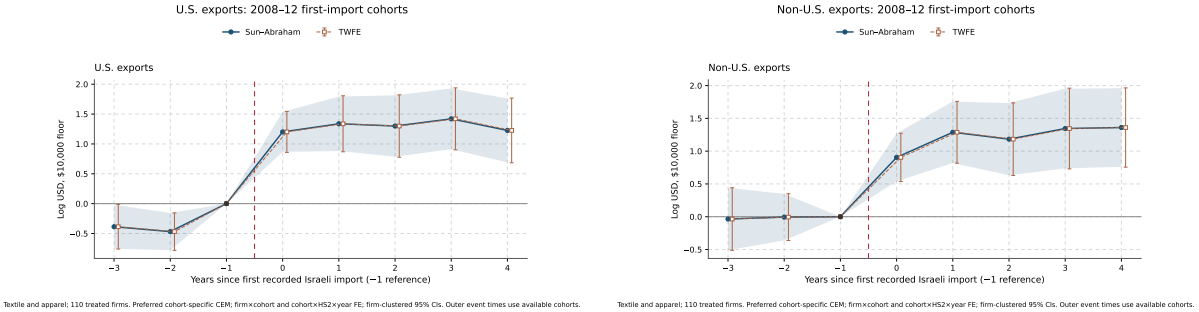


Figure 7: Total exports around the first recorded Israeli import

Export Growth across Destinations

Figure 8 examines whether the export expansion extended beyond the U.S. market directly targeted by the agreement. The results show substantial increases in exports to both the United States and other destinations following firms' first recorded imports from Israel. Thresholded log U.S. exports increase by 1.20 log points in the first-import year and 1.34 log points one year later. Exports to non-U.S. destinations also rise substantially, by 0.90 and 1.29 log points, respectively. In levels, the corresponding average increases over years zero and +1 amount to approximately $0.85 million in U.S. exports and $1.44 million in non-U.S. exports. These findings point to a broad expansion in firms' international sales, consistent with positive export spillovers beyond the market directly targeted by the agreement. Because export values below $10,000 are assigned a common floor, the log-point estimates capture both changes in export participation and increases in sales among existing exporters, rather than percentage changes among continuing exporters alone.

Importantly, the expansion in non-U.S. exports is not preceded by significant differential pre-trends, with the joint test of pre-event coefficients failing to reject zero ($p = 0.981$). For U.S. exports, the pre-event coefficients at -3 and -2 are negative relative to the omitted year -1, and jointly significant ($p = 0.011$), suggesting that firms had already begun expanding their U.S. sales before their first recorded Israeli import. This pattern indicates that entry into Israeli sourcing may be part of a broader adjustment process associated with expansion into the U.S. market, rather than marking its precise beginning. While this limits a causal interpretation of the U.S. post-event response, the substantial and persistent growth in non-U.S. exports, together with the absence of significant pre-trends in those markets, provides suggestive evidence of export spillovers associated with QIZ participation.



(a) U.S. exports (b) Non-U.S. exports

Figure 8: Exports by destination around the first recorded Israeli import

Number of Products and Destinations

The increase in non-U.S. exports was accompanied by an expansion in the number of products exported and destinations served, suggesting that export growth

also occurred along the extensive margin. Figure 9 shows that firms exported an additional 2.80 HS6 products and served 1.51 additional destinations in the first-import year, increasing to 3.40 products and 1.94 destinations one year later. The pre-event coefficients do not jointly reject zero for either outcome. These findings suggest that the increase in non-U.S. exports reflects not only higher sales in existing markets but also firms' expansion into new products and destinations, consistent with broader access to international markets following QIZ participation.

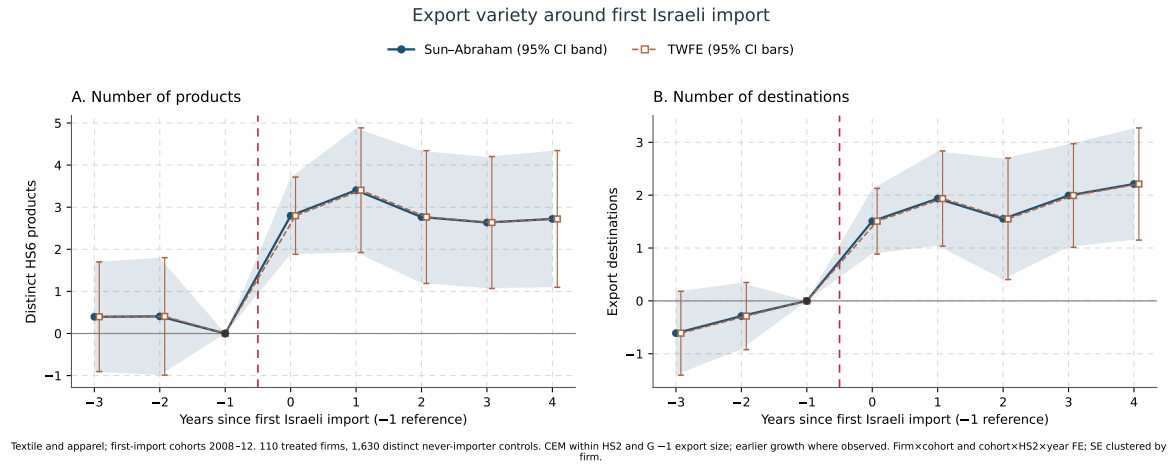


Figure 9: Numbers of exported products and destination markets around the first recorded Israeli import.

5.5 Robustness Checks

Figure 10 shows that the average U.S. export response over event years zero and +1 is 1.29 thresholded log points for textile and apparel firms, substantially larger than for other sectors. Food and agriculture show a smaller response, but include products also exported under QIZ preferences and are therefore only a partial placebo. The matched textile-and-apparel interaction in Appendix Figure D17 is positive after the event, although it also has pre-event movement.

Figure 11 compares first imports from Israel with first imports from six other countries. The other-origin regressions control for the full event-study path around first Israeli imports. The average year-0-to-+1 U.S. export coefficient is 1.27 log points for Israel, compared with 0.17–0.34 for the other origins. These origin-specific matches contain different firms, and positive responses around other first imports show that Israeli sourcing can coincide with a broader export expansion.

Appendix D reports the event paths for 2006–2015 cohorts (248 treated firms), dynamic coefficient tables for both cohort ranges, and an alternative matching comparison. Post-event export responses remain positive in the wider sample, although total- and U.S.-export leads reject zero. In the 2008–2012 sample, matching on the full pre-event U.S. export path flattens U.S. leads but uses the plotted years to choose controls; matching only on earlier history leaves a U.S. pretrend.

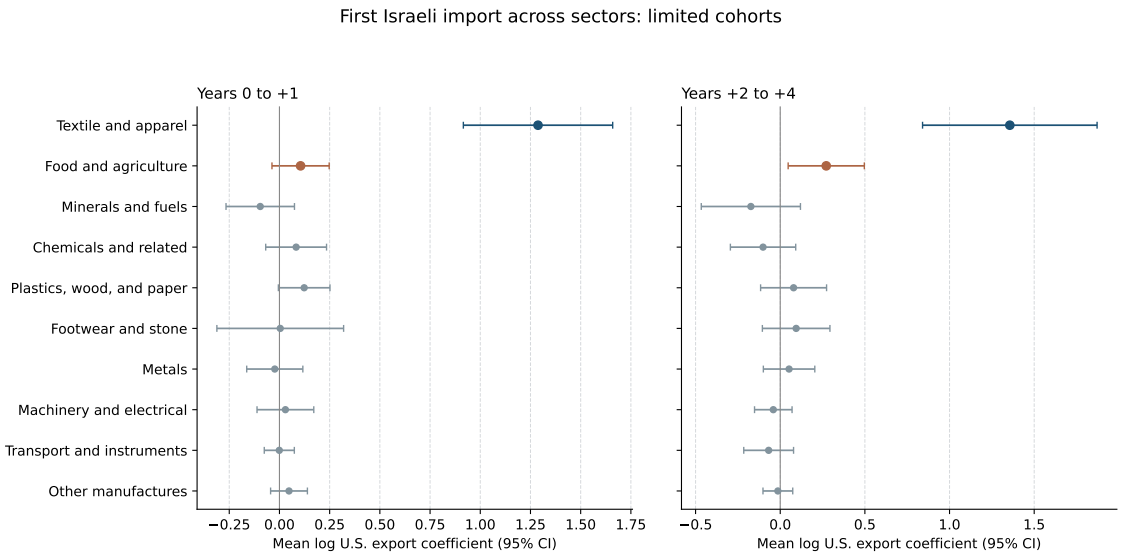


Figure 10: Post-event U.S. export responses across different sectors.

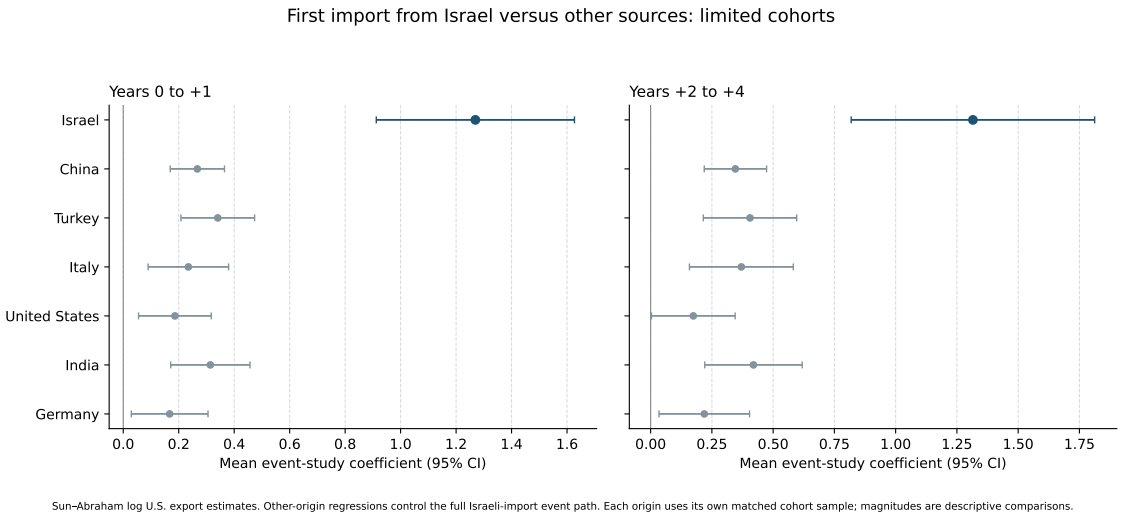


Figure 11: Post-event U.S. export responses around first imports from Israel and six other countries.

The Appendix also reports inverse-hyperbolic-sine outcomes, non-U.S. exports by region, annual export participation, heterogeneity by size, prior exporting and cohort timing, and sector and import-origin comparisons. These checks show the sensitivity of the firm-level patterns to sample and matching choices; remaining pretrends and endogenous import timing limit a causal interpretation.

6 Mechanisms

6.1 Empirical Approach

I examine several potential mechanisms behind the expansion of exports to non-U.S. destinations using successive waves of the World Bank Enterprise Surveys in Egypt. I focus on international certification, foreign ownership, foreign-licensed technology, and investment in fixed assets. Certification may help firms meet buyers' requirements, while foreign owners and technology can facilitate access to new destinations. Investment may also allow firms to expand their capacity to serve foreign markets. I estimate:

$$Y_{it} = \alpha_i + \lambda_{t,G_i} + \beta_N \left(E_{i,2004} \times Post_t\right) + \theta \left(E_{i,2004} \times Post_t \times G_i\right) + \varepsilon_{it}. \quad (6)$$

Where, Y_{it} denotes one of the four outcomes, and $E_{i,2004}$ is the average MFN tariff in the firm's 2004 industry, measured in percentage points. $Post_t$ equals one from 2005 onward, and G_i indicates exporting status. Firm fixed effects are denoted by α_i, while λ_{t,G_i} denotes exporter status by year fixed effects. The coefficient θ of interest measures the difference in an outcome which an exporting firm gains relative to other firms across different exposures to the QIZ agreement.

6.2 Results

Table 3 shows a positive certification response among exporting firms more exposed to the QIZ relative to other firms. The coefficient in column (1) is 1.612 percentage points, with a firm-clustered standard error of 0.690 and a bootstrap p-value of 0.011. A one-percentage-point higher baseline MFN tariff is therefore associated with a 1.61-percentage-point larger post-agreement increase in certification among exporting firms. This pattern is consistent with firms adopting international standards that may facilitate sales to destinations beyond the United States.

The remaining columns show no statistically significant difference in foreign ownership, foreign-licensed technology, or investment in fixed assets. The estimates therefore provide the clearest support for international certification as a potential channel of why the QIZ induces exports to other destinations. By certifying, firms reduce the costs of finding foreign customers for their products across all destinations, and hence may increase their exports across destinations.

The certification estimate remains positive with additional industry-by-survey-wave controls and alternative exporter definitions. Appendix Tables E1–E4 report

Table 3: MFN Exposure and Firm Adjustment

	International certification (1)	Foreign ownership share (2)	Foreign-licensed technology (3)	Fixed-asset investment (4)
MFN × Post × Exporter	**1.612**	**0.127**	**0.167**	**0.302**
	(0.690)	(0.290)	(0.882)	(1.494)
Bootstrap p-value	0.011	0.580	0.803	0.797
Observations	2,120	2,118	2,091	1,744
Firms	655	655	647	541
Firm fixed effects	Yes	Yes	Yes	Yes
Exporter-status × wave FE	Yes	Yes	Yes	Yes

Notes: Each column reports θ from Equation (6). Outcomes are expressed on a 0–100 scale; foreign ownership is the foreign-owned share, and the other outcomes are indicators. Firm-clustered standard errors are in parentheses; bootstrap p-values use 999 restricted Rademacher draws.

these alternative specifications.

7 Model

This section develops a quantitative model to assess the effects of the QIZ agreement on Egyptian welfare and the intensity of trade between Egypt and Israel. Increasing bilateral trade is a key geopolitical motivation behind the U.S.'s support for the agreement. Building on the heterogeneous-firm framework of Melitz (2003), the model features multiple sectors with sector-specific tariffs, two export destinations, and three foreign sourcing origins: the United States, Israel, and the rest of the world. Crucially, firms exporting to the U.S. can endogenously choose to export under the QIZ program or not and pay the MFN tariffs.

7.1 Environment

Consider a small open economy (Egypt) with S manufacturing sectors and a fixed supply of labor L. Each sector has a continuum of differentiated varieties produced under monopolistic competition. Firms differ in productivity and choose whether to export to the United States and to a composite non-U.S. market. Exporters to the U.S. can pay the MFN tariff or can obtain duty-free access through the QIZ program, conditional on satisfying its sourcing requirements and paying a fixed cost. Labor is mobile across domestic activities and earns a wage w. Foreign input prices and foreign demands are exogenous.

7.2 Preferences

Household preferences are Cobb–Douglas across sectors and CES across varieties within each sector:

$$U = \prod_{s=1}^{S} \left(\frac{C_s}{\alpha_s}\right)^{\alpha_s}, \qquad C_s = \left[\int_{\mathcal{V}_s} c_s(v)^{(\sigma_s-1)/\sigma_s}\, dv\right]^{\sigma_s/(\sigma_s-1)}, \qquad \sum_s \alpha_s = 1. \quad (7)$$

The set of varieties $\mathcal{V}_s$, contains both domestic and imported varieties, $\alpha_s > 0$ is the share of expenditure per sector s by consumers, and $\sigma_s > 1$ is the elasticity of substitution across varieties within the sector s. Given household income Y and variety price $p_s(v)$, demand for variety v under CES is:

$$c_s(v) = \alpha_s Y P_s^{\sigma_s-1} p_s(v)^{-\sigma_s}, \qquad P_s = \left[\int_{\mathcal{V}_s} p_s(v)^{1-\sigma_s}\, dv\right]^{1/(1-\sigma_s)}. \quad (8)$$

The aggregate price index is $P = \prod_s P_s^{\alpha_s}$. Imported varieties are supplied at fixed prices taken as given. Their contribution to the CES price index is therefore fixed, while expenditure on imports adjusts with income and domestic prices.

7.3 Technology

Following the input-sourcing structure in Blaum et al. (2018), a firm i with productivity φ_i combines labor, ℓ, with a CES bundle of intermediate inputs from different origins, X_s. It's production function is thus:

$$y_s = \varphi_i \ell_s^{1-\gamma_s} X_s^{\gamma_s}, \qquad X_s = \left[\sum_{o\in\mathcal{O}} \beta_{o,s}^{1/\eta_s} x_{o,s}^{(\eta_s-1)/\eta_s}\right]^{\eta_s/(\eta_s-1)} \quad (9)$$

where the origins $o = \{E, I, U, R\} \in \mathcal{O}$. denote Egypt, Israel, the United States, and the rest of the world. The parameter $\gamma_s \in (0, 1)$ determines intermediate-input intensity, while $\eta_s > 1$ governs substitution of inputs across the different input origins, and $\beta_{o,s}$ is the share of inputs from an origin o such that $\sum_o \beta_{o,s} = 1$. Egyptian intermediates are assumed to be produced competitively with one unit of labor per unit of output, giving $p_{E,s} = w$. Thus, I do not model input output linkages across sectors in the baseline model. Foreign input prices $p_{I,s}, p_{U,s}, p_{R,s}$ are fixed in foreign-numeraire units and taken as given.
With unrestricted sourcing, unit cost at productivity one is

$$a_{M,s}(w) = \left(\frac{w}{1-\gamma_s}\right)^{1-\gamma_s} \left(\frac{P_{X,s}}{\gamma_s}\right)^{\gamma_s}, \qquad P_{X,s} = \left[\sum_o \beta_{o,s} p_{o,s}^{1-\eta_s}\right]^{1/(1-\eta_s)}. \quad (10)$$

A firm with productivity φ has marginal cost $a_{M,s}/\varphi$. This technology is used for domestic sales, non-U.S. exports, and U.S. exports under MFN treatment. The subscript M refers to ordinary production; tariffs are applied separately.

Potential entrants pay $wf_{e,s}$ before drawing productivity from G_s, a Pareto distribution with shape $\theta_s > 0$ on $[\underline{\varphi}_s, \overline{\varphi}_s]$. Each entrant produces one variety and can sell domestically without a fixed operating cost, thus all no firm which has entered exits the market. The mass of entrants therefore equals the mass of producing firms, denoted M_s. Productivity remains fixed after entry. The upper support is finite, allowing the thick revenue tails in the census without infinite aggregate sales. The distribution and aggregation formulas are given in Appendix F.

7.4 Trade and the QIZ agreement

Serving a foreign market requires paying a fixed cost in terms of labor: $f_{R,s}$ for the non-U.S. market and $f_{U,s}$ for the United States. Let $m \in \{M, Q\}$ denote the U.S. regime. Under MFN, firms source freely and pay tariff τ_s. Under QIZ, the tariff is zero, but firms must satisfy the qualifying-sourcing rules and incur an additional QIZ-specific fixed cost $wf_{Q,s}$. Firms choosing the QIZ agreement have to register with a QIZ unit and so this QIZ-specific fixed cost captures this idea.

Under the QIZ agreement, firms must source a minimum share of the value of their output domestically, as is common under rules of origin requirements, and an additional minimum share from a third country which is Israel, a requirement specific to the QIZ agreement. Firms must also satisfy a broader content requirement that allows sourcing from Egypt, Israel, and the United States to be combined.

Let $\bar{e}$, r, and $\bar{q}$ denote the minimum content shares required under the QIZ agreement for Egyptian, Israeli, and combined qualifying content, respectively. Egyptian content includes expenditures on domestic labor and Egyptian intermediate inputs, while Israeli content consists of expenditures on intermediate inputs sourced from Israel. The combined requirement counts both Egyptian and Israeli content, as well as inputs sourced from the United States.

These requirements are expressed as shares of the value of output rather than production costs. Let $C = w\ell + \sum_o p_{o,s} x_o$ denote total variable expenditure and $\mu_s = \sigma_s/(\sigma_s - 1)$ the CES markup in sector s. I approximate the value of output by $\mu_s C$, so that each content requirement imposes a minimum level of qualifying expenditure relative to this value. A firm producing under the QIZ regime therefore solves the following unit-cost minimization problem:

$$
\begin{aligned}
a_{Q,s}(w,r) = \min_{\ell, \boldsymbol{x} \geq 0} \; & w\ell + \sum_o p_{o,s} x_o \\
\text{subject to} \quad & \ell^{1-\gamma_s} X_s(\boldsymbol{x})^{\gamma_s} \geq 1, \\
& w\ell + p_{E,s} x_E \geq \bar{e}\, \mu_s C, \\
& p_{I,s} x_I \geq r \mu_s C, \\
& w\ell + p_{E,s} x_E + p_{I,s} x_I + p_{U,s} x_U \geq \bar{q}\, \mu_s C.
\end{aligned}
\tag{11}
$$

Egyptian content includes direct labor and Egyptian inputs. The combined requirement additionally includes Israeli and U.S. inputs. In the baseline, $(\bar{e}, r, \bar{q}) = (0.117, 0.105, 0.35)$.

Constant returns imply that a firm's marginal cost is $a_{Q,s}/\varphi$. The sourcing requirements apply only to QIZ sales, whereas participating firms use unrestricted sourcing for their other production. Because the requirements restrict feasible input combinations, $a_{Q,s} \geq a_{M,s}$. A reduction in a binding Israeli requirement lowers QIZ unit cost.

Let $A_{j,s}$ summarize demand in destination j in sector s, $\Delta_{j,s}$ be the iceberg trade cost to destination j in sector s, and $T^m_{j,s}$ the gross tariff factor under the MFN program. Constant-markup pricing implies producer revenue

$$R^m_{j,s}(\varphi) = B^m_{j,s}\varphi^{\sigma_s - 1}, \qquad B^m_{j,s} = A_{j,s}(T^m_{j,s})^{-\sigma_s}(\mu_s \Delta_{j,s} a_{m,s})^{1-\sigma_s}. \tag{12}$$

For U.S. sales, $T^M_{U,s} = 1 + \tau_s$ and $T^Q_{U,s} = 1$. Non-U.S. sales use the unrestricted technology and face a specific tariff.

The QIZ revenue advantage relative to the MFN for a firm with productivity φ and a sector s is:

$$\rho_s(w, r) \equiv \frac{R^Q_{U,s}(\varphi)}{R^M_{U,s}(\varphi)} = (1 + \tau_s)^{\sigma_s} \left(\frac{a_{Q,s}(w, r)}{a_{M,s}(w)} \right)^{1-\sigma_s}. \tag{13}$$

Tariff savings raise ρ_s, while the cost of satisfying the sourcing requirement lowers it. Export revenues exclude the tariff paid to the U.S. government. Therefore, golding the producer price fixed, removing the tariff lowers the buyer's price and increases quantity demanded—and therefore the firm's revenue—by $(1 + \tau_s)^{\sigma_s}$.

7.5 Export participation and regime choice

Operating profits are revenue divided by σ_s. Net export profits to the rest of the world, to the U.S under MFN, and to the U.S under QIZ respectively are:

$$\begin{aligned} \pi_{R,s}(\varphi) &= \frac{R_{R,s}(\varphi)}{\sigma_s} - w f_{R,s}, \\ \pi_{M,s}(\varphi) &= \frac{R^M_{U,s}(\varphi)}{\sigma_s} - w f_{U,s}, \\ \pi_{Q,s}(\varphi) &= \frac{\rho_s R^M_{U,s}(\varphi)}{\sigma_s} - w(f_{U,s} + f_{Q,s}). \end{aligned} \tag{14}$$

A firm serves the non-U.S. market if $\pi_{R,s} \geq 0$. For the United States, it chooses the most profitable of MFN, QIZ, and not exporting. Its value after the productivity draw is

$$\Pi_s(\varphi) = \frac{R_{D,s}(\varphi)}{\sigma_s} + \max\{0, \pi_{R,s}(\varphi)\} + \max\{0, \pi_{M,s}(\varphi), \pi_{Q,s}(\varphi)\}. \tag{15}$$

Higher productivity raises sales and allows firms to cover larger fixed costs. In the absence of QIZ, the export cutoffs are

$$\varphi_{R,s} = \left(\frac{\sigma_s w f_{R,s}}{B_{R,s}} \right)^{1/(\sigma_s - 1)}, \qquad \varphi_{M,s} = \left(\frac{\sigma_s w f_{U,s}}{B_{U,s}^M} \right)^{1/(\sigma_s - 1)}. \tag{16}$$

QIZ is chosen only if it is profitable and dominates MFN. If $\rho_s \leq 1$, no firm uses the program: its variable profits are no higher and its fixed cost is larger. For $\rho_s > 1$, the participation cutoff is

$$\varphi_s^* = \max \left\{ \left[\frac{\sigma_s w (f_{U,s} + f_{Q,s})}{\rho_s B_{U,s}^M} \right]^{1/(\sigma_s - 1)}, \left[\frac{\sigma_s w f_{Q,s}}{(\rho_s - 1) B_{U,s}^M} \right]^{1/(\sigma_s - 1)} \right\}. \tag{17}$$

The first threshold solves $\pi_{Q,s} = 0$; the second solves $\pi_{Q,s} = \pi_{M,s}$. Firms above both choose QIZ. If the cutoff exceeds $\overline{\varphi}_s$, participation is zero even though the variable-revenue advantage is positive.

Writing $\kappa_s = f_{Q,s}/f_{U,s}$, MFN and QIZ exporters coexist when $1 < \rho_s < 1 + \kappa_s$, provided both participation intervals intersect the productivity support. Intermediate-productivity U.S. exporters use MFN, while the most productive use QIZ. When $\rho_s \geq 1 + \kappa_s$, all profitable U.S. exporters choose QIZ. These selection conditions follow from comparing the profit functions in Equation (14).

7.6 Equilibrium

An equilibrium consists of a common wage, sectoral firm masses, prices, and market participation such that households and firms optimize, entry is unrestricted, and goods and labor markets clear.

Entry and income. Free entry equates expected operating profits, $\Pi_s(\varphi)$, net of fixed export costs but before establishment costs, to the entry cost:

$$\int \Pi_s(\varphi)\, dG_s(\varphi) = w f_{e,s}, \qquad s = 1, \ldots, S. \tag{18}$$

Here, G_s is the productivity distribution and $f_{e,s}$ is establishment labor. Since all entrants produce, M_s also denotes the mass of producing firms. Expected profits net of entry costs are zero, so household income and welfare are

$$Y = wL + \mathcal{B}, \qquad W = \frac{Y}{P}, \qquad P = \prod_s P_s^{\alpha_s}. \tag{19}$$

Labor supply L and the nominal transfer $\mathcal{B}$ from outside manufacturing are fixed across equilibria. Welfare measures purchasing power over the manufacturing consumption basket.

Goods markets. Households spend $\alpha_s Y$ on sector s, allocating expenditure between Egyptian and imported varieties according to CES demand. Let $H_{D,s}$ and $H_{F,s}$ denote their respective contributions to the sectoral price index:

$$\begin{aligned} H_{D,s} &= M_s \int \left(\frac{\mu_s a_{M,s}(w)}{\varphi} \right)^{1-\sigma_s} dG_s(\varphi), \\ P_s^{1-\sigma_s} &= H_{D,s} + H_{F,s}, \qquad \lambda_{D,s} = \frac{H_{D,s}}{H_{D,s} + H_{F,s}}, \\ R_{D,s} &= \alpha_s Y \lambda_{D,s}. \end{aligned} \tag{20}$$

The subscripts D and F identify Egyptian and foreign varieties. Egyptian prices equal the markup μ_s times ordinary unit cost $a_{M,s}(w)/\varphi$; foreign prices and $H_{F,s}$ are fixed. Each firm supplies its domestic and foreign demand at its chosen prices. Aggregating domestic sales gives $R_{D,s}$, while expenditure $\alpha_s Y(1-\lambda_{D,s})$ purchases imported final goods.

Labor market. Let $N_{R,s}$, $N_{U,s}$, and $N_{Q,s}$ denote the masses of non-U.S., U.S., and QIZ exporters. Correspondingly, $X_{R,s}$ is non-U.S. export revenue, while $X_{M,s}$ and $X_{Q,s}$ are U.S. revenues under MFN and QIZ. Variable expenditures under ordinary sourcing (M) and QIZ sourcing (Q) are

$$V_{M,s} = \frac{R_{D,s} + X_{R,s} + X_{M,s}}{\mu_s}, \qquad V_{Q,s} = \frac{X_{Q,s}}{\mu_s}.$$

For technology $m \in \{M, Q\}$, define $d_{m,s} = t^m_{L,s} + t^m_{E,s}$, where the two terms are expenditure shares on direct labor and Egyptian inputs. Since Egyptian inputs are produced entirely with labor, $d_{m,s}V_{m,s}/w$ is total variable labor demand. Labor-market clearing requires

$$L = \sum_s \left[M_s f_{e,s} + f_{R,s} N_{R,s} + f_{U,s} N_{U,s} + f_{Q,s} N_{Q,s} + \frac{d_{M,s}V_{M,s} + d_{Q,s}V_{Q,s}}{w} \right]. \tag{21}$$

All fixed costs are measured in labor units. QIZ exporters pay the ordinary U.S. access cost $f_{U,s}$ and the additional compliance cost $f_{Q,s}$.

Counterfactual solution. I solve the counterfactuls in exact changes relative to observed QIZ (Dekle et al., 2008). Define $\widehat{z} = z'/z$, with primes identifying counterfactual values, and let $b = \mathcal{B}/Y$ be the baseline transfer share. Then

$$\widehat{Y} = (1-b)\widehat{w} + b, \qquad \widehat{W} = \frac{\widehat{Y}}{\prod_s \widehat{P}_s^{\alpha_s}}.$$

The S free-entry conditions and labor-market clearing determine the wage and sectoral firm-mass changes, with prices, sourcing, and export participation recomputed at each candidate equilibrium. Appendix F defines the equilibrium, derives

the exact-change system, and describes the solution algorithm.

7.7 The role of the Israeli input requirement

The Israeli input requirement creates a tradeoff between sourcing intensity and participation: requiring more Israeli content from each firm can discourage firms from using the QIZ program. Holding wages and foreign demand fixed, relaxing a binding requirement lowers QIZ unit costs, $a_{Q,s}$, raises its relative revenue advantage, ρ_s, and reduces the productivity cutoff for participation. This may expand exporting under incumbent QIZ firms and induce new firms to utilize the QIZ program, including in previously inactive sectors.

Consequently, a lower requirement can increase Israeli imports. Total Israeli input purchases are

$$M_I = \sum_s \left(t_{I,s}^M V_{M,s} + t_{I,s}^Q V_{Q,s} \right), \tag{22}$$

where $t_{I,s}^m$ is the Israeli share of variable production costs, $V_{m,s}$, under ordinary or QIZ sourcing, $m \in \{M, Q\}$. The legal requirement instead uses appraised output value. Under our approximation that this value equals producer revenue, $X_{Q,s} = \mu_s V_{Q,s}$, a binding requirement implies

$$t_{I,s}^Q = \mu_s r, \qquad M_{I,s}^Q = t_{I,s}^Q V_{Q,s} = r X_{Q,s}.$$

A relaxation of the input requirement reduces Israeli purchases per dollar of QIZ exports but expands QIZ sales. Where the requirement binds and exports are positive and differentiable, Israeli purchases increase if

$$-\frac{d \log X_{Q,s}}{d \log r} > 1. \tag{23}$$

The requirement can therefore undermine its own sourcing objective when the loss of export participation outweighs the additional content demanded from remaining users.

8 Quantitative Analysis

I calibrate the model to the Egyptian economy and use it to evaluate the welfare effects of the agreement and alternative Israeli input requirements. The quantitative application contains fifteen manufacturing sectors. The observed QIZ equilibrium is the baseline; counterfactuals hold foreign demand, foreign prices, technology, and fixed costs constant while allowing sourcing, export participation, entry, and wages to adjust.

8.1 Calibration

The calibration combines data from the 2017 Egyptian economic census, firm-level customs records, U.S. tariffs and trade data, and Egypt's 2016/17 input-output and supply-use tables. All data sources are harmonized to the model's fifteen manufacturing sectors. Table 4 summarizes the calibration strategy, distinguishing parameters estimated directly from the data, parameters calibrated to match baseline moments, and maintained assumptions based on the literature. The detailed construction of the calibration targets are reported in Appendix F.

Table 4: Calibration: Data, Matched Moments, and Assumptions

Parameter	Interpretation	Source, moment, or restriction
A. Directly measured or estimated from data		
γ_s	Intermediate-input cost share	2017 economic census
θ_s	Productivity distribution shape	2017 census revenue distribution
$\underline{\varphi}_s/\overline{\varphi}_s$	Relative productivity support	2017 census revenue distribution
$(1+\tau_s)^{\sigma_s}$	Sectoral tariff-benefit factor	2016 U.S. HTS8 tariffs and export data
α_s	Sectoral consumption expenditure share	2016/17 IO and supply-use tables
$\lambda_{D,s}$	Baseline domestic expenditure share	2016/17 IO and supply-use tables
Y	Baseline household expenditure	2016/17 IO and supply-use tables
B. Internally calibrated to baseline moments		
$\beta_{o,s}$	Input-origin preference weights	Input expenditure shares by origin
$A_{R,s}, A_{U,s}$	Foreign-demand	Export revenues by destination
$f_{R,s}, f_{U,s}$	Fixed export costs	Destination exporter counts and revenues
$f_{Q,s}$	QIZ Fixed cost	QIZ shares of U.S. export revenue
C. Parameters		
σ_s	Substitution across varieties	Maintained: 6
η_s	Substitution across input origins	Maintained: 3
$p_I/p_U,\ p_R/p_U$	Relative foreign input prices	Maintained: 1.25, 1
$f_{Q,s}$	QIZ cost: non QIZ sector	Common fixed cost

Panel A reports parameters measured or estimated directly from the data. I construct the intermediate-input cost share, γ_s, as the ratio of intermediate expenditure to total variable production costs, using weighted census totals. The parameters of the truncated Pareto productivity distribution are estimated from the distribution of firm revenues in the census. Conditional on the maintained elasticity of substitution, the estimated revenue-tail parameter is mapped to the productivity distribution according to $\theta_s = (\sigma_s - 1)\theta_s^{\text{rev}}$. The sectoral tariff-benefit factor is constructed as the export-weighted average of product-level $(1+\tau_p)^{\sigma_s}$. Finally, household expenditure shares α_s, domestic expenditure shares $\lambda_{D,s}$, and aggregate household expenditure Y are obtained from the input-output and supply-use table

for Egypt in 2016/2017, excluding intermediate sales from final household demand.

Panel B reports parameters calibrated to match baseline equilibrium moments. Conditional on the assumed input substitution elasticity and relative input prices, the origin preference weights $\beta_{o,s}$ are chosen to reproduce domestic and foreign input expenditure shares observed in the census and customs data. Foreign-demand coefficients and fixed export costs are jointly calibrated to match export revenues and exporter counts by destination. Exporter counts discipline the productivity cutoffs, while export revenues discipline foreign demand, with fixed costs recovered from the corresponding zero-profit conditions. For food, textiles, and apparel, the QIZ fixed costs are calibrated to match the share of U.S. export revenue receiving QIZ treatment in each sector.

Panel C summarizes the maintained assumptions and parameter restrictions. I set the elasticity of substitution across varieties to $\sigma_s = 6$, the elasticity of substitution across input origins to $\eta_s = 3$, close to Blaum et al. (2019) estimation of 2.78, and the relative prices of Israeli and rest-of-world inputs to U.S. inputs to 1.25 and 1, respectively. In sectors with negligible baseline QIZ utilization, the data identify a lower bound on compliance costs rather than a unique value. I therefore assign a common finite cost just above the bound consistent with nonparticipation, allowing firms in these sectors to enter the program in counterfactual equilibria.

8.2 Model Fit

I assess the model's fit by comparing its predictions with both targeted baseline moments and untargeted patterns of firm participation and industry-level responses to the agreement. By construction, the calibration matches export revenues and exporter counts by destination, input expenditure shares, and the share of each sector's U.S. export revenue receiving QIZ preferential treatment. The latter shares are 13.0% in food, 38.1% in textiles, and 98.2% in apparel.

I first examine whether the model reproduces broader patterns of firm participation that are not directly targeted in the calibration. Among U.S. exporters, 80.4% also serve non-U.S. destinations in the customs data, compared with 100% in the model. Similarly, QIZ users in the three active sectors account for 22.4% of all U.S. exporters in the model, compared with 15.8% identified through Israeli imports in the data. These comparisons indicate that the model captures the prevalence of joint export-market participation and QIZ utilization, although it tends to predict higher participation rates.

I next examine whether the model reproduces the industry-level responses associated with exposure to the agreement. To construct comparable outcomes, I aggregate product-level trade flows and UNIDO employment to the model's fifteen manufacturing sectors. Israeli intermediate purchases are allocated across fourteen purchasing industries using the 2008/09 input-output table. Industry exposure is measured by the average 2004 U.S. MFN tariff, weighted by each product's share of the industry's Egyptian world exports. The input-output allocation assigns Israeli inputs to their purchasing industries rather than the industries supplying the

imported products.[0]

In the data, I estimate PPML regressions over 1998–2020 with industry and year fixed effects, interacting baseline tariff exposure with an indicator for the post-agreement period. I construct the corresponding model estimates by comparing industry outcomes under the no-QIZ and observed-QIZ equilibria, using the same exposure measures and including industry and policy-state fixed effects. Table 5 reports the implied percentage responses, $100[\exp(\beta) - 1]$, associated with a one pp increase in baseline tariff exposure. Appendix F.6 provides details.

Table 5: Untargeted Industry-Level Responses

Outcome	Industries	Data (%)	Model (%)
U.S. exports	15	6.73	10.47
Non-U.S. exports	15	1.94	0.45
Israeli input purchases	14	4.84	9.85
Employment / labor	15	0.80	0.30

Notes: Responses per one percentage point of industry MFN exposure, using fixed 2004 world-export weights. Israeli purchases use the 2008/09 IO allocation. None of these responses is targeted in calibration.

The model reproduces the main qualitative patterns observed in the data. In particular, industries with greater exposure to the agreement experience larger increases in U.S. exports and Israeli input purchases, while responses in non-U.S. exports and employment are modest. Importantly, the model reproduces this ordering without targeting any of these empirical responses in the calibration. Quantitatively, it predicts larger responses in U.S. exports and Israeli input purchases and smaller responses in non-U.S. exports and employment than those estimated in the data. Given the imprecision of the industry-level estimates, these comparisons are best interpreted as evidence of the model's ability to capture the relative strength and direction of the responses rather than their exact magnitudes.

8.3 The Economic Effects of QIZ

Table 6 reports the aggregate economic effects of the QIZ agreement by comparing the observed equilibrium with a counterfactual without the QIZ program available. Hence, firms can still export to the United States under MFN tariffs and freely source inputs from Israel.

The agreement generates substantial increases in the trade flows it directly targets. U.S. exports rise by 71.4%, while Israeli input imports increase by 169.3%. However, total exports increase by only 2.2%, reflecting the relatively small initial share of the U.S. market in Egyptian exports and the concentration of QIZ participation in textiles and apparel. The increase in export profitability raises labor demand and wages, increasing production costs across sectors. As a result,

[0]Israeli input purchases by industry are not directly observed and are allocated using the 2008/09 input-output coefficients. Since these coefficients postdate the agreement, the comparison is conditional on the allocation and may partly reflect changes induced by the QIZ program.

Table 6: Economic Effects of the QIZ Agreement

	No QIZ	Observed QIZ	Change (%)
Welfare and prices (observed QIZ = 100)			
Manufacturing-basket welfare	99.9803	100.0000	0.0197
Wage	99.8397	100.0000	0.1606
Consumer price index	99.9111	100.0000	0.0890
Trade (EGP billion)			
U.S. exports	5.305	9.092	71.40
Non-U.S. exports	107.495	106.226	−1.18
Total exports	112.800	115.318	2.23
Israeli imports	0.391	1.053	169.26
QIZ exporter	0	134.9	—

Notes: Changes are relative to no QIZ. Welfare is real income over the manufacturing consumption basket.

exports to non-U.S. destinations decline by 1.2%, partially offsetting the gains in the U.S. market.

Despite the substantial expansion in trade, the agreement generates a modest welfare gain of 0.0197%, measured by real household income over the manufacturing consumption basket. This reflects both the limited share of economic activity directly benefiting from preferential access and the costs of Israeli input sourcing and QIZ compliance. Moreover, higher wages increase household income but also raise domestic prices, limiting the gains in purchasing power. The welfare effect can be decomposed as:

$$\Delta \log W = \Delta \log Y - \sum_s \alpha_s \Delta \log P_s. \tag{24}$$

Household income increases by 0.1087 log-percentage points, while higher consumer prices offset 0.0890 log-percentage points, yielding a net welfare gain of only 0.0197%.

Overall, the agreement substantially expands U.S. exports and Israeli input purchases, but these gains translate into limited aggregate welfare improvements because they are concentrated in a small part of the economy and partly offset by an increase in wages due to general equilibrium effects.

8.4 Alternative input requirements

In this counterfactual, I vary the Israeli input requirement from 20% to zero, holding the Egyptian and combined requirements at 11.7% and 35%. The novelty of the QIZ agreeement is that it links market access to geopolitical objectives through these input requirements. Therefore, determining how the input requirement influences different margins is essential to understanding the design of the policy.

Each input requirement counterfactual is solved as a new equilibrium, allowing wages, producer entry, and QIZ participation to adjust. Figure 12 reports changes in U.S. exports, welfare, and Israeli imports relative to the 10.5% baseline as we vary the input requirement from 20% to 0%.

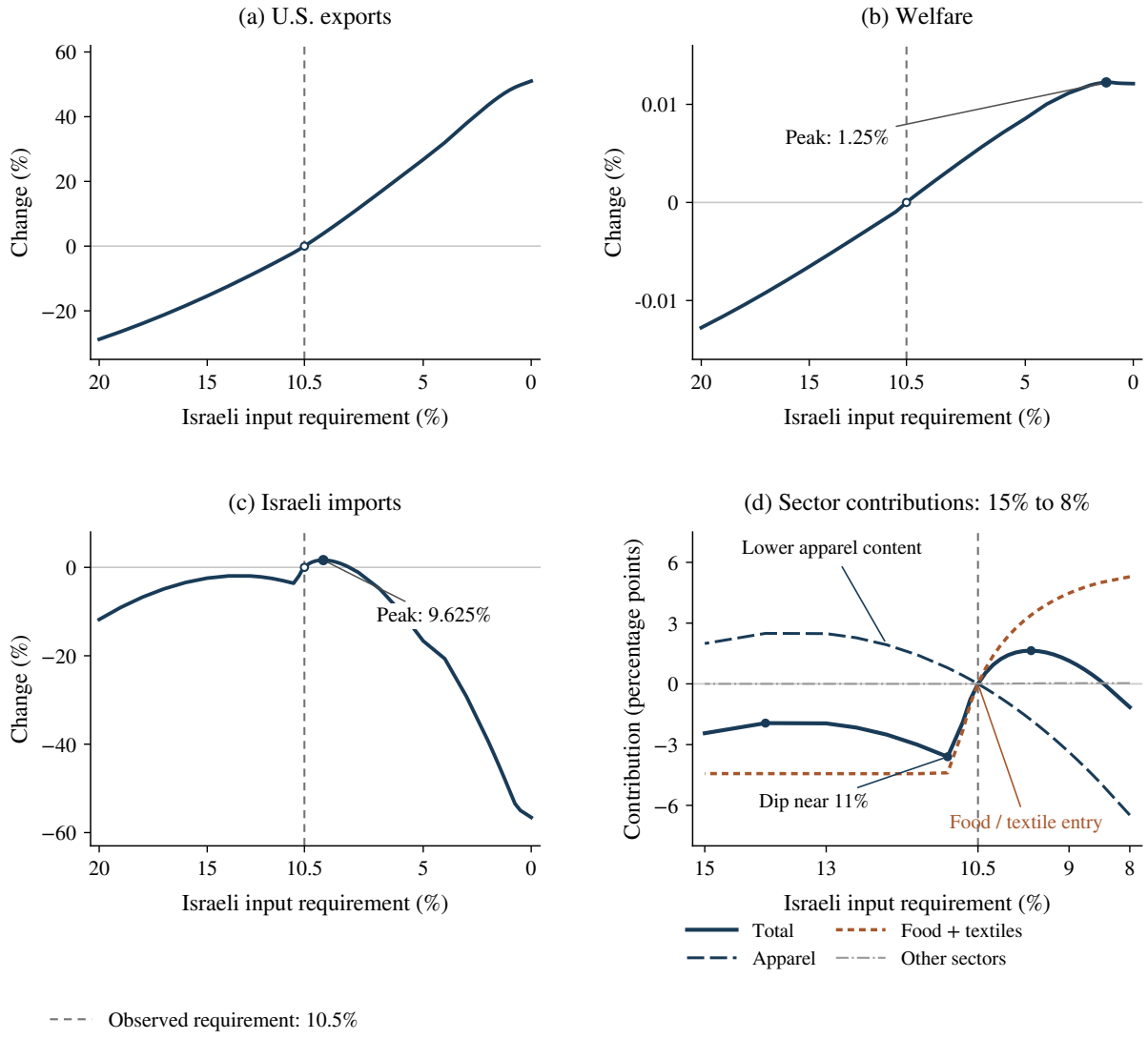


Figure 12: Economic Effects of the Israeli Input Requirement

Notes: Changes are relative to observed QIZ at 10.5%. Panel (d) enlarges the 15–8% range and looks at sector contributions to Israeli imports.

Relaxing the input requirement expands U.S. exports generally as shown in panel (a), but Israeli imports exhibit more than one turning point because sectors start using the QIZ agreement at different requirements. Panel (c) shows the aggregate response of israeli imports to different input requirements, while panel (d) isolates the sectoral contributions to the import. Starting from 20%, relaxing the requirement initially increases participation in apparel, the only QIZ-using sector due to its high initial MFN tariff. Between roughly 14% and 11%, however, lower Israeli content outweighs apparel's export expansion and thus aggregate imports decrease. Its QIZ-related purchases fall from EGP 648 million to EGP 628 million, producing a decrease in aggregate imports.

The subsequent recovery is driven by different sectors beginning to utilize the QIZ program. As the requirement falls below roughly 11%, textiles and food firms begin participating. Their additional purchases outweigh the continuing decline in apparel's purchases, lifting total Israeli imports to their evaluated maximum at 9.625%. Reducing the requirement from 10.5% to 9% raises U.S. exports by 6.8% and Israeli purchases by 1.1%. Below 9.625%, further export expansion no longer offsets lower sourcing intensity, and aggregate purchases decline again.

The requirements maximizing Israeli purchases and Egyptian welfare differ substantially. Among the counterfactuals, Egyptian welfare peaks at a requirement of 1.25% and stays stable after that, and this is well below the input requirement maximizing Israeli-sales. At a zero Israeli requirement, U.S. exports rise by 51.0% and welfare by 0.0121% relative to the baseline QIZ, whereas Israeli imports fall by 56.6%. The QIZ agreement therefore creates a divergence between the U.S geopolitical objectives of expanding bilateral trade ties between Egypt and Israel, Israel's objective of maximizing sales to Egypt, and Egypt's objective of maximizing exports and welfare.

8.5 The Geopolitical Value of the QIZ

I use the counterfactual trade flows in a bargaining exercise to answer two questions: what geopolitical valuation of bilateral trade between Egypt and Israel would justify U.S. support for the observed agreement, and how much higher would that valuation need to be for a lower input requirement to become politically feasible?

Suppose the United States values Egypt–Israel input trade but dislikes additional Egyptian exports to its market because it affects its trade balance negative. The U.S value function would then be given by:

$$U_U(r) = \lambda M_I(r) - X_U(r), \tag{25}$$

where $M_I(r)$ denotes Israeli input sales to Egypt and $X_U(r)$ Egyptian exports to the United States. The parameter λ measures how much additional market access the U.S. would accommodate per dollar of additional bilateral input trade. between Egypt and Israel. I further assume that Egypt only values its exports ot the U.S., while Israel only values its sales to Egypt.

For each counterfactual, I calculate the changes in both Egyptian exports to the U.S. and imports from Israel relative to the relevant outside option. U.S. approval of a change in the input requirement requires that:

$$\lambda \, \Delta M_I \geq \Delta X_U \qquad \Longrightarrow \qquad \lambda \geq \frac{\Delta X_U}{\Delta M_I}, \tag{26}$$

provided that both Egypt and Israeli accept this change, i.e. that $\Delta M_I > 0$ and $\Delta X_U > 0$. The counterfactual input requirements therefore can provide a geopolitical valuation that the U.S. puts on trade between Egypt and Israel in order to approve the agreement.

Table 7: Counterfactual Trade Gains and Political Approval

Policy comparison	ΔX_U	ΔM_I	Minimum λ
10.5% versus no QIZ	3,787.7	661.7	5.72
10% versus 10.5%	199.4	15.0	13.30
9.625% versus 10.5%	353.9	17.3	20.49
9% versus 10.5%	618.8	12.0	51.53

Notes: Trade changes are EGP million.

The first comparison in Table 7 reveals that conditional on the assumed U.S. valuations, U.S. support for the observed agreement requires that $\lambda \geq 5.72$. The remaining two comparisons evaluate renegotiation of the baseline input requirement of 10.5% to other lower values. Both Egypt and Israel gain from all these of the alternative requirement shown in the table under the assumed objectives, but U.S. approval requires substantially higher geopolitical valuation of trade between Egypt and Israel. Negotiating a 10% requirement, for example, requires $\lambda \geq 13.30$, whereas a a 9% requirement requires $\lambda \geq 51.53$.

The exercise shows that the existing program creates substantial bilateral trade, but further relaxation adds relatively little Israeli imports for additional Egyptian market access. A valuation sufficient to sustain the existing QIZ may therefore be insufficient to approve a lower requirement, even though both Egypt and Israel would benefit from it. This may explain why negotiations to reduce the requirement have been repeatedly deferred, despite recurring proposals (EnterpriseAM, 2026).

9 Conclusion

In this paper, I study how geopolitically motivated trade agreements affect trade and firm outcomes, and how their design shapes their effectiveness. Using the QIZ agreement in Egypt, I show that preferential access to the U.S. market increased Egyptian exports and imports from Israel, while generating export spillovers to other destinations. Survey evidence is consistent with international certification as a potential mechanism for these wider gains.

A calibrated heterogeneous-firm model shows that the agreement generated positive but modest welfare gains, with utilization of the agreement concentrated in few irms and few sectors. The model reveals a central trade-off: stricter Israeli input requirements increase sourcing from Israel among participating firms but discourage other firms from utilizing the agreement. Consequently, relaxing the requirement can increase Egyptian exports and welfare without reducing imports from Israel. Yet the bargaining exercise shows that even reforms generating gains in both trade flows may fail to secure U.S. approval. These findings highlight a fundamental tension in geopolitically motivated trade agreements: the conditions intended to strengthen trade relationships may undermine their effectiveness, while mutually beneficial reforms may be politically infeasible.

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A Program Geography and Exposure

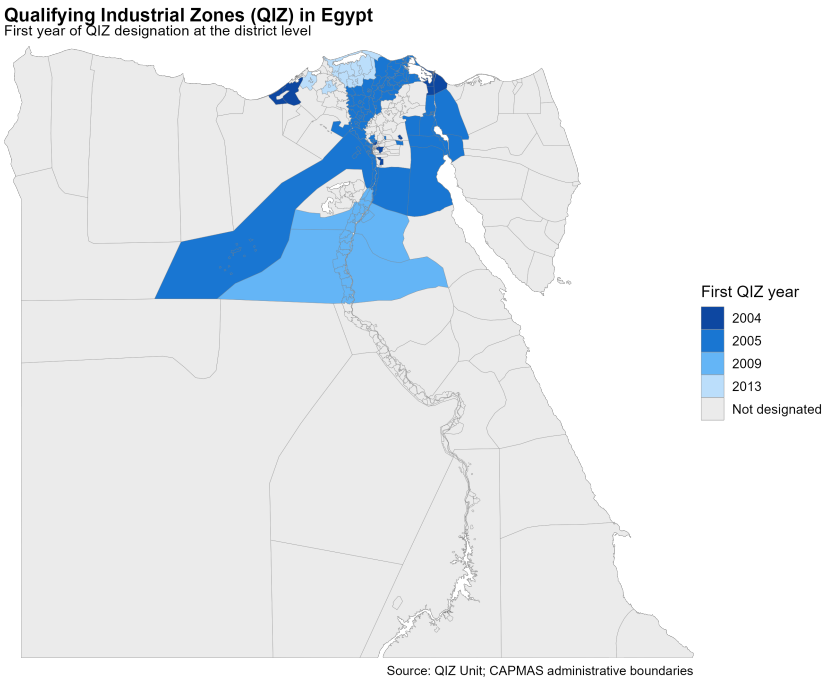


Figure A1: Map of Egyptian QIZs

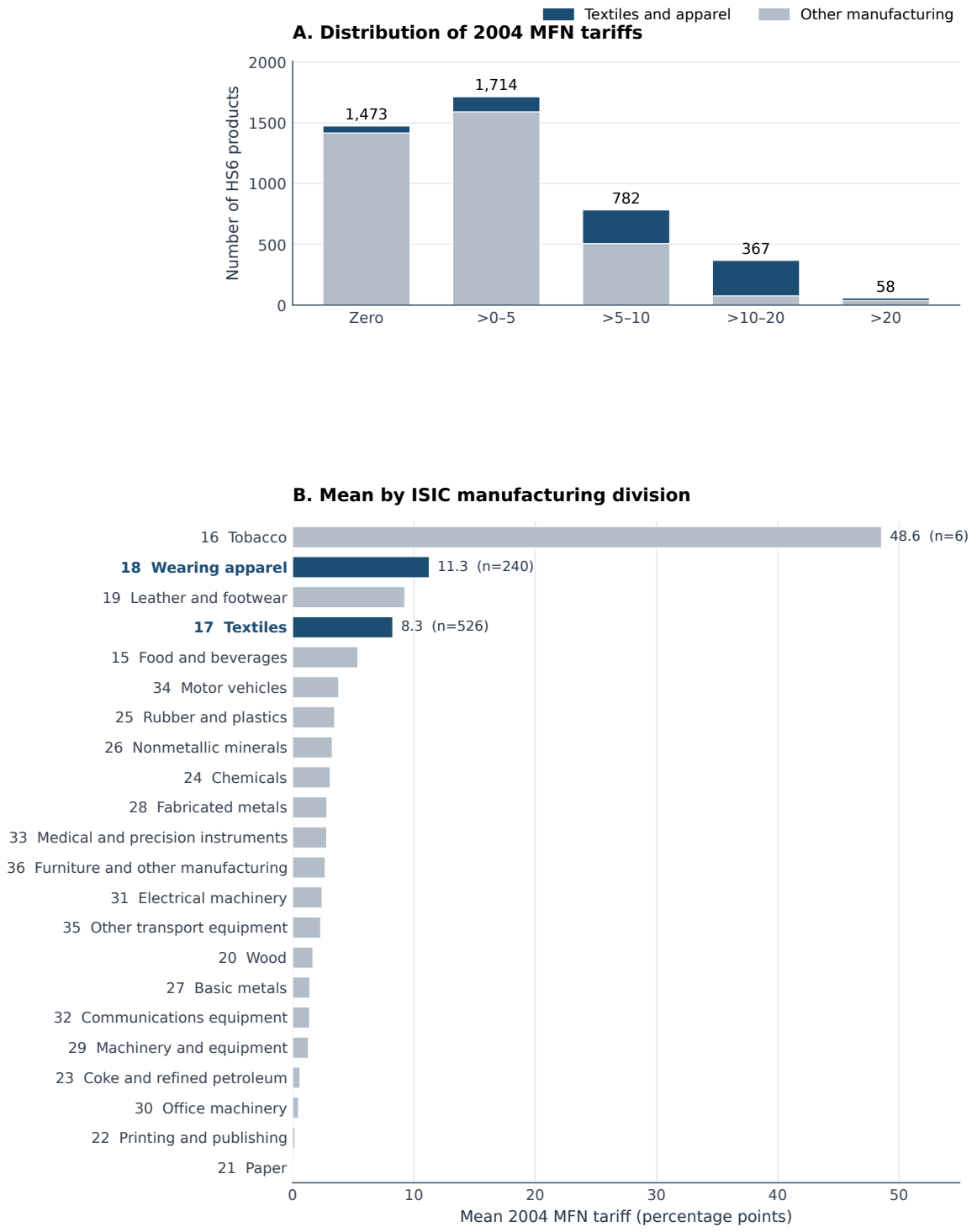


Figure A2: Distribution of 2004 U.S. MFN tariffs in the manufacturing export-exposure sample. The 4,394 HS6 products are each counted once. Panel A shows tariff bands; Panel B shows unweighted means by ISIC manufacturing division. Textiles and apparel (ISIC 17–18) are highlighted. The tobacco mean is based on six products.

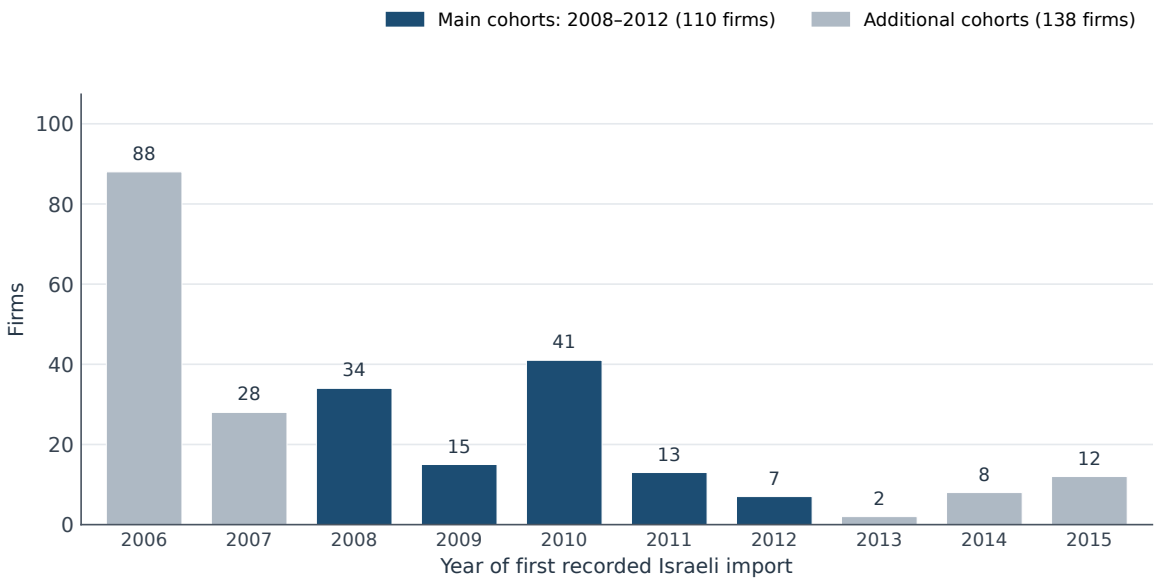


Figure A3: First recorded Israeli imports among textile and apparel exporters. The 2006–2015 cohorts used in the wider event study include 248 firms; the main 2008–2012 cohorts (110 firms) are highlighted. Firms already importing from Israel in 2005 are excluded because their first-import year is not observed.

A.1 Robustness of the U.S. Export DDD Estimates

Figures A4–A6 present the three export robustness event studies. They vary the MFN tariff year and clustering level while retaining the preferred fixed effects and manufacturing sample. Each panel reports the relative export response to one percentage point of MFN exposure and reports the bootstrap joint pretrend p-value. Pointwise confidence intervals remain conventional clustered intervals. The associated pooled coefficients and bootstrap pretrend tests appear in columns (2)–(4) of Table 1.

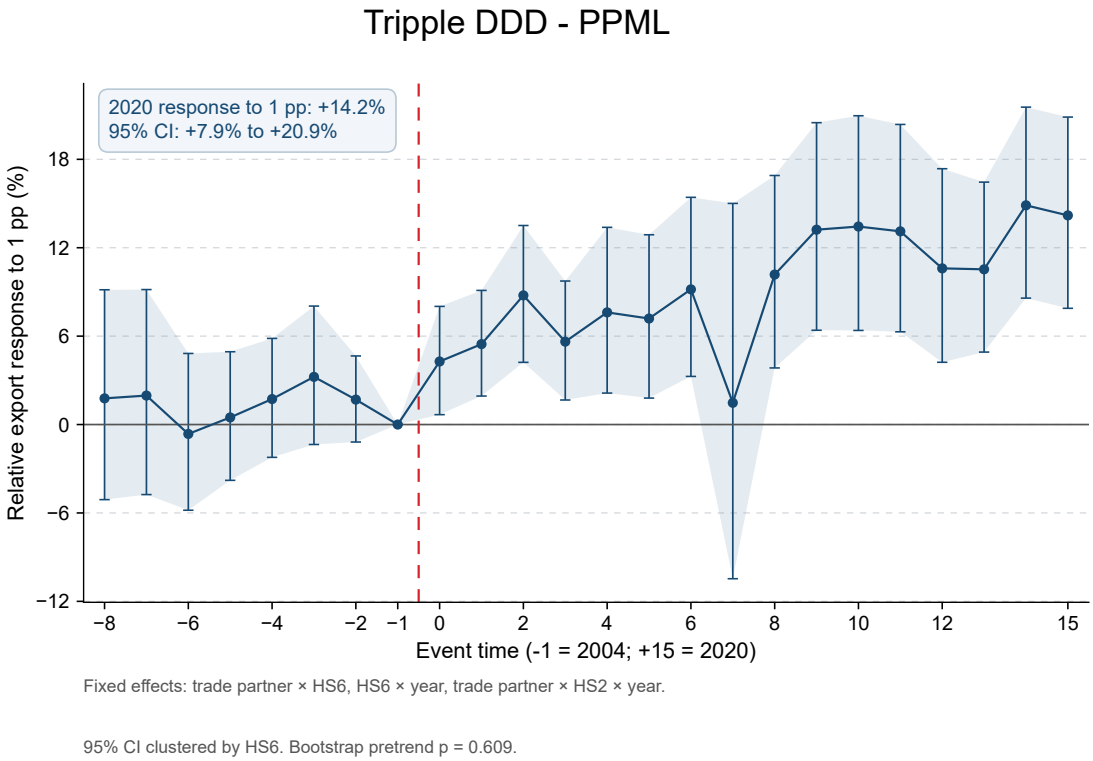


Figure A4: Manufacturing Export DDD Robustness: 2003 MFN Exposure, HS6 Clustering

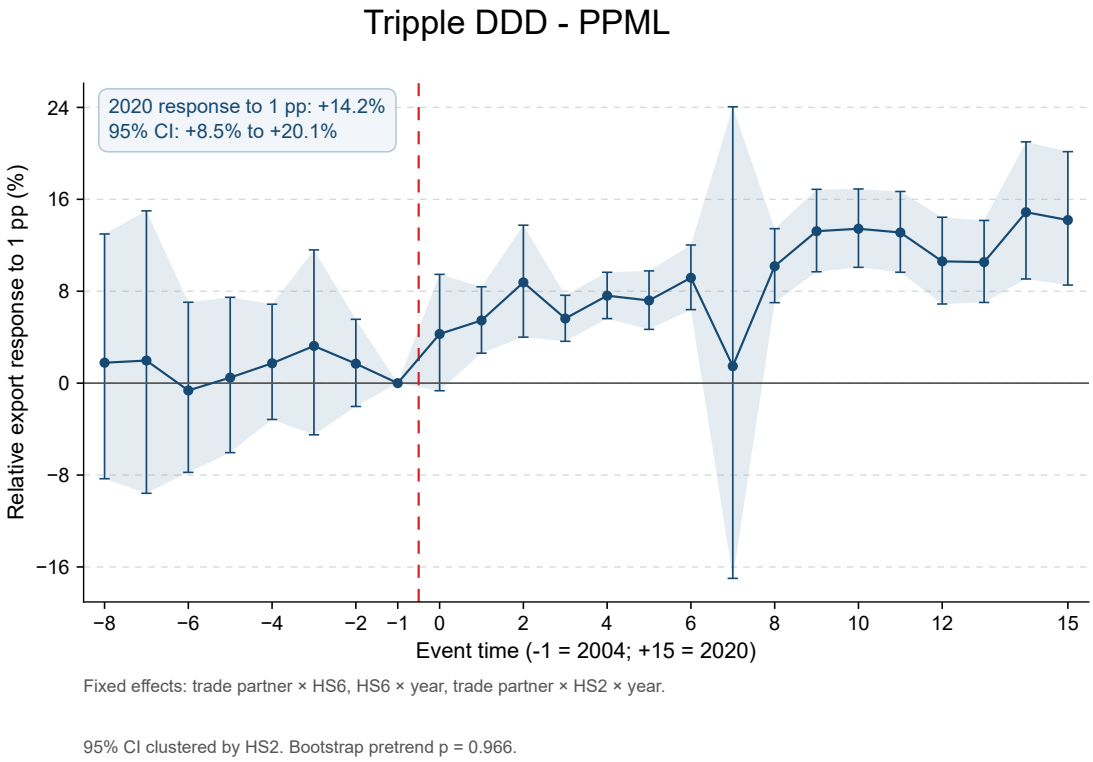


Figure A5: Manufacturing Export DDD Robustness: 2003 MFN Exposure, HS2 Clustering

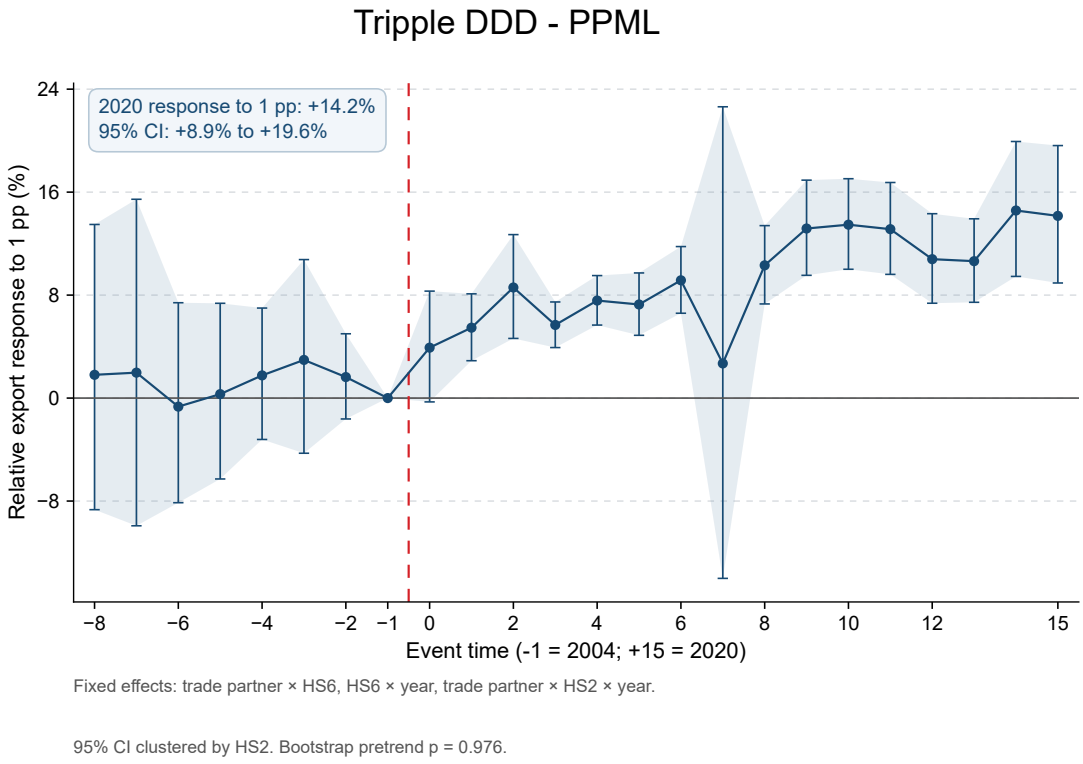


Figure A6: Manufacturing Export DDD Robustness: 2004 MFN Exposure, HS2 Clustering

A.2 Imports from Israel: Trade-Partner-by-Year Controls

Figure A7 presents the 2000/01 input-output event study with trade-partner-by-year controls, complementing the specification with sector controls in Figure 5. By 2020, the estimated relative import response reaches 316.7% per exposure percentage point. The Webb bootstrap joint pretrend p-value is 0.545, while the conventional Wald test rejects ($p < 0.001$).

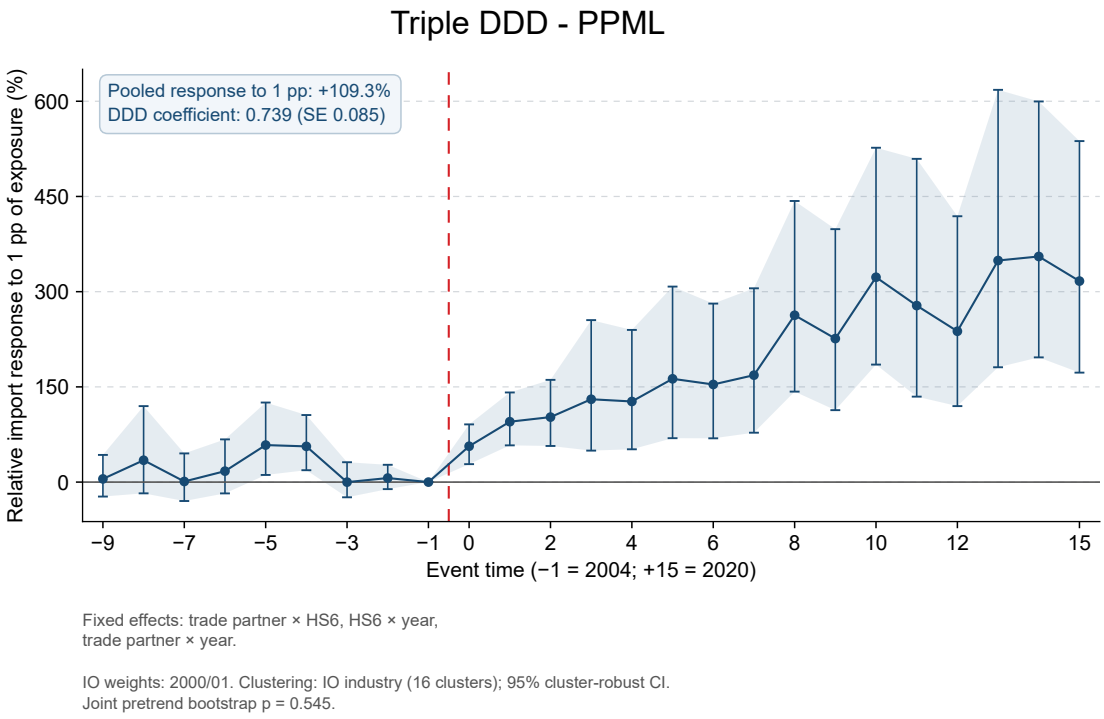


Figure A7: Imports from Israel: 2000/01 IO Exposure, Trade-Partner-by-Year Controls

A.3 Imports from Israel: 2008/09 Input-Output Weights

Table A1 and Figures A8–A9 report the import robustness results using the 2008/09 input-output table. The pooled coefficients without industry trends are 0.4127 with trade-partner-by-year controls and 0.7498 with trade-partner-by-broad-sector-by-year controls, implying relative responses of 51.1% and 111.7% per exposure percentage point. The Webb pretrend p-values are 0.816 and 0.532, while both conventional Wald tests reject at 5%. With differential industry trends, the pooled coefficients become -0.0247 ($p = 0.467$) and 0.0501 ($p = 0.678$). These results reinforce the sensitivity to trend assumptions. Because the 2008/09 weights are measured after implementation, they provide a robustness comparison rather than a predetermined baseline.

Table A1: Imports from Israel: Robustness Using 2008/09 IO Exposure

	Baseline (1)	Sector FE (2)	Sector FE + trends (3)
DDD coefficient	**0.4127*****	**0.7498*****	**0.0501**
	(0.0895)	(0.1651)	(0.1193)
Response to 1 pp exposure	+51.1%	+111.7%	+5.1%
Fixed effects and trends			
Trade partner × HS6	Yes	Yes	Yes
HS6 × year	Yes	Yes	Yes
Trade partner time effects	Year	Sector × year	Sector × year
Egypt × industry linear trends	No	No	Yes
Observations	103,005	102,756	102,753
Years	1996–2020	1996–2020	1996–2020
IO-industry clusters	29	29	29
Bootstrap pretrend p-value	0.816	0.532	–

Notes: Sector FE denotes trade partner × broad sector × year effects; column (3) adds Egypt × industry linear trends. Standard errors clustered by IO industry in parentheses; $^{***}p < 0.01$. Bootstrap p-values test joint pretrends (Webb weights); – denotes not applicable.

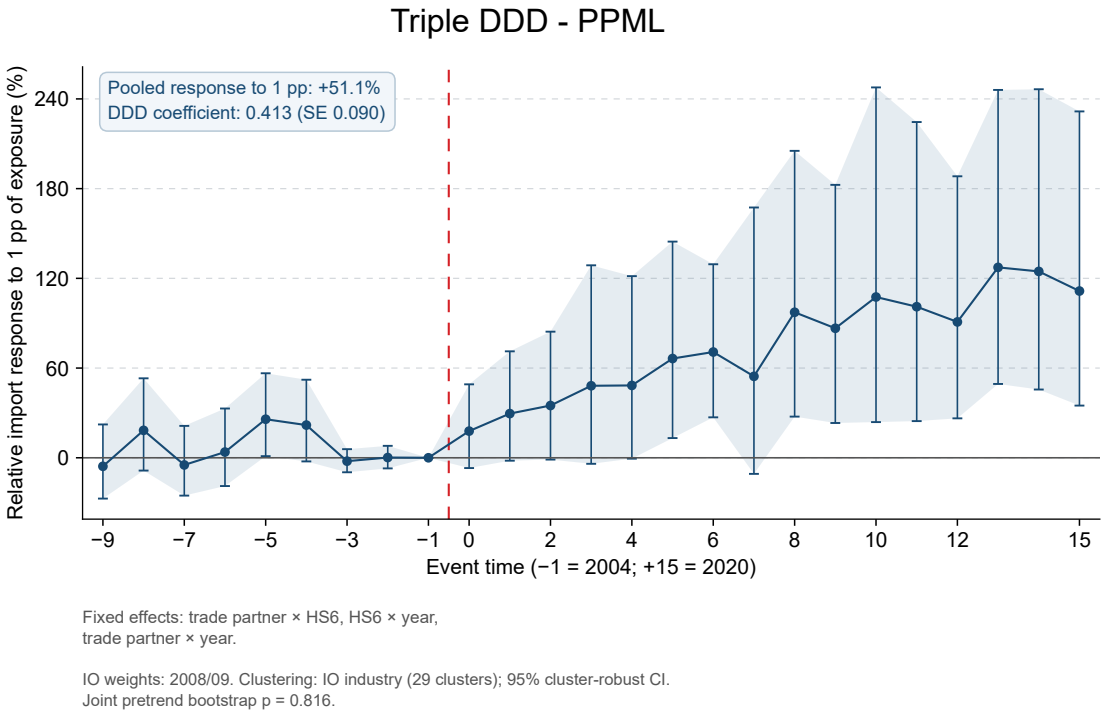


Figure A8: Imports from Israel: 2008/09 IO Exposure, Trade-Partner-by-Year Controls

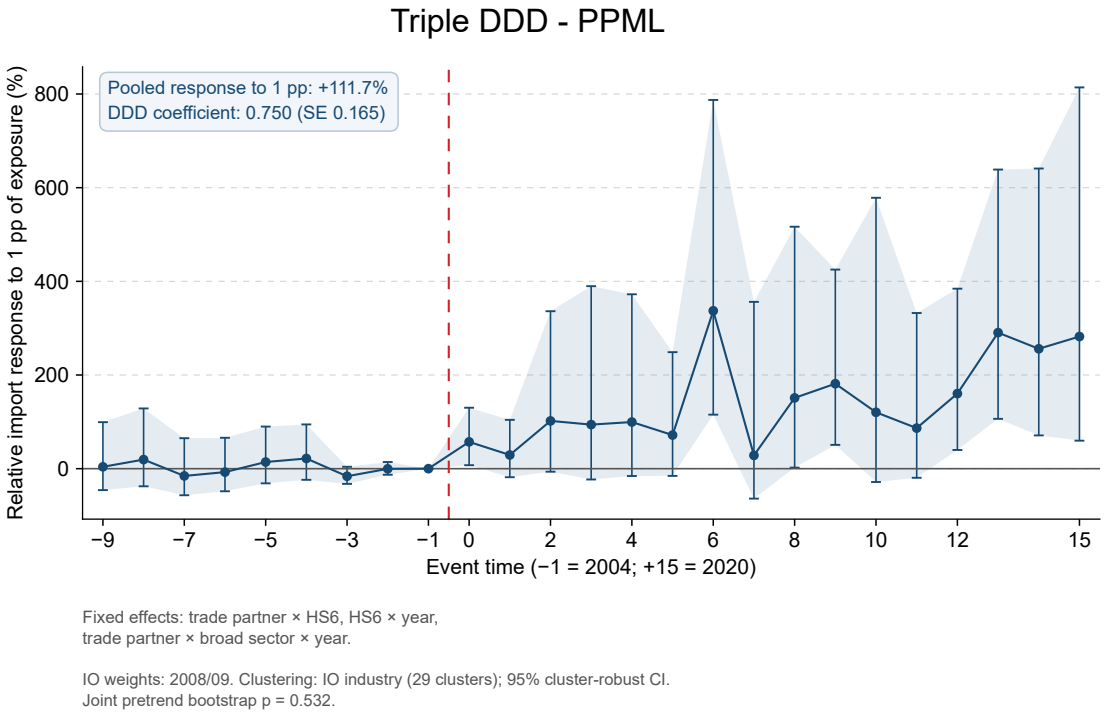


Figure A9: Imports from Israel: 2008/09 IO Exposure, Trade-Partner-by-Sector-by-Year Controls

B Log Outcomes with Zeros

For the firm-level log-outcome specifications, I use a transformation of exports designed for settings with many zero or very small trade flows. Following Chen and Roth (2024), I assign export values below 10,000 a common floor and take logs of values at or above that threshold. Changes entirely below the floor do not affect the transformed outcome, while a firm moving from zero to exports above the floor does contribute. The estimates therefore include sufficiently large entries into exporting as well as changes among continuing exporters; export levels and annual export participation provide complementary views of adjustment.

$$y_{it} = \begin{cases} \log(10{,}000), & \text{if Exports}_{it} < 10{,}000, \\ \log(\text{Exports}_{it}), & \text{if Exports}_{it} \geq 10{,}000. \end{cases} \tag{27}$$

C Additional Industry-Level Results

The following exercise complements the output, employment, and total-wage results in Section 5.3.

C.1 Labor Productivity

I first examine whether higher labor productivity could account for firms' expansion into non-U.S. markets. Using the UNIDO data, I measure labor productivity as value added per worker and estimate its response to pre-agreement industry exposure. Figure C1 shows no sustained increase after the agreement. Most post-agreement coefficients are negative, and the pooled estimate is −0.060 log points with a standard error of 0.049, which is not statistically distinguishable from zero.

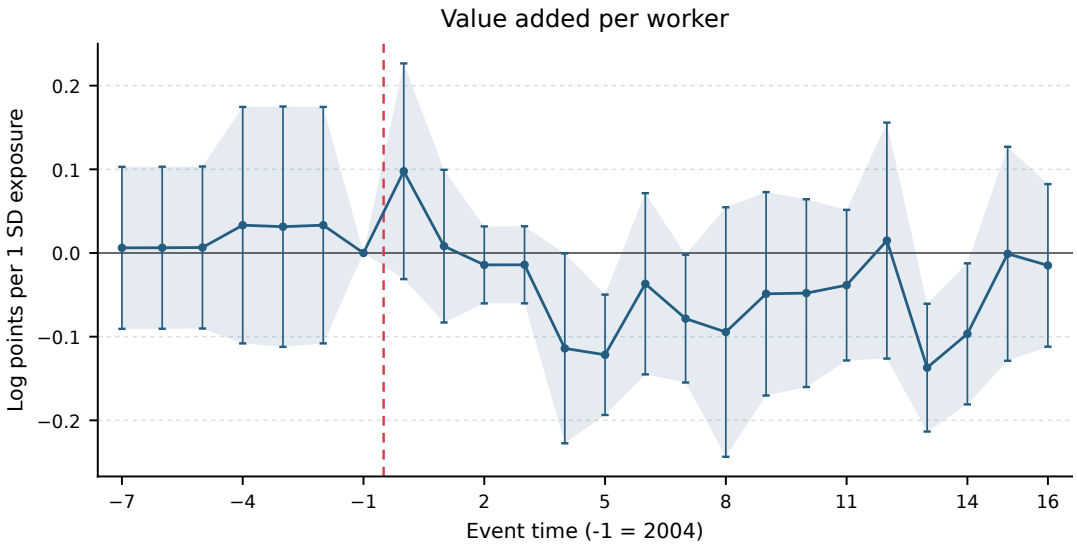


Figure C1: Industry-Level Response of Labor Productivity

Notes: Fixed effects: industry and year. Pointwise 95% confidence intervals clustered by three-digit industry. Exposure is measured in standard deviations; 2004 is the reference year.

The increase in manufacturing output documented above was therefore accompanied by an expansion in employment without a corresponding sustained increase in value added per worker. This evidence does not support broad industry-level labor-productivity gains as the main explanation for non-U.S. export growth. It cannot rule out productivity improvements among individual exporters, which may be obscured by changes in industry composition.

D Additional Firm-Level Results

These figures use the rebuilt 2005–2016 customs panel and first recorded Israeli import. The main sample has 2008–2012 first-import cohorts (110 treated textile and apparel firms); the wider 2006–2015 sample includes 248 treated firms. Unless stated otherwise, estimates use the cohort-specific CEM weights, firm-by-cohort and cohort-by-HS2-by-year fixed effects, firm-clustered standard errors, and event year -1 as the reference. The figures describe associations around recorded Israeli sourcing.

D.1 Annual Export Participation

An additional margin of adjustment is whether firms export in a given year. I examine an annual indicator for positive exports to any destination. I estimate its event-study path for the same matched textile and apparel firms used in the main firm analysis, with the first recorded Israeli import defining event year zero. The sample includes firms that exported at least once during 2005–2016, including firms with no exports before the event.

Figure D1 shows that the probability of exporting to any destination rises by 24.1 percentage points in the first-import year and 29.2 percentage points one year later in the Sun–Abraham specification. The two displayed pre-event coefficients are jointly indistinguishable from zero ($p = 0.433$).

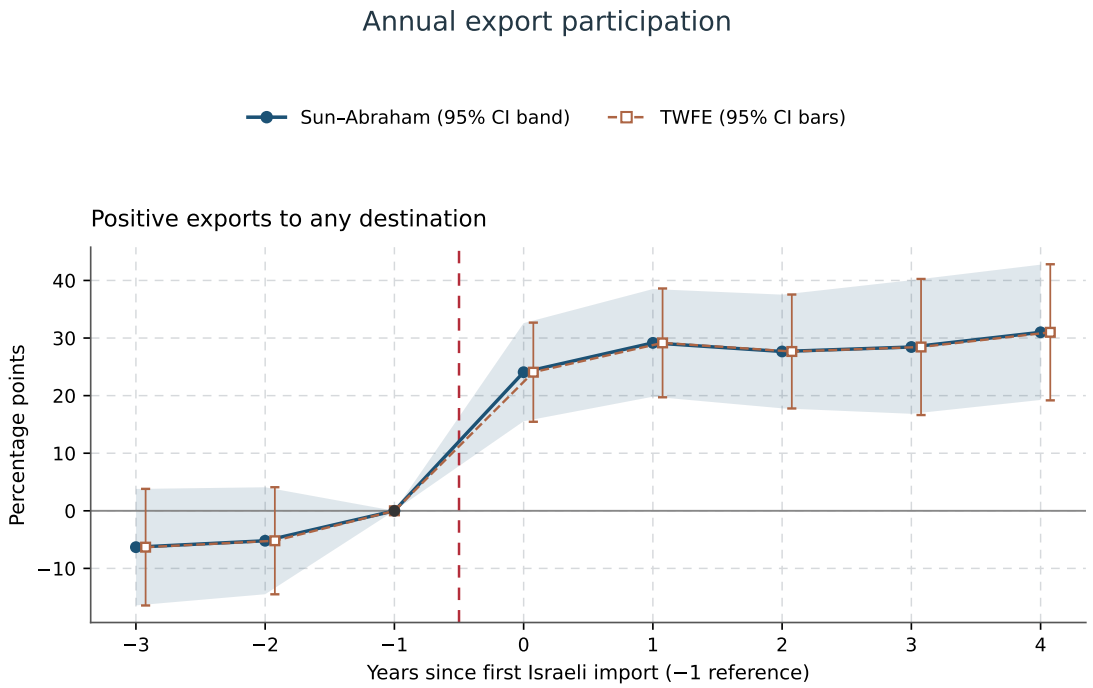


Figure D1: Annual Export Participation to Any Destination around the First Recorded Israeli Import

Notes: The outcome equals one when the firm records positive exports to any destination in the indicated year. The sample contains 110 treated textile and apparel firms with first Israeli imports in 2008–2012 and 1,630 distinct never-Israel-importer controls, all of whom exported at least once during 2005–2016. Estimates use the main coarsened-exact-matching weights, firm-by-cohort and cohort-by-HS2-by-year fixed effects, and firm-clustered 95% confidence intervals. Coefficients are in percentage points; event year −1 is the reference.

This outcome measures annual export activity among firms observed exporting at some point in the customs panel. It does not measure entry into exporting from the universe of Egyptian firms or the creation of new firms. The increase in export participation is consistent with a broader expansion of trade following Israeli sourcing, while endogenous selection into Israeli sourcing and the absence of firm-level QIZ registration data limit a causal interpretation.

D.2 Cohort Range and Matching

Figures D2–D5 repeat the main outcomes for the wider 2006–2015 set of first-import cohorts. This raises the treated sample to 248 firms. The dynamic estimates for both cohort ranges appear in Tables D1 and D2.

Figure D6 compares the preferred CEM with two nearest-neighbor matches that retain all 110 treated firms. Matching only through event year −3 leaves the U.S. export lead test significant ($p < 0.001$). Matching through year −1 flattens those leads ($p = 0.255$), but uses the plotted pre-event outcomes to select controls and produces a destination-count pretrend ($p = 0.024$). The flatter U.S. leads therefore show in-sample balance rather than independent evidence for parallel trends.

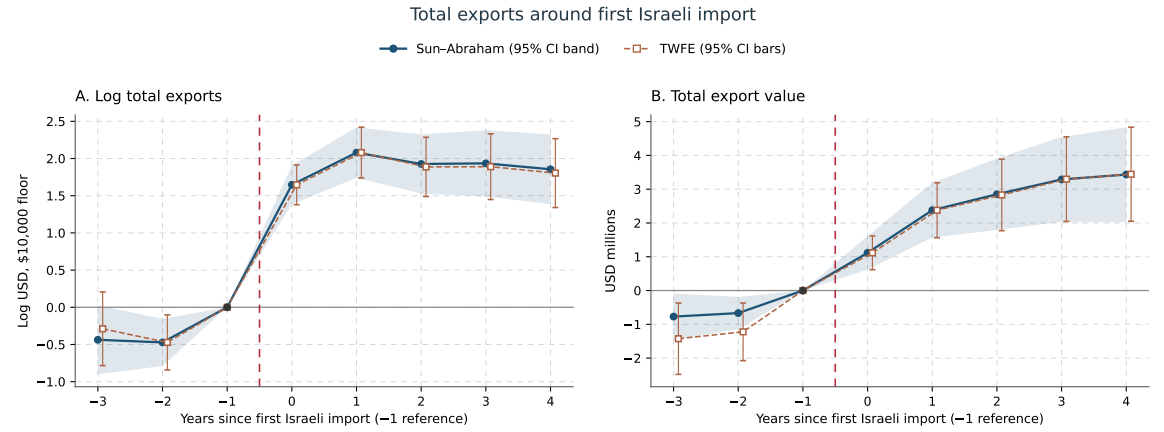


Figure D2: Log and level total exports, 2006–2015 first-import cohorts.

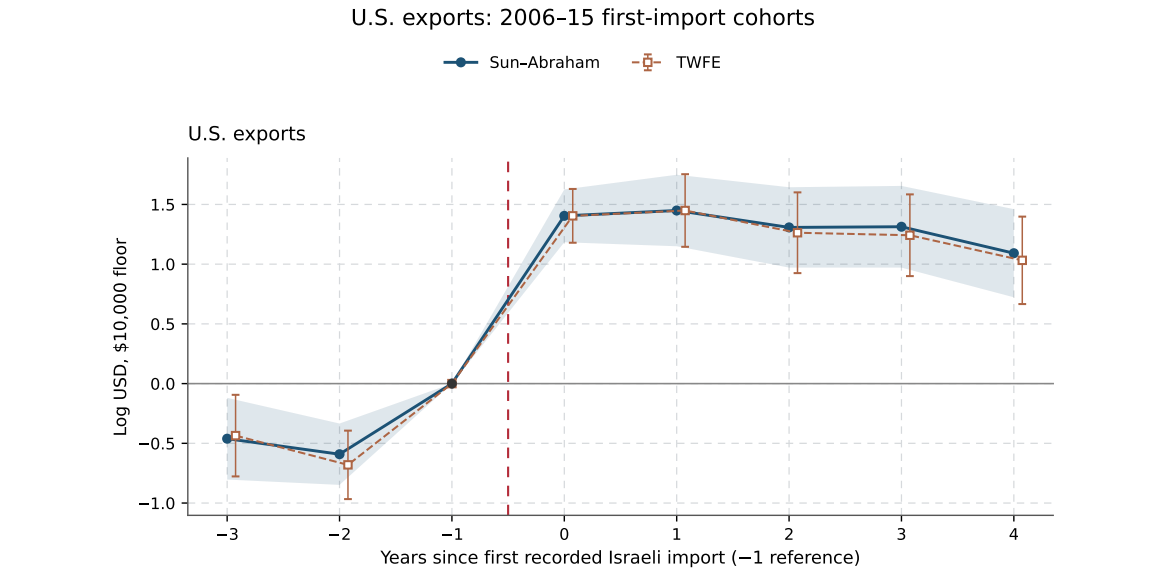
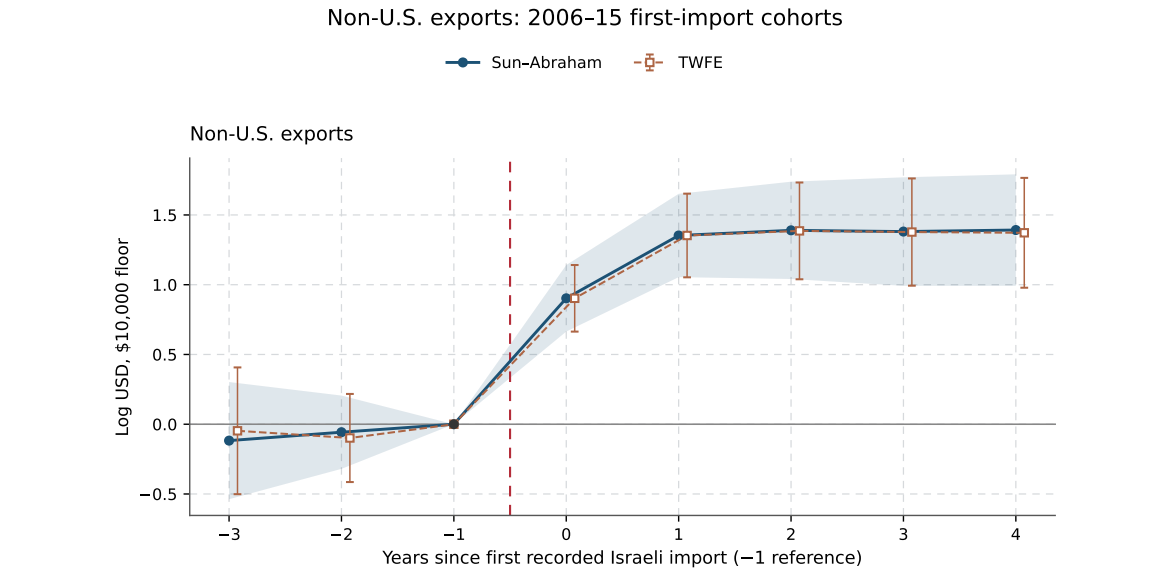
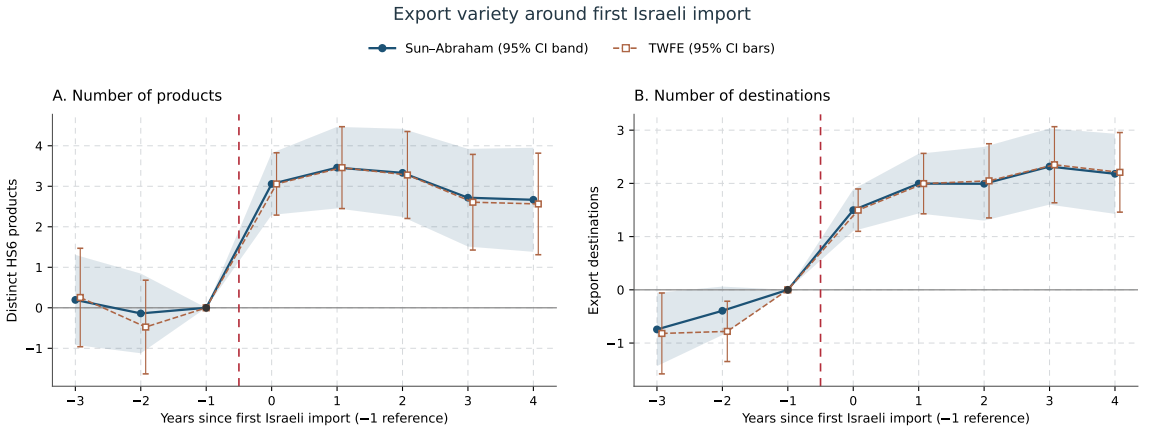


Figure D3: Thresholded log U.S. exports, 2006–2015 first-import cohorts.



Textile and apparel; 248 treated firms. Preferred cohort-specific CEM; firm×cohort and cohort×HS2×year FE; firm-clustered 95% CIs. Outer event times use available cohorts.

Figure D4: Thresholded log non-U.S. exports, 2006–2015 cohorts.



Textile and apparel; first-import cohorts 2006–15. 248 treated firms, 1,846 distinct never-importer controls. CEM within HS2 and G −1 export size; earlier growth where observed. Firm×cohort and cohort×HS2×year FE; SE clustered by firm. Outer event times use available cohorts.

Figure D5: Product and destination counts, 2006–2015 cohorts.

Table D1: Sun–Abraham dynamic outcomes: 2008–2012 cohorts

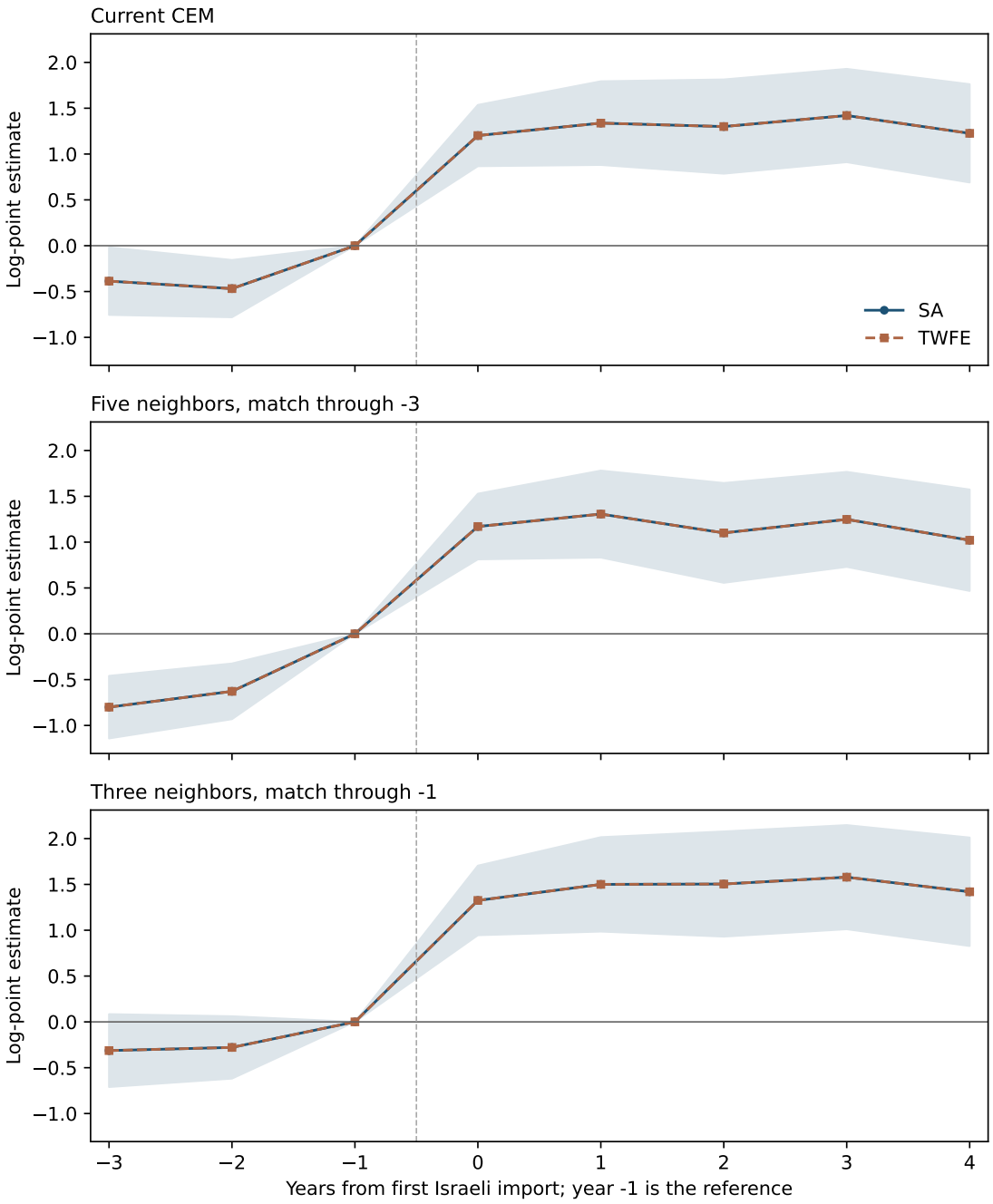
Event time	Log U.S.	U.S. USDm	Log non-U.S.	Non-U.S. USDm	Products	Destinations
-3	-0.386	-0.182	-0.035	-0.488	0.397	-0.611
	(0.188)	(0.156)	(0.243)	(0.334)	(0.665)	(0.400)
-2	-0.468	-0.228	-0.006	-0.434	0.406	-0.287
	(0.159)	(0.129)	(0.181)	(0.278)	(0.711)	(0.320)
$+0$	1.202	0.414	0.905	1.077	2.797	1.507
	(0.171)	(0.186)	(0.187)	(0.403)	(0.465)	(0.313)
$+1$	1.337	1.287	1.288	1.801	3.404	1.936
	(0.233)	(0.336)	(0.239)	(0.565)	(0.751)	(0.455)
$+2$	1.300	1.678	1.183	1.807	2.766	1.552
	(0.262)	(0.438)	(0.281)	(0.644)	(0.801)	(0.583)
$+3$	1.420	2.107	1.345	2.423	2.636	1.993
	(0.260)	(0.559)	(0.310)	(0.784)	(0.796)	(0.495)
$+4$	1.226	1.948	1.360	2.406	2.721	2.211
	(0.274)	(0.533)	(0.306)	(0.902)	(0.827)	(0.538)
Mean 0 to $+1$	1.270	0.851	1.096	1.439	3.100	1.721
	(0.182)	(0.232)	(0.204)	(0.470)	(0.565)	(0.360)
Mean $+2$ to $+4$	1.315	1.911	1.296	2.212	2.708	1.919
	(0.253)	(0.488)	(0.285)	(0.730)	(0.780)	(0.498)
Joint leads p $(-3,-2)$	0.011	0.111	0.981	0.280	0.820	0.270

Notes: Estimates and firm-clustered standard errors in parentheses. First customs-recorded Israeli import is event 0; event -1 is omitted. Cohort-by-event interactions are aggregated by treated-cohort size. Windows average the indicated interactions over a common set of eligible cohorts. Firm-by-cohort and cohort-by-HS2-by-year fixed effects. CEM within HS2 and year -1 total-export size, with earlier total-export growth when available. Log outcomes use $\log(\max\{x, 10{,}000\})$; export levels are USD millions. Requires the booktabs package.

Table D2: Sun–Abraham dynamic outcomes: 2006–2015 cohorts

Event time	Log U.S.	U.S. USDm	Log non-U.S.	Non-U.S. USDm	Products	Destinations
-3	-0.461	-0.402	-0.117	-0.370	0.195	-0.744
	(0.175)	(0.177)	(0.214)	(0.281)	(0.567)	(0.351)
-2	-0.591	-0.366	-0.057	-0.302	-0.139	-0.394
	(0.131)	(0.133)	(0.134)	(0.192)	(0.502)	(0.232)
$+0$	1.404	0.461	0.903	0.656	3.057	1.497
	(0.113)	(0.130)	(0.122)	(0.203)	(0.390)	(0.200)
$+1$	1.449	1.056	1.353	1.321	3.460	1.996
	(0.153)	(0.193)	(0.152)	(0.322)	(0.514)	(0.286)
$+2$	1.308	1.302	1.390	1.548	3.331	1.993
	(0.172)	(0.250)	(0.178)	(0.405)	(0.555)	(0.354)
$+3$	1.313	1.550	1.381	1.738	2.718	2.316
	(0.175)	(0.329)	(0.198)	(0.479)	(0.615)	(0.365)
$+4$	1.092	1.528	1.392	1.905	2.666	2.180
	(0.188)	(0.316)	(0.204)	(0.557)	(0.656)	(0.384)
Mean 0 to $+1$	1.427	0.758	1.128	0.988	3.259	1.746
	(0.122)	(0.135)	(0.131)	(0.248)	(0.413)	(0.228)
Mean $+2$ to $+4$	1.246	1.480	1.395	1.736	2.898	2.155
	(0.169)	(0.288)	(0.187)	(0.463)	(0.595)	(0.348)
Joint leads p $(-3,-2)$	< 0.001	0.018	0.860	0.291	0.717	0.106

Notes: Estimates and firm-clustered standard errors in parentheses. First customs-recorded Israeli import is event 0; event -1 is omitted. Cohort-by-event interactions are aggregated by treated-cohort size. Windows average the indicated interactions over a common set of eligible cohorts. Firm-by-cohort and cohort-by-HS2-by-year fixed effects. CEM within HS2 and year -1 total-export size, with earlier total-export growth when available. For the wide sample, outer event times use only cohorts observed at those times; two 2006 firms use adjacent within-HS2 size bins to preserve treated support. Log outcomes use $\log(\max\{x, 10{,}000\})$; export levels are USD millions. Requires the booktabs package.



110 treated firms. All models use never-Israel controls, dominant HS2 matching, firm-by-cohort and cohort-by-HS2-by-year fixed effects. The bottom panel uses years -3 through -1 in matching; its lead test is in-sample.

Figure D6: Alternative matches for thresholded log U.S. exports, 2008–2012 first-import cohorts. The middle panel leaves years -2 and -1 unused in matching; the bottom panel matches on U.S. exports through year -1. Shading shows firm-clustered 95% confidence intervals for Sun–Abraham estimates. The outcome is $\log(\max\{x, 10{,}000\})$.

D.3 Inverse-Hyperbolic-Sine Export Outcomes

The inverse-hyperbolic-sine transformation retains zero export values. Figures D7 and D8 report U.S. and non-U.S. exports for the wider and main cohort ranges. I interpret these estimates alongside the thresholded logs and export levels because the transformation depends on the units of measurement near zero.

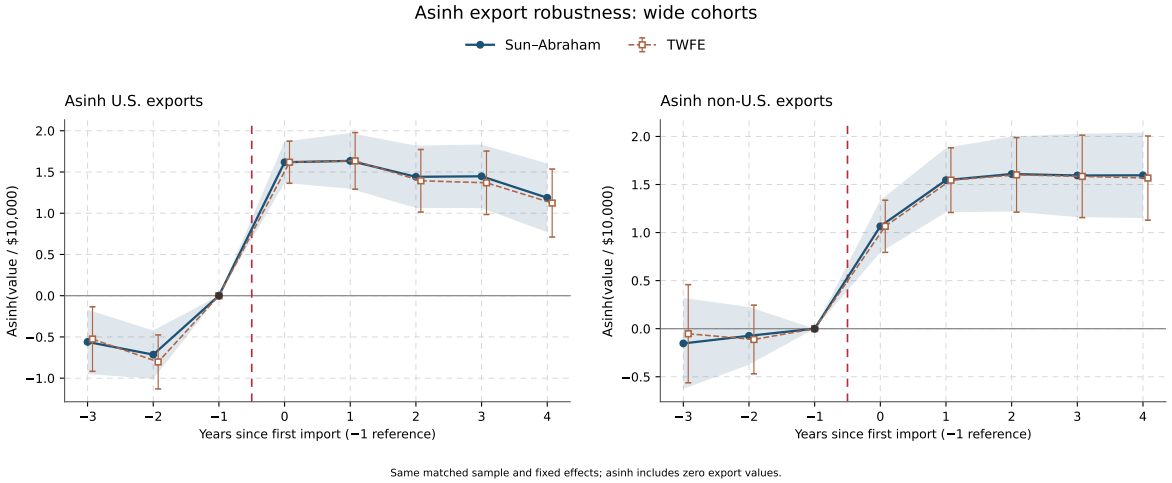


Figure D7: Inverse-hyperbolic-sine U.S. and non-U.S. exports, 2006–2015 cohorts.

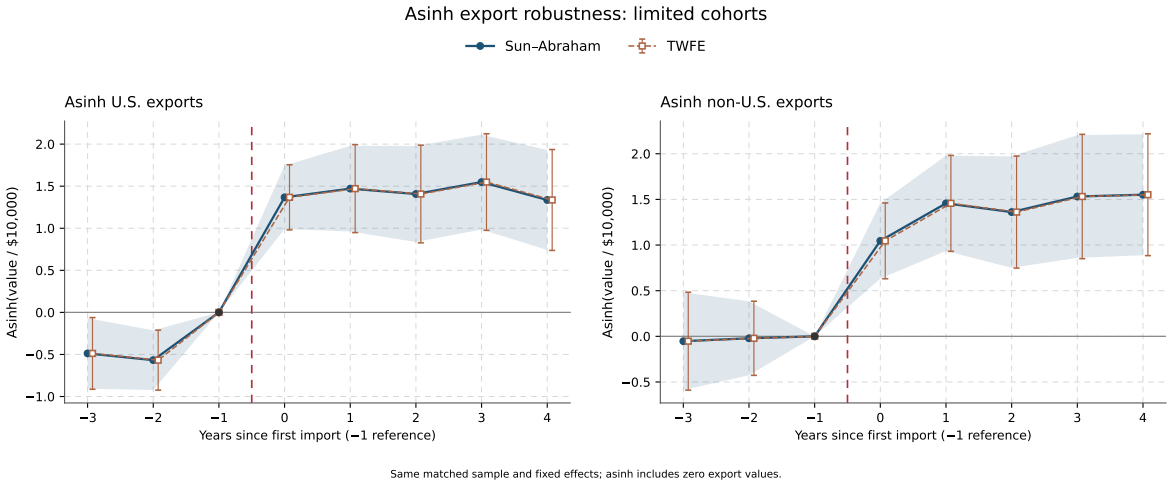


Figure D8: Inverse-hyperbolic-sine U.S. and non-U.S. exports, 2008–2012 cohorts.

D.4 Exports by Destination Region

The supplied region crosswalk covers 94.5% of textile and apparel non-U.S. export value in Europe, Africa, and Asia. Unmapped destinations are excluded from the three regional outcomes.

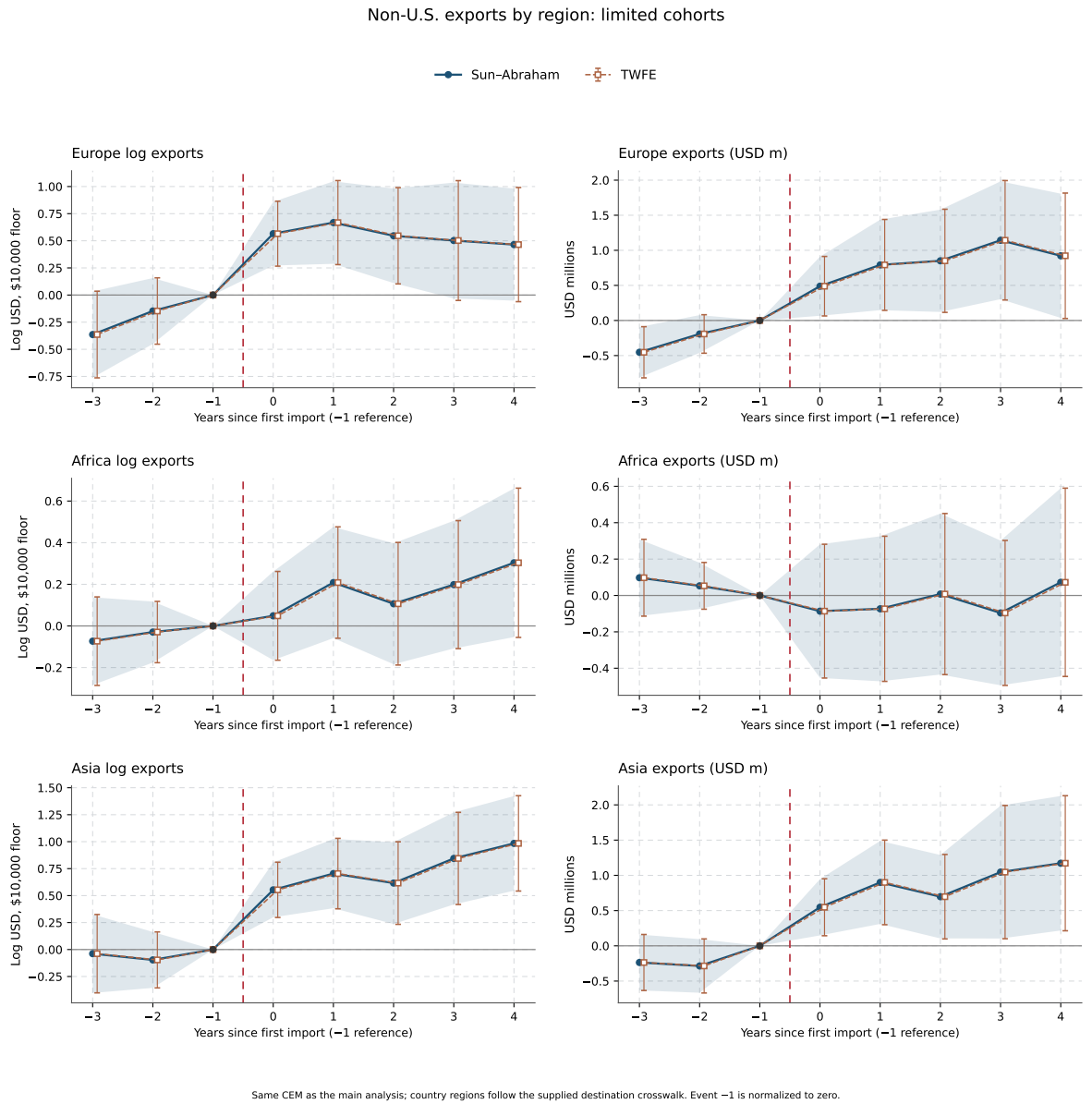
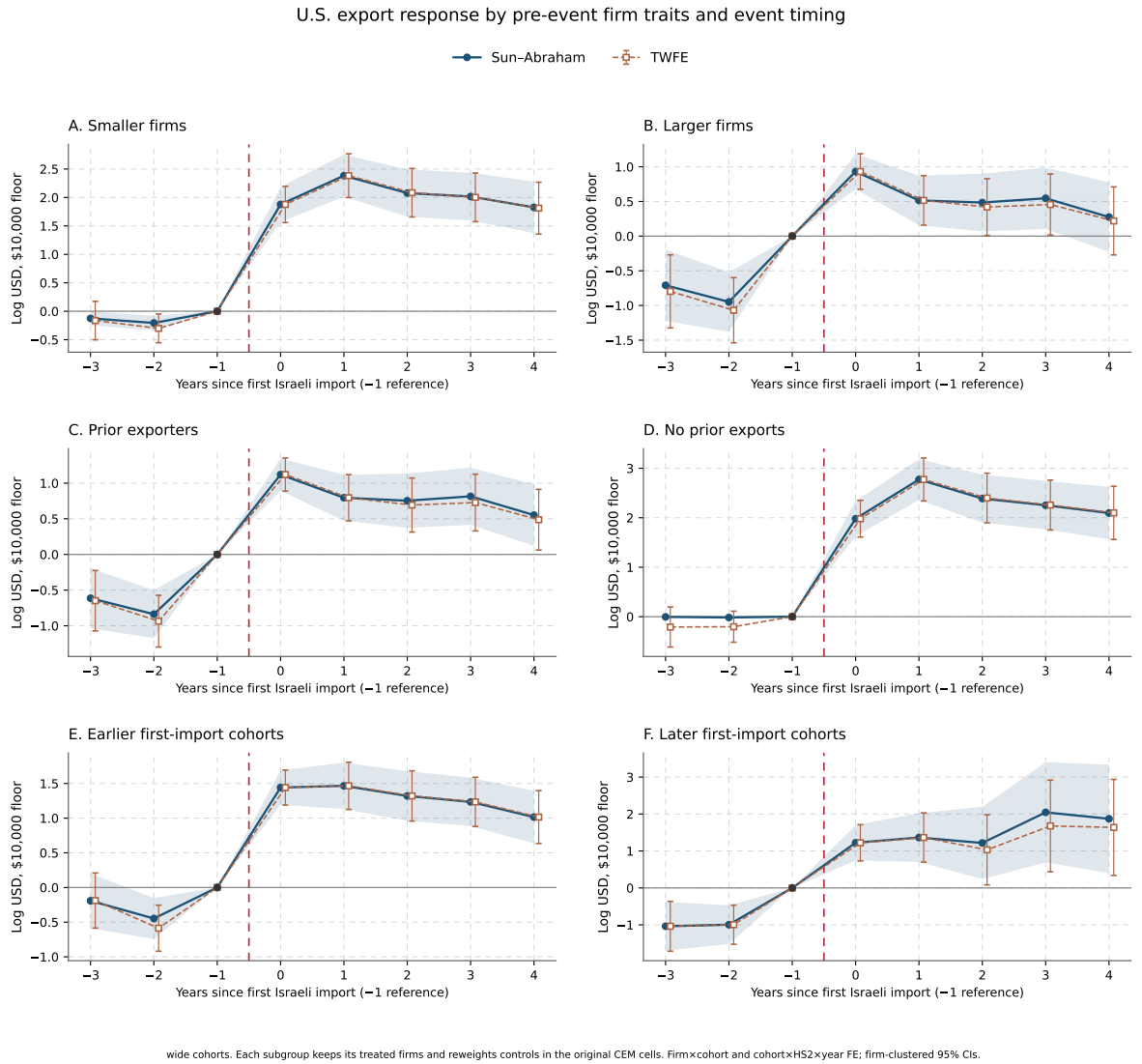


Figure D9: Non-U.S. exports to Europe, Africa, and Asia, 2008–2012 cohorts.

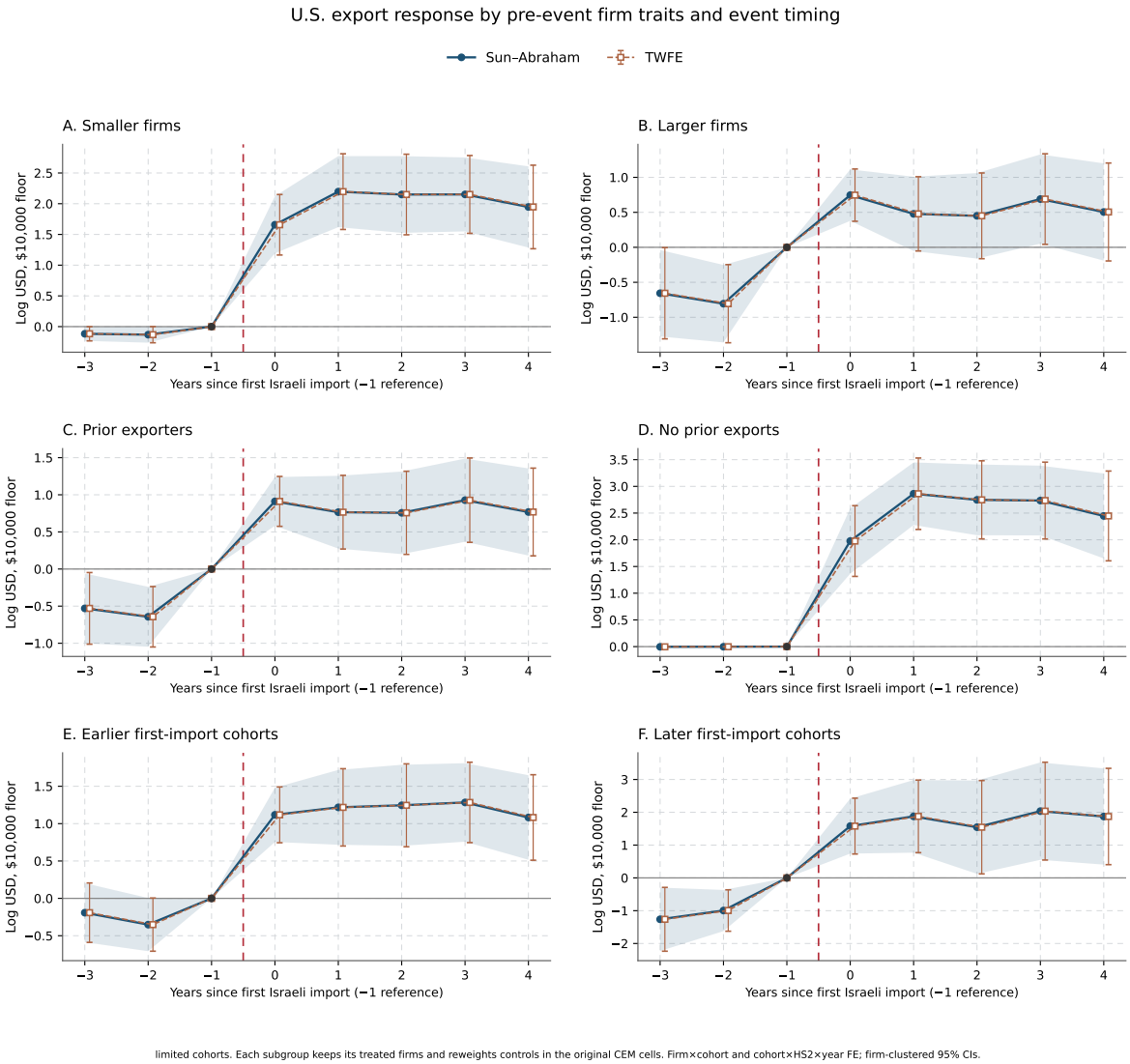
D.5 Firm Heterogeneity

The prior-export split checks whether a treated firm had any positive total exports before its own event year; it does not identify firm birth. Size is split at the median treated year-−1 total exports. Early and late cohorts are split within each cohort range.



wide cohorts. Each subgroup keeps its treated firms and reweights controls in the original CEM cells. Firm×cohort and cohort×HS2×year FE; firm-clustered 95% CIs.

Figure D10: U.S. export responses by size, prior exporting, and first-import cohort timing, 2006–2015 cohorts.



limited cohorts. Each subgroup keeps its treated firms and reweights controls in the original CEM cells. Firm×cohort and cohort×HS2×year FE; firm-clustered 95% CIs.

Figure D11: U.S. export responses by size, prior exporting, and first-import cohort timing, 2008–2012 cohorts.

D.6 Sector and Import-Origin Comparisons

Sector estimates match within each broad HS2 group and cap controls at five per treated firm within a CEM cell. The food-and-agriculture group (HS2 01–24) includes products that could also have used QIZ preferences, so it is not a clean untreated placebo. The textile and apparel association is much larger, but its differential event path has pre-event movement.

The other-origin regressions use separate origin-specific CEM samples and include the full first-Israeli-import event path as a control. The Israel comparison uses the main matched sample; the figure does not test equality across these different populations.

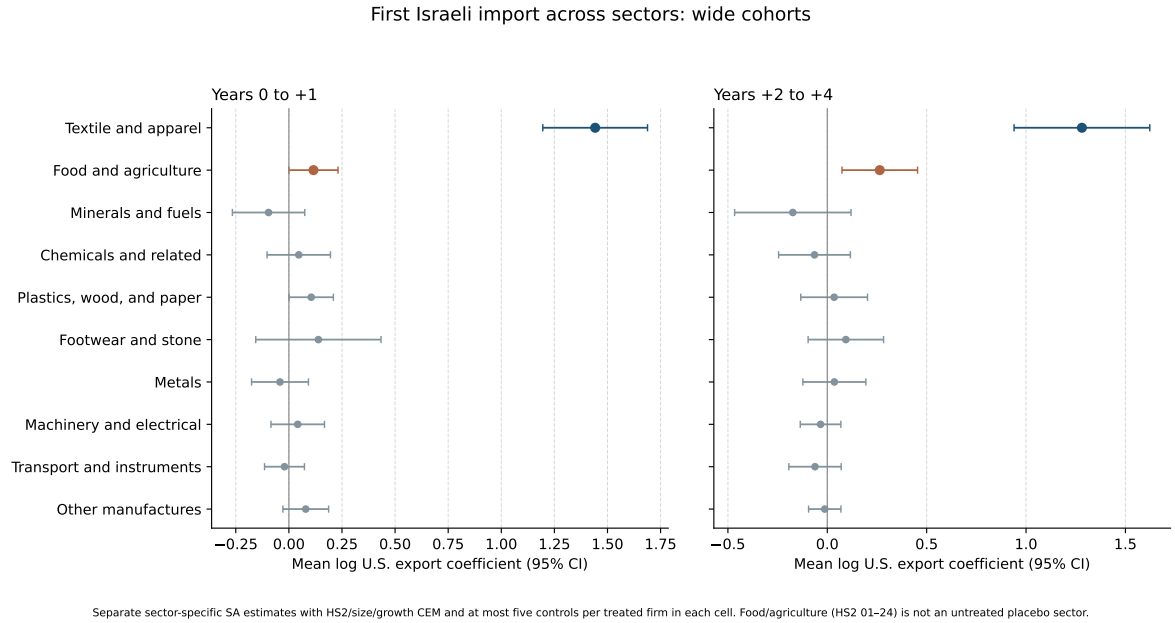


Figure D12: Post-event log U.S. export averages by sector, highlighting textile and apparel and food/agriculture, 2006–2015 cohorts.

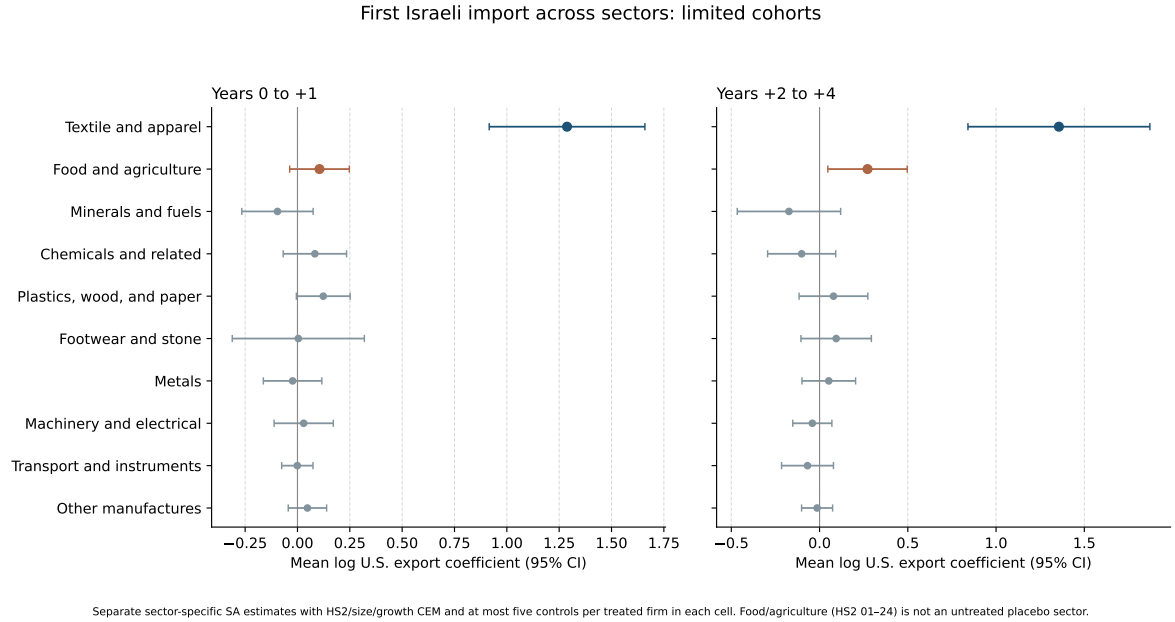


Figure D13: Post-event log U.S. export averages by sector, 2008–2012 cohorts.

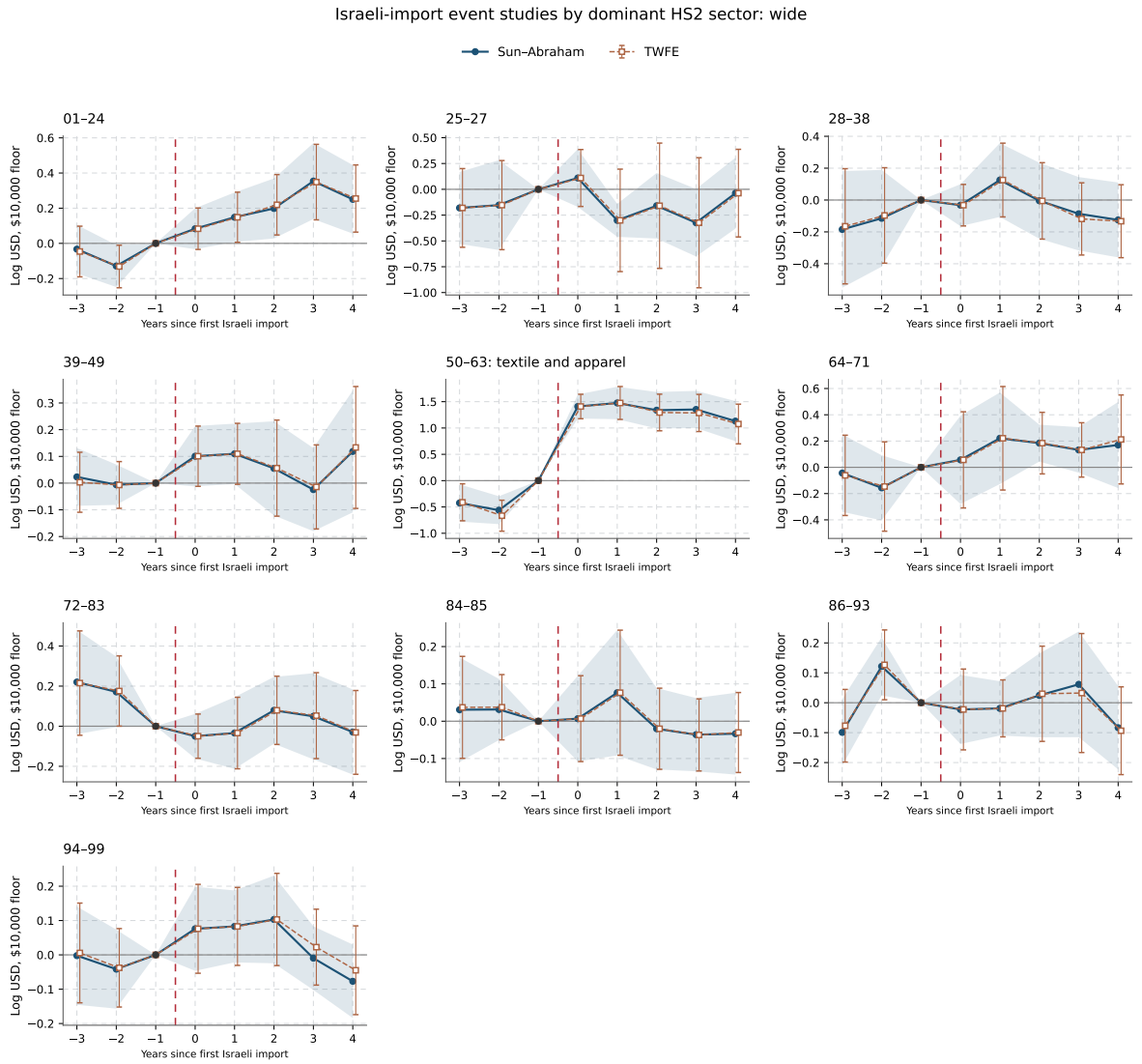


Figure D14: Dynamic first-Israeli-import event studies across broad HS2 sector groups, 2006–2015 cohorts.

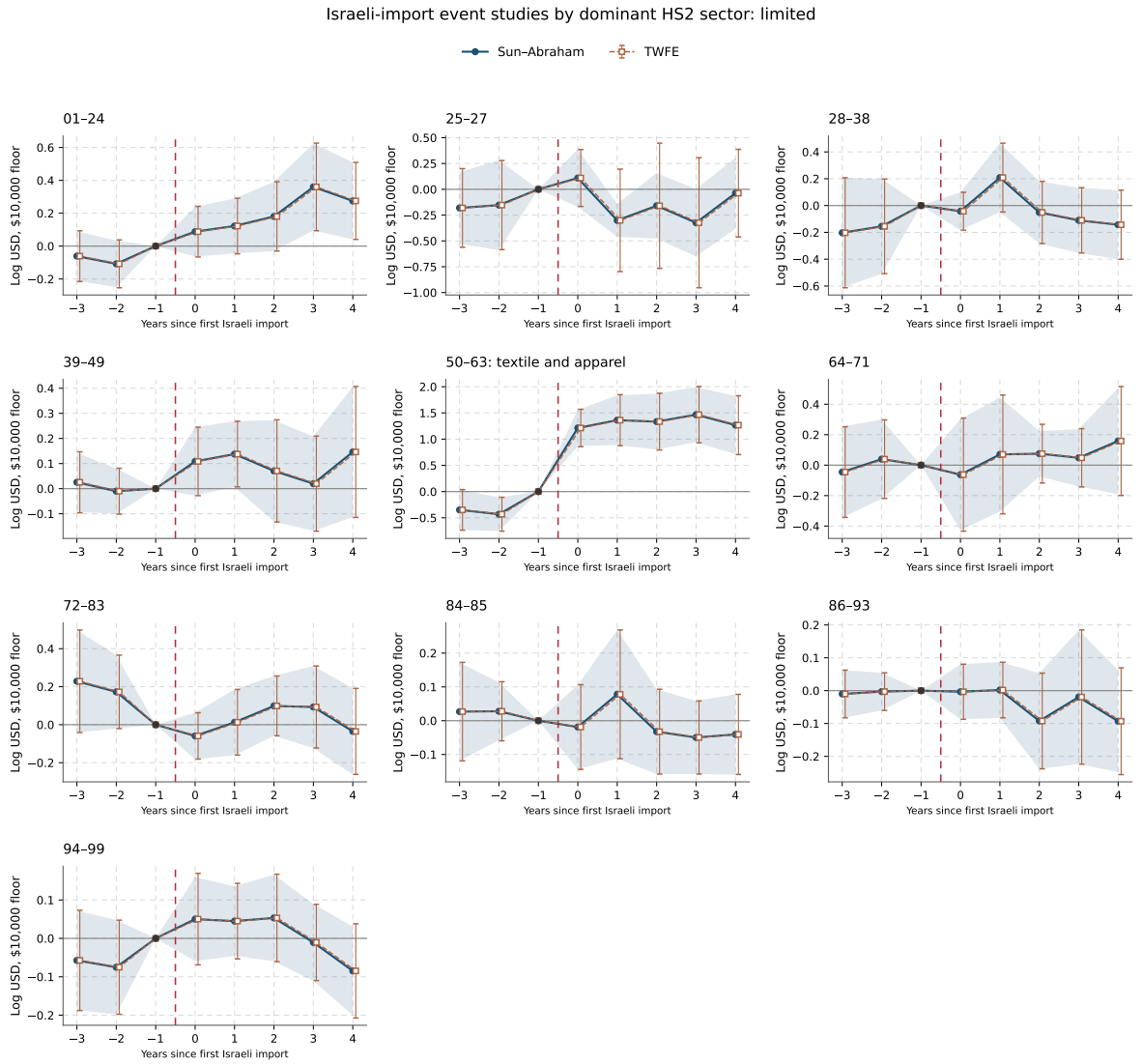


Figure D15: Dynamic first-Israeli-import event studies across broad HS2 sector groups, 2008–2012 cohorts.

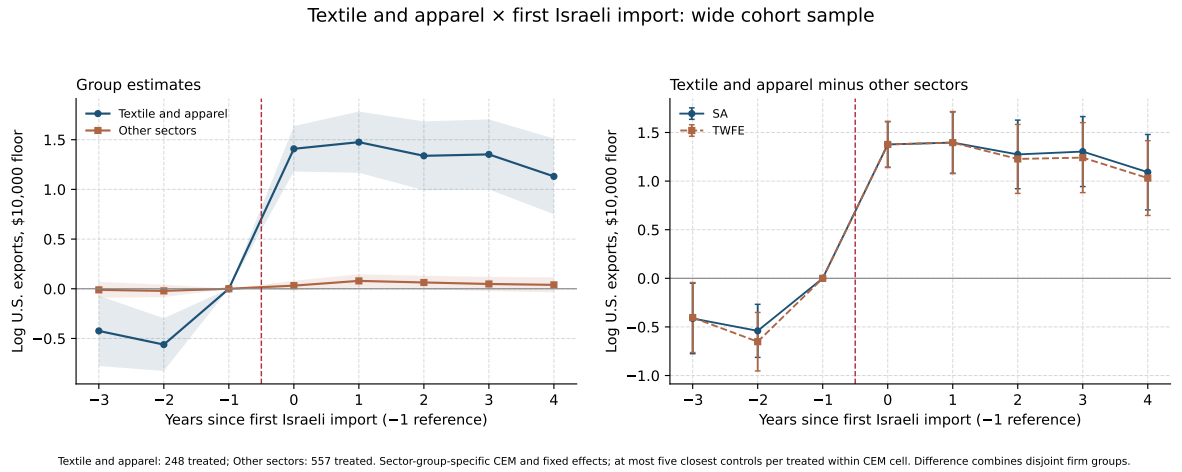


Figure D16: Textile and apparel versus other sectors: matched first-Israeli-import event contrast, 2006–2015 cohorts.

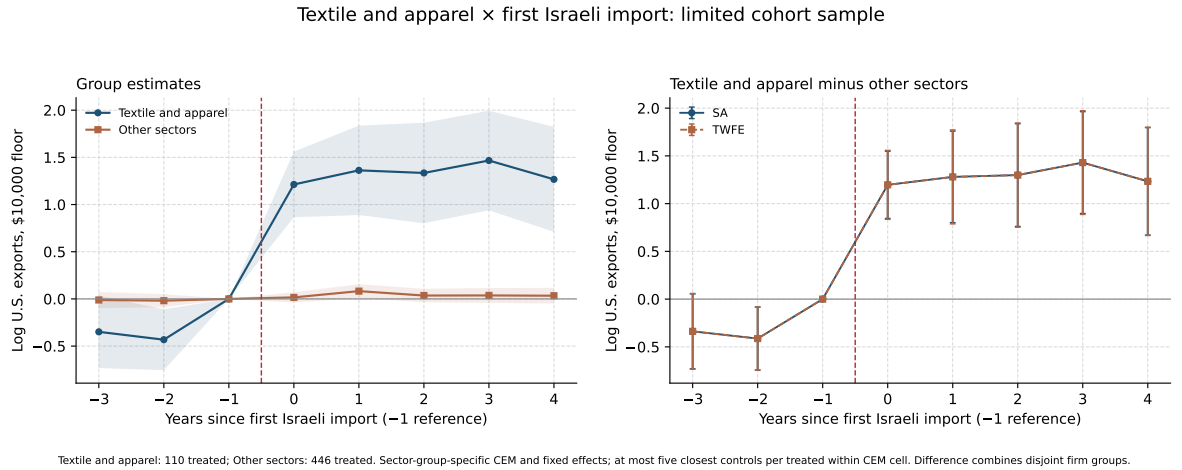


Figure D17: Textile and apparel versus other sectors: matched first-Israeli-import event contrast, 2008–2012 cohorts.

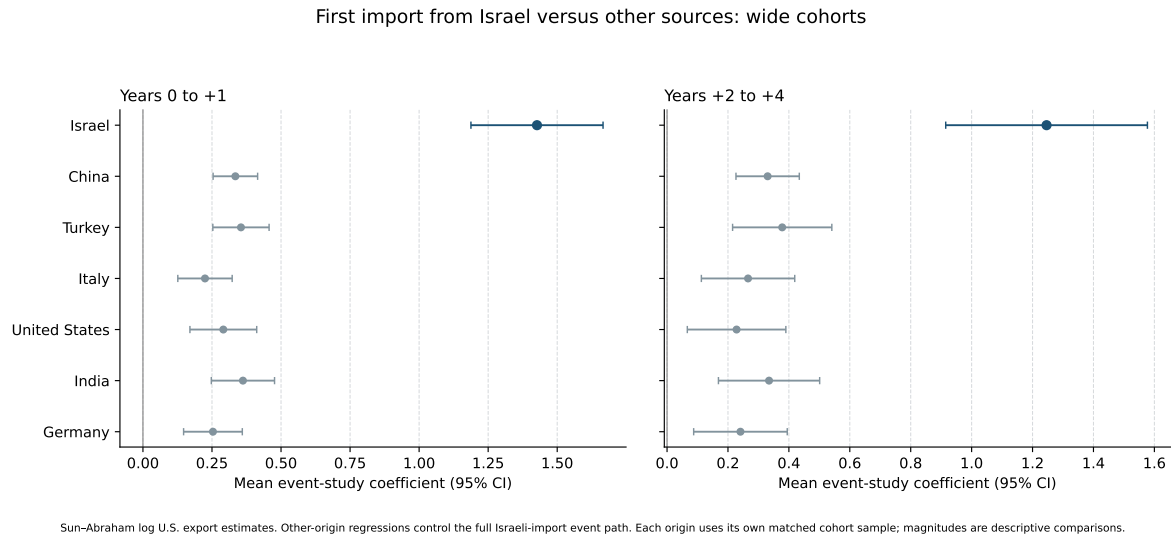


Figure D18: Post-event log U.S. export averages around first imports from Israel and six other source countries, 2006–2015 cohorts.

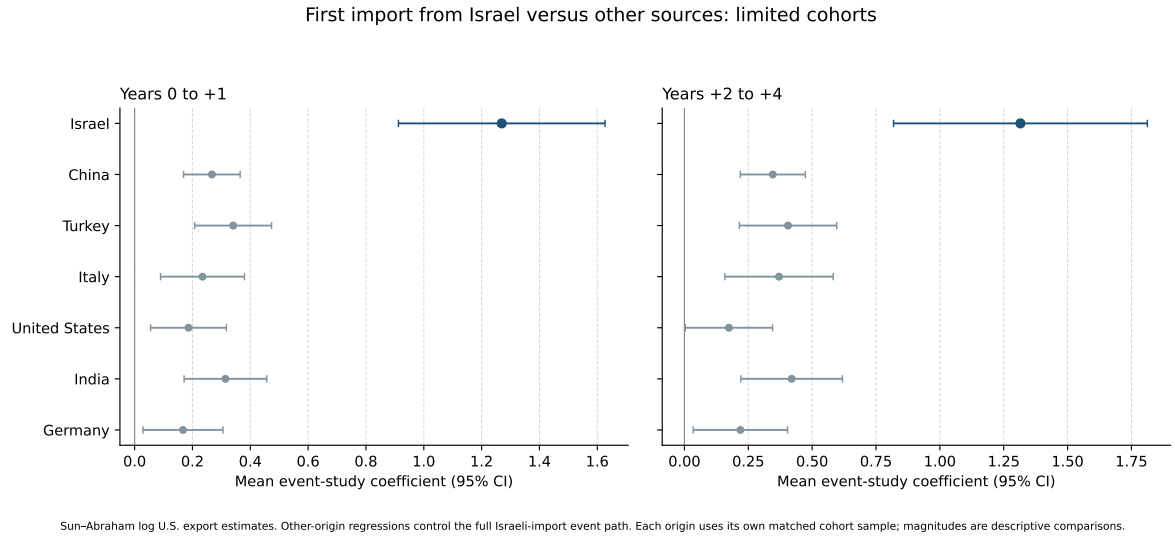


Figure D19: Post-event log U.S. export averages around first imports from Israel and six other source countries, 2008–2012 cohorts.

E Survey Mechanisms: Additional Specifications

Tables E1–E4 extend the survey exercise in Section 6. Columns (1) and (2) report the firm-panel and repeated cross-sectional specifications. Columns (3) and (4) add industry-by-survey-wave fixed effects to each estimator. These additional effects absorb the common MFN-by-post response, leaving the exporter–other-firm difference identified. Each table varies the timing and definition of exporter status and reports both firm- and industry-clustered bootstrap tests. For 2004 status, both estimators use the same linked sample and fixed baseline industry exposure. The ever-exporter cross-section instead uses all retained observations and assigns the 2004 tariff using the industry reported in each wave. Differences across the ever-panel and full cross-section may therefore reflect sample composition as well as specification.

Table E1 shows that certification responds positively across these comparisons. For ever direct-or-indirect exporters, adding industry-by-wave effects to the panel yields a coefficient of 1.428 with a firm-bootstrap p-value of 0.026. The corresponding cross-sectional coefficients are 0.993 without these effects and 1.138 with them, with firm-bootstrap p-values of 0.091 and 0.052. The industry-bootstrap p-values for the two panel estimates are 0.053 and 0.060, respectively, indicating weaker precision when allowing for common industry dependence. Direct-only and 2004 exporter definitions also retain positive certification estimates.

Tables E2–E4 show no consistent positive exporter differential for the other outcomes. In particular, the foreign-ownership differential is negative for 2004 exporters, while the ever-exporter estimates are small and statistically indistinguishable from zero. Foreign technology and investment also provide no statistically significant positive exporter differential. The survey evidence therefore supports a cautious interpretation centered on certification.

Table E1: Survey Mechanisms: International Certification

	(1) Panel	(2) Cross-section	(3) Panel	(4) Cross-section
A. Ever direct or indirect exporters				
MFN × Post × Exporter	1.612	0.993	1.428	1.138
	(0.690)	(0.560)	(0.735)	(0.559)
Firm-bootstrap p-value	0.011	0.091	0.026	0.052
Industry-bootstrap p-value	0.053	0.327	0.060	0.250
Observations	2,120	7,662	2,120	7,662
Firm clusters	655	4,969	655	4,969
Industry clusters	34	48	34	48
B. Ever direct exporters				
MFN × Post × Exporter	2.258	1.444	2.090	1.560
	(0.789)	(0.634)	(0.827)	(0.634)
Firm-bootstrap p-value	0.002	0.026	0.002	0.016
Industry-bootstrap p-value	0.021	0.319	0.031	0.294
Observations	2,116	7,658	2,116	7,658
Firm clusters	654	4,968	654	4,968
Industry clusters	34	48	34	48
C. Direct or indirect exporters in 2004				
MFN × Post × Exporter	2.443	2.430	2.443	2.606
	(1.151)	(0.910)	(1.208)	(0.940)
Firm-bootstrap p-value	0.012	0.006	0.008	0.003
Industry-bootstrap p-value	0.042	0.040	0.041	0.063
Observations	2,134	2,134	2,134	2,134
Firm clusters	659	659	659	659
Industry clusters	35	35	35	35
D. Direct exporters in 2004				
MFN × Post × Exporter	2.563	2.541	2.621	2.792
	(1.259)	(0.999)	(1.335)	(1.046)
Firm-bootstrap p-value	0.015	0.009	0.014	0.005
Industry-bootstrap p-value	0.058	0.101	0.075	0.119
Observations	2,134	2,134	2,134	2,134
Firm clusters	659	659	659	659
Industry clusters	35	35	35	35
Firm FE	Yes	No	Yes	No
Industry × exporter-status FE	No	Yes	No	Yes
Exporter-status × wave FE	Yes	Yes	Yes	Yes
Industry × wave FE	No	No	Yes	Yes

Notes: Coefficients are exporter–other-firm differences in outcome percentage points per one tariff percentage point. Firm-clustered standard errors are in parentheses. Both bootstrap rows use 999 restricted Rademacher draws. Industry clustering uses baseline ISIC3 for the linked sample and reported ISIC3 for the full cross-section. Panel samples retain firms observed in 2004 and later; the full ever-exporter cross-section includes firms first surveyed after 2004. For direct-only status, other firms can include indirect-only exporters. Ever status uses post-agreement information. All models use manufacturing, excluding dairy, tobacco, and footwear.

Table E2: Survey Mechanisms: Foreign Ownership Share

	(1) Panel	(2) Cross-section	(3) Panel	(4) Cross-section
A. Ever direct or indirect exporters				
MFN × Post × Exporter	0.127	0.110	0.066	0.263
	(0.290)	(0.280)	(0.297)	(0.294)
Firm-bootstrap p-value	0.580	0.675	0.787	0.374
Industry-bootstrap p-value	0.791	0.879	0.848	0.702
Observations	2,118	7,665	2,118	7,665
Firm clusters	655	4,979	655	4,979
Industry clusters	34	48	34	48
B. Ever direct exporters				
MFN × Post × Exporter	0.001	0.221	-0.031	0.370
	(0.330)	(0.311)	(0.341)	(0.321)
Firm-bootstrap p-value	0.995	0.507	0.920	0.247
Industry-bootstrap p-value	0.999	0.827	0.942	0.670
Observations	2,114	7,661	2,114	7,661
Firm clusters	654	4,978	654	4,978
Industry clusters	34	48	34	48
C. Direct or indirect exporters in 2004				
MFN × Post × Exporter	-0.576	-0.560	-0.645	-0.612
	(0.351)	(0.302)	(0.329)	(0.283)
Firm-bootstrap p-value	0.046	0.058	0.017	0.034
Industry-bootstrap p-value	0.062	0.064	0.029	0.037
Observations	2,132	2,132	2,132	2,132
Firm clusters	659	659	659	659
Industry clusters	35	35	35	35
D. Direct exporters in 2004				
MFN × Post × Exporter	-0.606	-0.576	-0.684	-0.633
	(0.386)	(0.333)	(0.367)	(0.319)
Firm-bootstrap p-value	0.056	0.072	0.025	0.045
Industry-bootstrap p-value	0.069	0.082	0.038	0.055
Observations	2,132	2,132	2,132	2,132
Firm clusters	659	659	659	659
Industry clusters	35	35	35	35
Firm FE	Yes	No	Yes	No
Industry × exporter-status FE	No	Yes	No	Yes
Exporter-status × wave FE	Yes	Yes	Yes	Yes
Industry × wave FE	No	No	Yes	Yes

Notes: Coefficients are exporter–other-firm differences in outcome percentage points per one tariff percentage point. Firm-clustered standard errors are in parentheses. Both bootstrap rows use 999 restricted Rademacher draws. Industry clustering uses baseline ISIC3 for the linked sample and reported ISIC3 for the full cross-section. Panel samples retain firms observed in 2004 and later; the full ever-exporter cross-section includes firms first surveyed after 2004. For direct-only status, other firms can include indirect-only exporters. Ever status uses post-agreement information. All models use manufacturing, excluding dairy, tobacco, and footwear.

Table E3: Survey Mechanisms: Foreign-Licensed Technology

	(1) Panel	(2) Cross-section	(3) Panel	(4) Cross-section
A. Ever direct or indirect exporters				
MFN × Post × Exporter	0.167	-0.341	-0.101	-0.596
	(0.882)	(0.713)	(0.963)	(0.733)
Firm-bootstrap p-value	0.803	0.650	0.900	0.417
Industry-bootstrap p-value	0.740	0.497	0.849	0.192
Observations	2,091	7,630	2,091	7,630
Firm clusters	647	4,961	647	4,961
Industry clusters	34	48	34	48
B. Ever direct exporters				
MFN × Post × Exporter	0.207	-0.723	-0.112	-1.116
	(0.981)	(0.804)	(1.066)	(0.830)
Firm-bootstrap p-value	0.805	0.364	0.894	0.172
Industry-bootstrap p-value	0.586	0.233	0.784	0.076
Observations	2,087	7,626	2,087	7,626
Firm clusters	646	4,960	646	4,960
Industry clusters	34	48	34	48
C. Direct or indirect exporters in 2004				
MFN × Post × Exporter	-0.476	-0.679	-0.426	-0.480
	(1.408)	(1.182)	(1.509)	(1.241)
Firm-bootstrap p-value	0.706	0.587	0.739	0.710
Industry-bootstrap p-value	0.433	0.336	0.366	0.368
Observations	2,105	2,105	2,105	2,105
Firm clusters	651	651	651	651
Industry clusters	35	35	35	35
D. Direct exporters in 2004				
MFN × Post × Exporter	-0.938	-1.117	-0.885	-0.886
	(1.528)	(1.282)	(1.636)	(1.347)
Firm-bootstrap p-value	0.479	0.390	0.503	0.515
Industry-bootstrap p-value	0.166	0.155	0.108	0.150
Observations	2,105	2,105	2,105	2,105
Firm clusters	651	651	651	651
Industry clusters	35	35	35	35
Firm FE	Yes	No	Yes	No
Industry × exporter-status FE	No	Yes	No	Yes
Exporter-status × wave FE	Yes	Yes	Yes	Yes
Industry × wave FE	No	No	Yes	Yes

Notes: Coefficients are exporter–other-firm differences in outcome percentage points per one tariff percentage point. Firm-clustered standard errors are in parentheses. Both bootstrap rows use 999 restricted Rademacher draws. Industry clustering uses baseline ISIC3 for the linked sample and reported ISIC3 for the full cross-section. Panel samples retain firms observed in 2004 and later; the full ever-exporter cross-section includes firms first surveyed after 2004. For direct-only status, other firms can include indirect-only exporters. Ever status uses post-agreement information. All models use manufacturing, excluding dairy, tobacco, and footwear.

Table E4: Survey Mechanisms: Fixed-Asset Investment

	(1) Panel	(2) Cross-section	(3) Panel	(4) Cross-section
A. Ever direct or indirect exporters				
MFN × Post × Exporter	0.302	1.155	-0.317	1.200
	(1.494)	(1.008)	(1.597)	(1.045)
Firm-bootstrap p-value	0.797	0.268	0.807	0.252
Industry-bootstrap p-value	0.753	0.301	0.769	0.300
Observations	1,744	7,485	1,744	7,485
Firm clusters	541	4,923	541	4,923
Industry clusters	32	48	32	48
B. Ever direct exporters				
MFN × Post × Exporter	0.567	1.162	0.152	1.254
	(1.646)	(1.118)	(1.722)	(1.154)
Firm-bootstrap p-value	0.665	0.291	0.912	0.287
Industry-bootstrap p-value	0.470	0.209	0.875	0.182
Observations	1,741	7,482	1,741	7,482
Firm clusters	540	4,922	540	4,922
Industry clusters	32	48	32	48
C. Direct or indirect exporters in 2004				
MFN × Post × Exporter	-0.228	0.084	-1.242	-0.762
	(2.073)	(1.693)	(2.225)	(1.818)
Firm-bootstrap p-value	0.880	0.957	0.487	0.666
Industry-bootstrap p-value	0.880	0.955	0.416	0.616
Observations	1,754	1,754	1,754	1,754
Firm clusters	544	544	544	544
Industry clusters	33	33	33	33
D. Direct exporters in 2004				
MFN × Post × Exporter	0.004	0.405	-0.870	-0.273
	(2.154)	(1.749)	(2.293)	(1.862)
Firm-bootstrap p-value	0.998	0.812	0.639	0.886
Industry-bootstrap p-value	0.999	0.742	0.586	0.881
Observations	1,754	1,754	1,754	1,754
Firm clusters	544	544	544	544
Industry clusters	33	33	33	33
Firm FE	Yes	No	Yes	No
Industry × exporter-status FE	No	Yes	No	Yes
Exporter-status × wave FE	Yes	Yes	Yes	Yes
Industry × wave FE	No	No	Yes	Yes

Notes: Coefficients are exporter–other-firm differences in outcome percentage points per one tariff percentage point. Firm-clustered standard errors are in parentheses. Both bootstrap rows use 999 restricted Rademacher draws. Industry clustering uses baseline ISIC3 for the linked sample and reported ISIC3 for the full cross-section. Panel samples retain firms observed in 2004 and later; the full ever-exporter cross-section includes firms first surveyed after 2004. For direct-only status, other firms can include indirect-only exporters. Ever status uses post-agreement information. All models use manufacturing, excluding dairy, tobacco, and footwear.

F Equilibrium, Exact-Hat Algebra, and Numerical Solution

This appendix completes the equilibrium definition in Section 7 and states the system used for the counterfactuals. The baseline is the observed QIZ equilibrium. Foreign demand, foreign input prices, tariffs, productivity distributions, and fixed costs remain unchanged; the policy changes the Israeli content requirement or removes the QIZ option. No upgrading or certification decision is included in this benchmark.

F.1 Equilibrium definition and market clearing

Given the content requirements $(\bar{e}, r, \bar{q})$ and whether QIZ is available, an equilibrium consists of a wage w, positive producer masses $\{M_s\}_{s=1}^{S}$, input allocations, prices, household demand, and export participation such that:

1. Households maximize the preferences in Equation (7) subject to income $Y = wL + \mathcal{B}$. They spend $\alpha_s Y$ on sector s and allocate that expenditure across Egyptian and imported varieties according to CES demand.

2. Firms minimize ordinary production costs and, where available, QIZ production costs subject to all three content constraints. They charge the constant markup $\mu_s = \sigma_s/(\sigma_s - 1)$ and select the most profitable export options in Equation (15).

3. Free entry equates expected profits before establishment costs to $wf_{e,s}$ in every producing sector.

4. Each variety's output meets its domestic and foreign demand, including any iceberg shipment costs. Labor used in establishment, fixed export activities, final production, and Egyptian intermediate production sums to the endowment L.

Every entrant produces domestically because there is no fixed domestic operating cost. The productivity distribution is unchanged across equilibria. Consequently, domestic producer exit following an unfavorable draw is not a separate selection margin.

For clarity, domestic demand for the variety of a firm with productivity φ is

$$q_{D,s}(\varphi) = \alpha_s Y P_s^{\sigma_s - 1} \left(\frac{\mu_s a_{M,s}}{\varphi} \right)^{-\sigma_s}. \tag{28}$$

Here, D denotes the domestic destination, P_s is its sectoral consumption price index, and $a_{M,s}$ is ordinary unit cost before dividing by productivity. Integrating

price times quantity across Egyptian firms yields

$$\begin{aligned} H_{D,s} &= M_s \int \left(\frac{\mu_s a_{M,s}}{\varphi}\right)^{1-\sigma_s} dG_s(\varphi), \\ P_s^{1-\sigma_s} &= H_{D,s} + H_{F,s}, \\ R_{D,s} &= \alpha_s Y \frac{H_{D,s}}{H_{D,s} + H_{F,s}}. \end{aligned} \tag{29}$$

The terms $H_{D,s}$ and $H_{F,s}$ are the Egyptian and foreign contributions to the CES price index. Thus, goods-market clearing equates aggregate domestic sales to expenditure allocated to Egyptian varieties, not to all sectoral expenditure. The remainder purchases imported final goods. Foreign varieties are available at fixed prices, so their contribution $H_{F,s}$ is fixed, but their sales adjust.

Let $N_{R,s}$, $N_{U,s}$, and $N_{Q,s}$ denote the masses of non-U.S., U.S., and QIZ exporters. Let $X_{R,s}$, $X_{M,s}$, and $X_{Q,s}$ be their export revenues, where M and Q identify the mutually exclusive U.S. regimes. All U.S. exporters pay $f_{U,s}$; QIZ users additionally pay $f_{Q,s}$. Aggregate free entry is

$$\begin{aligned} M_s w f_{e,s} = \frac{R_{D,s} + X_{R,s} + X_{M,s} + X_{Q,s}}{\sigma_s} \\ - w\left(f_{R,s}N_{R,s} + f_{U,s}N_{U,s} + f_{Q,s}N_{Q,s}\right). \end{aligned} \tag{30}$$

This follows by integrating maximized firm profits over productivity and multiplying by M_s. It does not require every firm's realized profits to equal its establishment cost.

Define variable production costs and their domestic resource shares as

$$\begin{aligned} V_{M,s} &= \frac{R_{D,s} + X_{R,s} + X_{M,s}}{\mu_s}, \\ V_{Q,s} &= \frac{X_{Q,s}}{\mu_s}, \\ d_{m,s} &= t_{L,s}^m + t_{E,s}^m, \qquad m \in \{M, Q\}. \end{aligned} \tag{31}$$

The superscript m identifies the sourcing technology; subscripts L and E identify direct labor and Egyptian inputs. Egyptian inputs are produced with one unit of labor per unit of output and sell at w. Their labor is therefore included through $t_{E,s}^m$, giving

$$\begin{aligned} L = \sum_s \Bigg[M_s f_{e,s} + f_{R,s}N_{R,s} + f_{U,s}N_{U,s} + f_{Q,s}N_{Q,s} \\ + \frac{d_{M,s}V_{M,s} + d_{Q,s}V_{Q,s}}{w} \Bigg]. \end{aligned} \tag{32}$$

These are physical labor requirements. In particular, $d_{Q,s}$ is calculated from actual expenditure shares, not replaced by the wage elasticity of constrained unit cost. The two can differ because the value-based sourcing constraints themselves depend

on input prices. Egyptian intermediate sales are not added to household final demand.

F.2 Exact changes in costs, demand, and income

For a positive baseline quantity z, define $\widehat{z} = z'/z$, with a prime denoting its counterfactual value. I normalize the baseline wage to one. Labor supply L and the nominal transfer $\mathcal{B}$ are fixed, so, with $b = \mathcal{B}/Y$,

$$\begin{aligned} \widehat{Y} &= (1-b)\widehat{w} + b, \\ \widehat{P} &= \prod_s \widehat{P}_s^{\alpha_s}, \\ \widehat{W} &= \frac{\widehat{Y}}{\widehat{P}}. \end{aligned} \tag{33}$$

The first equality follows by dividing $Y' = w'L + \mathcal{B}$ by baseline income. The remaining equalities follow from Cobb–Douglas expenditure allocation and the definition of real income. These are exact equilibrium changes, not a first-order approximation (Dekle et al., 2008).

Let $\delta_{o,s}$ denote origin o's baseline share of intermediate expenditure, rather than its share of total production cost:

$$\delta_{o,s} = \frac{\beta_{o,s} p_{o,s}^{1-\eta_s}}{\sum_h \beta_{h,s} p_{h,s}^{1-\eta_s}}. \tag{34}$$

The indices $o, h \in \{E, I, U, R\}$ identify input origins. Dividing the ordinary CES unit-cost function by its baseline value gives

$$\widehat{a}_{M,s} = \widehat{w}^{\,1-\gamma_s} \left[\sum_o \delta_{o,s} \widehat{p}_{o,s}^{1-\eta_s} \right]^{\gamma_s/(1-\eta_s)}. \tag{35}$$

Here, $\widehat{p}_{E,s} = \widehat{w}$, while all foreign input-price hats equal one.

QIZ cost is not updated with a fixed compliance wedge. At each trial wage and policy, I solve the constrained sourcing problem anew. To state its numerical implementation, let $\boldsymbol{t} = (t_L, t_E, t_I, t_U, t_R)$ denote shares of total variable cost and $\nu_s = (\eta_s - 1)/\eta_s$. Output per unit of expenditure at productivity one is

$$F_s(\boldsymbol{t}; w', \boldsymbol{p}') = \left(\frac{t_L}{w'} \right)^{1-\gamma_s} \left[\sum_o \beta_{o,s}^{1/\eta_s} \left(\frac{t_o}{p'_{o,s}} \right)^{\nu_s} \right]^{\gamma_s/\nu_s}. \tag{36}$$

Substituting $\ell = t_L/w'$ and $x_o = t_o/p'_{o,s}$ into production gives this expression.

Constant returns imply

$$
\begin{aligned}
(a'_{Q,s})^{-1} = \max_{\boldsymbol{t}\geq 0} \quad & F_s(\boldsymbol{t}; w', \boldsymbol{p}') \\
\text{subject to} \quad & \sum_{h\in\{L,E,I,U,R\}} t_h = 1, \\
& t_L + t_E \geq \mu_s \bar{e}, \\
& t_I \geq \mu_s r', \\
& t_L + t_E + t_I + t_U \geq \mu_s \bar{q}.
\end{aligned}
\tag{37}
$$

The factors μ_s convert requirements expressed relative to appraised output value into cost shares. No constraint is assumed to bind in advance.

Because foreign demand and tariffs remain fixed, the revenue coefficients in Equation (12) satisfy

$$
\begin{aligned}
\widehat{B}_{R,s} &= \widehat{a}_{M,s}^{1-\sigma_s}, \\
\widehat{B}_{U,s}^{M} &= \widehat{a}_{M,s}^{1-\sigma_s}, \\
\rho'_s &= (1+\tau_s)^{\sigma_s} \left(\frac{a'_{Q,s}}{a'_{M,s}} \right)^{1-\sigma_s}.
\end{aligned}
\tag{38}
$$

The sectoral tariff factor is the maintained export-weighted average of product-level factors. When QIZ is available in both equilibria, $\rho'_s/\rho_s = (\widehat{a}_{Q,s}/\widehat{a}_{M,s})^{1-\sigma_s}$. When it is removed, the QIZ option is excluded rather than assigned a revenue hat.

Since every entrant produces domestically and G_s is fixed, the Egyptian contribution to the price index changes by

$$
\widehat{H}_{D,s} = \widehat{M}_s \widehat{a}_{M,s}^{1-\sigma_s}. \tag{39}
$$

Using the baseline domestic expenditure share $\lambda_{D,s}$ and fixed $H_{F,s}$ then yields

$$
\begin{aligned}
\widehat{P}_s &= \left[1 - \lambda_{D,s} + \lambda_{D,s} \widehat{H}_{D,s} \right]^{1/(1-\sigma_s)}, \\
\widehat{R}_{D,s} &= \widehat{Y} \frac{\widehat{H}_{D,s}}{1 - \lambda_{D,s} + \lambda_{D,s} \widehat{H}_{D,s}}.
\end{aligned}
\tag{40}
$$

Thus, final imports remain endogenous even though foreign prices and the foreign-variety contribution to the price index are fixed.

F.3 Export participation and finite-support aggregation

Write $k_s = \sigma_s - 1$. The ordinary export cutoffs in Equation (16) update as

$$\begin{aligned} \varphi'_{R,s} &= \varphi_{R,s} \left(\frac{\widehat{w}}{\widehat{B}_{R,s}} \right)^{1/k_s}, \\ \varphi'_{M,s} &= \varphi_{M,s} \left(\frac{\widehat{w}}{\widehat{B}^M_{U,s}} \right)^{1/k_s}. \end{aligned} \tag{41}$$

Fixed costs are unchanged in labor units, but their monetary cost changes with the wage.

For $\rho_s > 1$, let $\varphi_{Q0,s}$ denote the productivity at which QIZ profits first cover all U.S. fixed costs, and $\varphi_{QM,s}$ the productivity at which QIZ dominates MFN:

$$\begin{aligned} \varphi_{Q0,s} &= \left[\frac{\sigma_s w (f_{U,s} + f_{Q,s})}{\rho_s B^M_{U,s}} \right]^{1/k_s}, \\ \varphi_{QM,s} &= \left[\frac{\sigma_s w f_{Q,s}}{(\rho_s - 1) B^M_{U,s}} \right]^{1/k_s}, \\ \varphi^*_s &= \max\{\varphi_{Q0,s}, \varphi_{QM,s}\}, \\ \varphi_{U,s} &= \min\{\varphi_{M,s}, \varphi_{Q0,s}\}. \end{aligned} \tag{42}$$

Here, φ^*_s is the QIZ cutoff and $\varphi_{U,s}$ is the cutoff for exporting to the United States under either regime. If QIZ is unavailable or $\rho_s \leq 1$, set $\varphi^*_s = \varphi_{Q0,s} = \infty$ and use the MFN cutoff for U.S. entry. Counterfactual thresholds use the same formulas with primed wages, revenue coefficients, and revenue advantages. All fifteen sectors are evaluated; baseline nonparticipation does not preclude later QIZ adoption.

To integrate these decisions, define the survival probability $\mathcal{S}_s(c)$ and productivity moment $\mathcal{T}_s(c)$ above a cutoff c:

$$\begin{aligned} \mathcal{S}_s(c) &= \int_{\varphi \geq c} dG_s(\varphi), \\ \mathcal{T}_s(c) &= \int_{\varphi \geq c} \varphi^{k_s} dG_s(\varphi). \end{aligned} \tag{43}$$

For the truncated Pareto distribution, write $u_{0,s} = \underline{\varphi}_s / \overline{\varphi}_s$, $\zeta_s = \theta_s - k_s$, and

$$u_s(c) = \min\{1, \max\{u_{0,s}, c/\overline{\varphi}_s\}\}.$$

Integrating the Pareto density gives

$$\begin{aligned}
\mathcal{S}_s(c) &= \frac{u_s(c)^{-\theta_s} - 1}{u_{0,s}^{-\theta_s} - 1}, \\
\mathcal{T}_s(c) &= \frac{\theta_s \overline{\varphi}_s^{k_s}}{u_{0,s}^{-\theta_s} - 1} h_{\zeta_s}(u_s(c)), \\
h_\zeta(u) &= \begin{cases} (u^{-\zeta} - 1)/\zeta, & \zeta \neq 0, \\ -\log u, & \zeta = 0. \end{cases}
\end{aligned} \tag{44}$$

Both functions equal zero above the upper support. The logarithmic expression is the continuous limit at $\theta_s = k_s$. Finite support allows either sign of ζ_s; no restriction $\theta_s > \sigma_s - 1$ is imposed.

Aggregate foreign sales and exporter masses are therefore

$$\begin{aligned}
X_{R,s} &= M_s B_{R,s} \mathcal{T}_s(\varphi_{R,s}), \\
X_{M,s} &= M_s B_{U,s}^M [\mathcal{T}_s(\varphi_{M,s}) - \mathcal{T}_s(\varphi_s^*)]_+, \\
X_{Q,s} &= M_s B_{U,s}^M \rho_s \mathcal{T}_s(\varphi_s^*), \\
N_{R,s} &= M_s \mathcal{S}_s(\varphi_{R,s}), \\
N_{U,s} &= M_s \mathcal{S}_s(\varphi_{U,s}), \\
N_{Q,s} &= M_s \mathcal{S}_s(\varphi_s^*).
\end{aligned} \tag{45}$$

The notation $[x]_+ = \max\{x, 0\}$ ensures that MFN sales vanish when all U.S. exporters prefer QIZ. When the program is unavailable, set $X_{Q,s} = N_{Q,s} = 0$. These level expressions handle zero baseline use, entry, and regime switching without dividing by zero.

F.4 The system solved in exact hats

Let $\mathcal{E}_s = M_s w f_{e,s}$ denote baseline establishment expenditure. Dividing Equation (30) in the counterfactual by $\mathcal{E}_s$ gives

$$\begin{aligned}
\widehat{M}_s \widehat{w} = \frac{1}{\mathcal{E}_s} \Bigg[& \frac{R'_{D,s} + X'_{R,s} + X'_{M,s} + X'_{Q,s}}{\sigma_s} \\
& - w' \left(f_{R,s} N'_{R,s} + f_{U,s} N'_{U,s} + f_{Q,s} N'_{Q,s} \right) \Bigg].
\end{aligned} \tag{46}$$

The left side is counterfactual establishment expenditure relative to baseline expenditure. The right side is aggregate operating profits, after fixed export costs, expressed in the same units. Export levels and counts are recomputed using Equation (45), while domestic revenue follows Equation (40).

The unknown vector is

$$\boldsymbol{z} = \left(\log \widehat{w}, \log \widehat{M}_1, \ldots, \log \widehat{M}_S\right). \tag{47}$$

For each candidate vector, the sourcing problems, household demand, and participation formulas determine all other quantities. The S free-entry equations and counterfactual labor-market clearing then provide $S+1$ equations for $S+1$ unknowns. In the application, $S = 15$, so the nonlinear system has sixteen unknowns. Exact-hat algebra removes unnecessary scale parameters but does not eliminate the fixed costs governing participation.

F.5 Numerical algorithm and verification

I solve the system using the following procedure.

1. *Establish the benchmark.* Fix the calibrated parameters and baseline accounts. Verify that the observed policy evaluated at $\boldsymbol{z} = \mathbf{0}$ reproduces unit hats and clears the free-entry and labor equations.

2. *Update sourcing.* At a trial vector, exponentiate $\boldsymbol{z}$ to obtain positive wages and producer masses. Compute income and ordinary unit costs. If the unrestricted input allocation satisfies every QIZ rule, use it directly; otherwise solve Equation (37) by constrained optimization from multiple starting allocations, retaining the best feasible solution.

3. *Update participation and sales.* Recompute ρ'_s, compare MFN, QIZ, and nonexporting profits, and evaluate the finite-support integrals for every sector. Calculate domestic prices and household demand, foreign sales, exporter counts, and variable production costs.

4. *Clear entry and labor.* Form the free-entry residuals as the left side minus the right side of Equation (46). Form the labor residual as counterfactual labor demand divided by L, minus one. A trust-region nonlinear least-squares solver updates $\boldsymbol{z}$ until these residuals satisfy the acceptance criterion.

5. *Check and record the equilibrium.* Verify sourcing feasibility, free entry, labor clearing, and the external-account identity below. Compute welfare and trade outcomes only after these checks. Use neighboring policy solutions as starting values and check selected equilibria from alternative starts.

The implementation uses solver tolerances of 10^{-11} and accepts a solution only with successful solver termination and a maximum absolute normalized equilibrium residual below 3×10^{-7}. The saved policy grid has a maximum residual below 5.5×10^{-12}. These checks establish numerical accuracy for the reported solutions, not global equilibrium uniqueness. The log parameterization maintains positive producer masses; a policy inducing complete producer exit from a sector would require a complementarity formulation.

For an additional accounting check, let M^{final} denote imported final expenditure, M^{int} imported intermediate expenditure, and X_E total Egyptian exports. Then

$$\begin{aligned} M^{\text{final}} &= Y - \sum_s R_{D,s}, \\ M^{\text{int}} &= \sum_s \sum_{o \in \{I,U,R\}} \left(t^M_{o,s} V_{M,s} + t^Q_{o,s} V_{Q,s} \right), \\ X_E &= \sum_s (X_{R,s} + X_{M,s} + X_{Q,s}). \end{aligned} \tag{48}$$

Free entry equates aggregate markups to establishment and fixed export costs. Combining this with physical labor clearing gives aggregate producer revenue equal to $wL + M^{\text{int}}$. Subtracting domestic final sales and using household income therefore yields

$$M^{\text{final}} + M^{\text{int}} - X_E = Y - wL = \mathcal{B}. \tag{49}$$

This identity follows from the equilibrium conditions and is checked rather than imposed as an additional equation. The transfer is fixed; balanced trade is not assumed. U.S. duties are paid abroad and are excluded from Egyptian producer revenue and household income.

F.6 Industry-level validation

Table 5 compares untargeted industry exposure responses under a common measurement scheme. It uses sectoral DiD estimates rather than the product-level DDD specifications or firm first-import event studies elsewhere in the paper. The benchmark parameters are unchanged, and none of the four coefficients is used in calibration.

Aggregation and predetermined exposure. Egyptian U.S. and non-U.S. export values are summed from products into the fifteen model sectors. UNIDO employment is harmonized across industry revisions and summed into the same sectors, requiring the constituent industries to be observed. The selected employment series does not use author extrapolation or replace missing employment with zero; provider estimates retained in the source data remain included.

For each industry s, the exposure score is

$$\text{Exposure}_s = \frac{\sum_{p \in s} \tau_{p,2004} X^{\text{World}}_{p,2004}}{\sum_{p \in s} X^{\text{World}}_{p,2004}}, \tag{50}$$

where p indexes products, $\tau_{p,2004}$ is the actual U.S. MFN tariff in percentage points, and $X^{\text{World}}_{p,2004}$ is Egypt's export value to all destinations. Both tariffs and export weights are fixed before implementation. The score is not standardized and has no additional U.S.-orientation multiplier. Product records are used to construct

industry totals and baseline exposure, not to create product-level model responses.

Israeli inputs require distinguishing the industry supplying the traded product from the Egyptian industry using it. For supplier industry i and purchasing industry s, let $Z_{is}^{08/09}$ denote the corresponding input-output transaction value. The allocated empirical purchases are

$$\begin{aligned} a_{is}^{08/09} &= \frac{Z_{is}^{08/09}}{\sum_j Z_{ij}^{08/09}}, \\ \widetilde{M}_{I,s,t} &= \sum_i a_{is}^{08/09} M_{I,i,t}^{\mathrm{INT}}, \end{aligned} \tag{51}$$

where j ranges over all purchasing industries in the IO account and $M_{I,i,t}^{\mathrm{INT}}$ is Israel-reported intermediate-goods exports to Egypt from supplying industry i in year t. Thus $a_{is}^{08/09}$ is the share of supplier i's intermediate sales allocated to buyer s; it is not an input expenditure share within buyer s. The denominator retains nonmanufacturing users, rather than renormalizing their purchases onto manufacturing. The allocation uses neither MFN exposure nor post-treatment trade weights, but its fixed IO shares are measured after the agreement.

The 2008/09 table identifies fourteen groups consistent with the model: sectors G01–G13 separately, and furniture (G14) combined with other manufacturing, repair, and installation (G15). Israeli purchases are summed into these groups in each model equilibrium. The combined group's exposure is recomputed from the tariff numerator and baseline-export denominator in Equation (50), not obtained by averaging its component exposure scores. The IO allocation is used only to measure empirical purchases; it does not add an endogenous input-output production network to the model. Both the allocation proxy and its post-agreement vintage limit the interpretation of this comparison.

Data responses. For each outcome, I estimate the conditional mean

$$\mathbb{E}[Y_{s,t} \mid s, t, \mathrm{Exposure}_s] = \exp\big[\delta_s + \delta_t + \beta^{\mathrm{data}} \mathrm{Exposure}_s \mathbf{1}\{t \geq 2005\}\big] \tag{52}$$

by PPML over 1998–2020. Here $Y_{s,t}$ is the industry export value, allocated Israeli purchases, or employment; δ_s and δ_t are industry and year fixed effects, and the indicator equals one from 2005 onward. Each outcome has its own coefficient and fixed effects. The comparison uses fifteen industries for exports and employment and fourteen for Israeli purchases, yielding 345 and 322 industry-year observations, respectively. Baseline exposure is an industry's own weighted MFN tariff, not the downstream supplier-exposure index used in the detailed Israeli-import DDD.

The data coefficients, with sector-clustered standard errors in parentheses, are 0.06517 (0.02802) for U.S. exports, 0.01924 (0.06123) for non-U.S. exports, 0.04726 (0.02468) for Israeli purchases, and 0.00795 (0.03727) for employment. The corresponding 99,999-draw null-imposed Webb sector-cluster score-bootstrap p-values are 0.195, 0.821, 0.166, and 0.897. These estimates are imprecise and the series

exhibit pre-policy movement; their role here is descriptive validation, not a new causal estimate or evidence establishing parallel trends.

Model responses. Let $q = 0$ denote the no-QIZ equilibrium and $q = 1$ the observed-QIZ equilibrium. For each outcome, the measurement regression uses

$$\widetilde{Y}_{s,q} = \exp\left[c_s + c_q + \beta^{\text{model}}\text{Exposure}_s q\right], \tag{53}$$

as the PPML fitted mean, where c_s and c_q are industry and policy-state fixed effects. The dependent observations are the model's equilibrium levels, not its fitted means: total U.S. exports under either regime, non-U.S. exports, ordinary-plus-QIZ Israeli purchases by using industry, and physical sector labor. Any necessary industry combination is performed on levels separately in each equilibrium before estimation.

To match the observation weights of the sectoral pre–post window, the no-QIZ levels are repeated for the seven pre-period years and the observed-QIZ levels for the sixteen post-period years. Within each state the model outcomes are constant, so year effects reduce to policy-state effects. This is a static comparison of equilibria, not a simulated historical transition or an event-study path. The estimated model coefficients are 0.09962, 0.00446, 0.09396, and 0.00303. Applying the same transformation $100[\exp(\beta)-1]$ to both data and model gives the entries in Table 5. These are exposure gradients, not national aggregate effects or derivatives obtained by changing a sector's tariff alone.

Interpretation and checks. Model labor includes production, auxiliary Egyptian intermediate production, establishment, and fixed market-access activities. It is not factory headcounts, and its aggregate is fixed by labor-market clearing. A positive labor-exposure coefficient therefore measures reallocation toward exposed industries, not net national job creation. The Israeli comparison likewise conditions on an IO-based empirical allocation rather than directly observed purchases by industry.

Independent profiled and full Poisson estimators reproduce all four model coefficients to better than 10^{-9}. Industry sums preserve the original equilibrium totals, and baseline exposure numerators and denominators are verified at both aggregation levels. Giving the two equilibria equal weight produces responses of 10.78%, 0.45%, 9.79%, and 0.30%, close to the main results. The U.S. gradient is concentrated in apparel: omitting apparel from the measurement regression reduces its fifteen-industry response to 0.49%; this omission does not remove apparel from the general-equilibrium model. No sampling significance is attached to deterministic model coefficients, and no formal joint-fit test is claimed.